A Duality Framework for Deconfounding with Instrument Variables and Surrogate Control Variables

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1 Introduction

Modern scientific studies increasingly combine designed perturbations, high-dimensional measurements, and imperfect auxiliary information to study causal or structural relationships. In such settings, the variable of scientific interest is often observed rather than directly assigned, and hence may remain associated with latent factors such as biological state, environmental background, batch variation, or measurement artifacts. At the same time, the data often contain ancillary variables that carry partial deconfounding information. Some variables act as instrument-like sources of exogenous variation, whereas others act as surrogate controls or negative-control measurements that are informative about latent confounding. A central statistical challenge is therefore not only whether one can use an instrument or a surrogate control, but how these two imperfect deconfounding channels relate to one another and how they should be combined.

Single-cell Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) perturbation studies provide a concrete setting in which deconfounding is both scientifically important and statistically delicate. For a fixed target gene g, let z_i denote the single-guide RNA (sgRNA) or guide-pair indicator assigned to cell i, let x_i denote the measured expression of the target gene, and let y_i denote the expression of a candidate downstream gene. The scientific goal is to characterize how variation in the perturbed target-gene expression propagates to downstream expression. A direct regression of y_i on x_i, however, can remain confounded because both quantities may be associated with latent cell state, sequencing depth, cell-cycle effects, batch variation, and other unobserved factors. Thus, randomization of the perturbation does not by itself imply exogeneity of the measured molecular exposure x_i.

This application naturally contains two distinct sources of deconfounding information. The perturbation indicator z_i provides an instrument-like source of variation because it is designed to shift x_i while being assigned independently of pre-perturbation latent cellular state. At the same time, single-cell measurements may provide surrogate-control variables w_i, such as factors constructed from negative-control or housekeeping genes, that carry information about latent cell state and technical variation. The instrument channel may be low-dimensional or weak, while the surrogate-control channel may be richer but relies on the chosen controls adequately capturing confounding variation without introducing post-treatment adjustment. This motivates studying the two channels jointly rather than treating either as uniformly preferable.

We study a linear deconfounding framework in which instrumental variables and surrogate controls generate two projected moment restrictions for the same structural parameter. In the scalar-outcome model $y_i = x_i^\top \beta + \varsigma_i$, the IV channel uses $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$, which is implied by the stronger classical condition $\mathbb{E}(z_i\varsigma_i) = 0_q$, whereas the surrogate-control channel uses $\mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = 0_p$. In the multivariate-outcome setting, the same construction applies columnwise to $Y = XB + \Xi$. The central organizing observation is that IV retains the component of x_i explained by z_i, whereas SV retains the component orthogonal to the span of w_i. Both therefore produce p-dimensional projected moment blocks for the same coefficient vector, allowing their identification, first-order behavior, and joint weighting to be studied in a common framework.

1.1 Related work

Classical IV, projection, and GMM foundations. Classical instrumental-variable methods address endogeneity by combining instrument relevance with restrictions that separate instrument-induced variation from the structural disturbance (Anderson and Rubin, 1949; Theil, 1953; Sargan, 1958). Two-stage least squares implements this idea through projection onto the instrument span, while GMM provides a general moment-based formulation and an optimal-weighting theory for overidentified systems (Hansen, 1982). Our projected IV formulation inherits this moment and projection geometry but compresses the raw q-dimensional IV moment to the p-dimensional component relevant for the first-stage projection of x_i. Accordingly, the efficiency results in this paper concern the projected moment system rather than the larger raw-IV GMM class when $q > p$. Weak or poorly conditioned first-stage variation remains a distinct identification problem even when the IV moment restriction itself is valid (Stock and Yogo, 2002).

Proxy and proximal identification. Proxy-variable, negative-control, and proximal approaches use observed auxiliary variables to identify causal effects in the presence of latent confounding, often through bridge functions, completeness conditions, or related conditional-moment restrictions (Miao et al., 2018; Shi et al., 2020; Miao et al., 2024; Tchetgen Tchetgen et al., 2024). The auxiliary-variable framework of Miao et al. (2023) is particularly close conceptually because it accommodates IV-like and proxy-like auxiliary information within a common nonparametric identification framework. Our target and identification argument are different: we work with a linear structural coefficient and direct projected moment restrictions rather than

recovering a general causal functional through an unknown bridge. Thus, the present framework is narrower in functional form but focuses explicitly on the projection geometry, rank conditions, first-order scores, and weighting of the resulting IV and SV moment blocks.

Latent factors and combined deconfounding signals. Latent-factor and substitute-adjustment approaches use multivariate observed information to learn or account for latent confounding structure (Wang and Blei, 2019; Adams and Hansen, 2025). Our analysis does not require consistent recovery of the latent confounder itself; instead, it treats z_i and w_i as observed ancillary signals and studies the projected moment restrictions they induce. Recent work also combines multiple sources of deconfounding information, including negative controls for invalid instruments, data combination under persistent confounding, and surrogate information for efficient estimation (Dukes et al., 2025; Imbens et al., 2025; Kallus and Mao, 2025). Our contribution is a linear projected-moment formulation in which IV and SV information can be stacked, weighted, and studied under both exact and locally misspecified moment restrictions.

1.2 Summary of contributions

Our contributions are fourfold:

- **Dual projected-moment representation.** We place IV and surrogate-control adjustment in a common p-dimensional projected-moment representation. The IV block uses the instrument-explained component $\Pi_z x_i$, whereas the SV block uses the residualized component $\Pi_w^{\perp} x_i$. This representation makes explicit both the common structure of the two methods and the distinction between the classical raw-IV moment and the projected IV moment.
- **Single-source statistical targets and inference under imperfect validity.** For each channel, we characterize the statistical target under a fixed projected-moment violation and show when that target coincides with the structural coefficient β. We then establish consistency and asymptotic normality for the corresponding feasible projected estimators around their statistical targets, while separating these statistical results from the additional structural assumptions required for causal interpretation (Propositions 1 and 3 and theorems 2 and 4).
- **Complementary identification, information, and efficient combination.** We combine the IV and SV blocks in a $2p$-dimensional projected GMM system whose identification requires only full column rank of the stacked Jacobian, allowing the combined system to remain identified even when one block is individually rank deficient (Theorem 5). We show that projection preserves the identifying directions of the raw IV moment, while potentially discarding overidentifying information when $q > p$ (Propositions 12 and 13). When both single-source systems are individually identified and correctly specified, optimal projected-GMM weighting yields asymptotic variance no larger than either single-source estimator (Proposition 7). Under $n^{-1/2}$-local violations of the stacked moments, we further derive the corresponding local asymptotic bias and worst-case AMSE formulation and compare optimally weighted combined and single-source procedures under the stated misspecification class (Theorem 6 and proposition 9).

Notation

We use lower-case letters for unit-level variables and upper-case letters for sample arrays: $y_i \in \mathbb{R}$, $x_i \in \mathbb{R}^p$, $z_i \in \mathbb{R}^q$, $w_i \in \mathbb{R}^m$, and $u_i \in \mathbb{R}^r$ denote the outcome, regressors, instruments, surrogate controls, and latent confounders, while Y, X, Z, W, U denote the corresponding sample arrays.

Population L_2 projections are denoted by Π_z, Π_w, and $\Pi_w^{\perp}$, whereas for any sample matrix A with n rows, $\mathcal{C}(A)$ denotes its column space, P_A the Euclidean projection onto $\mathcal{C}(A)$, and $P_A^{\perp} := I_n - P_A$ the corresponding residual-maker. For any matrix A, $\ker(A) := \{v : Av = 0\}$ denotes its null space and $\operatorname{rank}(A) := \dim \mathcal{C}(A)$ its rank.

We use $\|\cdot\|_2$, $\|\cdot\|_{\mathrm{F}}$, and $\|\cdot\|_{\mathrm{op}}$ for the Euclidean, Frobenius, and operator norms, respectively, and $\operatorname{Cov}(a, b)$ for population cross-covariance.

We write $\xrightarrow{\mathrm{p}}$ and $\xrightarrow{\mathrm{d}}$ for convergence in probability and distribution, use $O_p(\cdot)$, $o_p(\cdot)$, and $\Theta_p(\cdot)$ in their standard senses, and write $a_n \lesssim b_n$ and $a_n \asymp b_n$ for one-sided and two-sided order equivalence up to constants.

The sets of $d \times d$ positive definite and positive semidefinite matrices are denoted by $\mathbb{S}_{++}^d$ and $\mathbb{S}_{+}^d$, respectively.

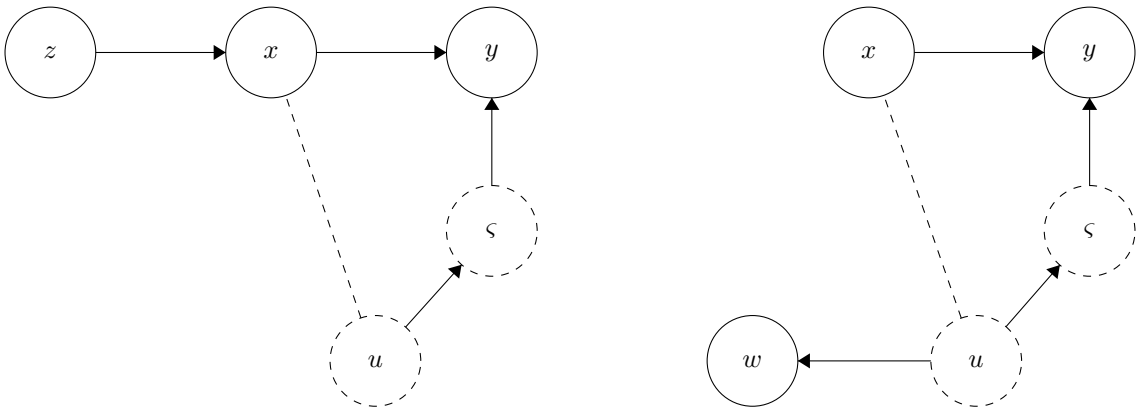


Figure 1: Causal diagrams under (more restricted) conditional independence assumptions.

2 Preliminaries

2.1 Ancillary variables for deconfounding

Consider the confounded linear regression model

$$y = x^\top \beta + \varsigma. \tag{2.1}$$

Here $y \in \mathbb{R}$, $x \in \mathbb{R}^p$, $\beta \in \mathbb{R}^p$, and $\varsigma \in \mathbb{R}$, with $\mathbb{E}(\varsigma) = 0$, $\mathrm{Var}(\varsigma) = \sigma^2 < \infty$, and $\Sigma_x \in \mathbb{S}^p_{++}$. The mean-zero normalization can be obtained by including an intercept or by centering x and y.

Under the conventional linear model framework, when $\mathbb{E}(x\varsigma) = 0$, the ordinary least squares estimator is consistent to β. In many empirical applications, however, the orthogonality condition between ς and x is violated because ς contains hidden confounding components $u \in \mathbb{R}^r$ that jointly associate with both the outcome and the covariates; see Figure 1.

With n i.i.d. observations, we consider the confounded linear model in matrix form,

$$Y = X\beta + \varsigma. \tag{2.2}$$

where $Y = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$ is the outcome vector, $X = (x_1, \ldots, x_n)^\top \in \mathbb{R}^{n\times p}$ is the regressor matrix with row vectors $x_i \in \mathbb{R}^p$, and $\varsigma = (\varsigma_1, \ldots, \varsigma_n)^\top \in \mathbb{R}^n$ is the error vector. The parameter vector $\beta \in \mathbb{R}^p$ is unknown.

A classical strategy for addressing endogeneity caused by hidden confounding is to introduce ancillary variables that provide information beyond the observed covariates. Broadly speaking, if such variables satisfy appropriate orthogonality or factorization conditions, they can be used to mitigate the bias induced by latent confounders. Two representative approaches have been extensively studied in the literature. The first relies on instrumental variables that satisfy orthogonality conditions with respect to the structural error term. The second exploits surrogate variables that act as noisy proxies for latent confounders.

2.2 Instrumental variable models

A canonical remedy is to introduce *ancillary control variables* that enable identification by enforcing orthogonality after an appropriate projection. The instrumental variables (IV) paradigm was formalized in the simultaneous equations literature (Anderson and Rubin, 1949; Theil, 1953; Sargan, 1958). With valid instruments Z, two-stage least squares (2SLS) can be viewed as replacing endogenous regressors by their projections onto the instrument span, yielding consistent estimation under standard moment restrictions (Hansen, 1982). Weak instruments are a well-known failure mode in this pipeline (Stock and Yogo, 2002).

IV methods address endogeneity by introducing variables that are associated with the endogenous regressors but, in the classical case, orthogonal to the structural error. Suppose we observe n i.i.d. samples $\{(y_i, x_i, z_i)\}_{i=1}^n$, where $y_i \in \mathbb{R}$, $x_i \in \mathbb{R}^p$, and $z_i \in \mathbb{R}^q$. Under the centered formulation, let $\Sigma_z := \mathbb{E}(z_i z_i^\top)$ and $\Sigma_{xz} := \mathbb{E}(x_i z_i^\top)$. When $\Sigma_z \in \mathbb{S}^q_{++}$, the population L_2-projection of x_i onto the linear span of z_i is $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$. We allow the projected component $\Pi_z x_i$ to remain associated with ς_i, and characterize the resulting statistical target below.

Proposition 1 (Identification of the statistical target under projected IV moments)**.** Under (2.1), let $\delta_{\mathrm{IV}} := \mathbb{E}\{(\Pi_z x_i)\varsigma_i\}$ and $Q_{\mathrm{IV}} := \mathbb{E}\{(\Pi_z x_i)x_i^\top\} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$. If $\Sigma_z \in \mathbb{S}_{++}^q$ and $Q_{\mathrm{IV}} \in \mathbb{S}_{++}^p$, the projected population moment equation $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top\theta)] = 0$ has the unique solution $\beta_{\mathrm{IV}} = \beta + Q_{\mathrm{IV}}^{-1}\delta_{\mathrm{IV}}$. In particular, $\delta_{\mathrm{IV}} = 0$ implies $\beta_{\mathrm{IV}} = \beta$.

Remark 1 (Dimension and identification of the projected IV moment)**.** Because

$$Q_{\mathrm{IV}} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx},$$

with $\Sigma_z \in \mathbb{S}_{++}^q$,

$$\mathrm{rank}(Q_{\mathrm{IV}}) = \mathrm{rank}(\Sigma_{zx}) \le \min(p, q).$$

Hence, identification of the full p-dimensional coefficient vector by the IV block alone requires $\mathrm{rank}(Q_{\mathrm{IV}}) = p$, and therefore necessarily $q \ge p$. In particular, when $q < p$, the projected IV moment is still written as a p-dimensional vector, but its Jacobian has rank at most $q < p$, so these p displayed equations contain fewer than p independent identifying directions and do not uniquely identify the full vector β.

The distinction between the classical and projected IV formulations is also important when $q \ge p$. The classical moment

$$\mathbb{E}[z_i(y_i - x_i^\top\theta)] = 0_q$$

contains q moment equations and, under $\mathrm{rank}(\Sigma_{zx}) = p$, is just-identified when $q = p$ and overidentified when $q > p$. By contrast,

$$\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top\theta)] = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}[z_i(y_i - x_i^\top\theta)]$$

is always a p-dimensional projected moment. When $Q_{\mathrm{IV}} \in \mathbb{S}_{++}^p$, this projected system has identifying rank p and therefore uniquely identifies the statistical target β_{IV}.

Under valid IV, $\mathbb{E}(z_i\varsigma_i) = 0_q$, the classical and projected formulations identify the same structural coefficient whenever the corresponding rank condition holds. When $q > p$, however, projection can discard overidentifying directions in the raw q-dimensional moment, so efficiency statements for the projected IV system should not be interpreted as efficiency statements for GMM based on the full classical IV moment.

Proposition 1 establishes identification at the population level. In practice, estimation is typically carried out using two-stage least squares (2SLS). Let $Z = (z_1, \ldots, z_n)^\top \in \mathbb{R}^{n\times q}$ be the instrument matrix. We require $\mathrm{rank}(Z) = q$ and $\mathrm{rank}(P_Z X) = p$, or equivalently $\mathrm{rank}(Z^\top X) = p$, which necessarily implies $p \le q \le n$ (emphasis). Throughout the fixed-dimensional asymptotic analysis, p and q are treated as fixed as $n \to \infty$.

At the sample level, $P_Z X$ is the fitted component of X obtained by projecting onto the instrument span. Regressing Y on this fitted design gives the 2SLS estimator

$$\widehat{\beta}_{\mathrm{IV}} := (X^\top P_Z X)^{-1} X^\top P_Z Y. \tag{2.3}$$

Remark 2 (On first-stage linear representations)**.** Some treatments of instrumental variables additionally posit an explicit linear first-stage representation $x_i = \eta^\top z_i + \epsilon_i, \mathbb{E}(z_i\epsilon_i^\top) = 0$. Such a representation is often useful as a working linear projection or as a structural interpretation of the first-stage regression. In the present projection-based formulation, however, the relevant object is the population L^2-projection $\Pi_z x_i$, which can be written as $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$ when $\Sigma_z \in \mathbb{S}_{++}^q$. Thus, for the statistical target result in Proposition 1 and for the moment-based derivation of 2SLS, the primitive requirements are the projected IV moment bias $\delta_{\mathrm{IV}} := \mathbb{E}[(\Pi_z x_i)\varsigma_i]$ together with the relevance condition $Q_{\mathrm{IV}} := \mathbb{E}[(\Pi_z x_i)x_i^\top] \in \mathbb{S}_{++}^p$. Accordingly, the decomposition $x_i = \eta^\top z_i + \epsilon_i$ should be viewed as an optional linear-projection representation rather than a necessary primitive assumption for 2SLS.

For IV, recall that $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$ and $Q_{\mathrm{IV}} = \mathbb{E}[(\Pi_z x_i)x_i^\top] = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$. Define $e_{\mathrm{IV},i} := y_i - x_i^\top\beta_{\mathrm{IV}} - z_i^\top\Sigma_z^{-1}\mathbb{E}[z_i(y_i - x_i^\top\beta_{\mathrm{IV}})]$, so that $\mathbb{E}(z_i e_{\mathrm{IV},i}) = 0$. The first-order linearization term of the projected 2SLS normal equation can then be written as $\psi_{\mathrm{IV},i} := (\Pi_z x_i)e_{\mathrm{IV},i} + x_i z_i^\top\Sigma_z^{-1}\mathbb{E}[z_i(y_i - x_i^\top\beta_{\mathrm{IV}})]$, and define $\Sigma_{\mathrm{IV}} := \mathrm{Var}(\psi_{\mathrm{IV},i})$.

Assumption 1 (Projected instrumental variables conditions)**.** Suppose the following conditions hold.

(i) Projected relevance. Suppose $\Sigma_z \in \mathbb{S}_{++}^q$ and $Q_{\mathrm{IV}} \in \mathbb{S}_{++}^p$.

(ii) Second moments and nondegeneracy. Suppose $\mathbb{E}[\|x_i z_i^\top\|_{\mathrm{op}}^2] < \infty$, $\mathbb{E}\|z_i e_{\mathrm{IV},i}\|_2^2 < \infty$, and $\Sigma_{\mathrm{IV}} \in \mathbb{S}_{++}^p$.

Theorem 2 (Statistical properties of projected 2SLS)**.** Under Assumption 1 (i), as $n \to \infty$, the following statements hold.

(1) Consistency: $\widehat{\beta}_{\mathrm{IV}} \xrightarrow{\mathrm{p}} \beta_{\mathrm{IV}}$.

(2) Asymptotic normality: If further Assumption 1 (ii) holds, then $\sqrt{n}(\widehat{\beta}_{\mathrm{IV}} - \beta_{\mathrm{IV}}) \xrightarrow{\mathrm{d}} \mathcal{N}(0, V_{\mathrm{IV}})$, where $V_{\mathrm{IV}} := Q_{\mathrm{IV}}^{-1}\Sigma_{\mathrm{IV}}Q_{\mathrm{IV}}^{-1}$.

Remark 3 (Second-moment conditions under valid IV)**.** The two moment conditions in Assumption 1(ii) control the two components of the equivalent first-order representation $\psi_{\mathrm{IV},i} = (\Pi_z x_i)e_{\mathrm{IV},i} + x_i z_i^\top \Sigma_z^{-1}\mathbb{E}[z_i(y_i - x_i^\top\beta_{\mathrm{IV}})]$. Specifically, $\mathbb{E}[\|x_i z_i^\top\|_{\mathrm{op}}^2] < \infty$ controls the second moment of the correction term involving $x_i z_i^\top$, while $\mathbb{E}\|z_i e_{\mathrm{IV},i}\|_2^2 < \infty$ controls the projected-score component because $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1}z_i$.

Under valid IV, $\mathbb{E}(z_i\varsigma_i) = 0$, so $\delta_{\mathrm{IV}} = 0$, $\beta_{\mathrm{IV}} = \beta$, and $\mathbb{E}[z_i(y_i - x_i^\top\beta_{\mathrm{IV}})] = 0$. Hence, the correction term vanishes identically and the condition $\mathbb{E}[\|x_i z_i^\top\|_{\mathrm{op}}^2] < \infty$ is no longer needed for asymptotic normality. Moreover, in this case $e_{\mathrm{IV},i} = \varsigma_i$, so the remaining condition becomes $\mathbb{E}\|z_i\varsigma_i\|_2^2 < \infty$. Since $\psi_{\mathrm{IV},i} = (\Pi_z x_i)\varsigma_i = \Sigma_{xz}\Sigma_z^{-1}z_i\varsigma_i$, this condition guarantees the finite second moment required for the multivariate central limit theorem, and Σ_{IV} reduces to $\mathrm{Var}\{(\Pi_z x_i)\varsigma_i\}$.

2.3 Surrogate controls and partialling out

Surrogate-based methods address endogeneity by exploiting observed variables informative about latent confounding (Miao et al., 2018; Adams and Hansen, 2025; Du et al., 2026). Suppose we observe n i.i.d. samples $\{(y_i, x_i, w_i)\}_{i=1}^n$, where $y_i \in \mathbb{R}$, $x_i \in \mathbb{R}^p$, and $w_i \in \mathbb{R}^m$. Under the centered formulation, let $\Sigma_w := \mathbb{E}(w_i w_i^\top)$ and $\Sigma_{xw} := \mathbb{E}(x_i w_i^\top)$. When $\Sigma_w \in \mathbb{S}_{++}^m$, the population projection and residualization are $\Pi_w x_i = \Sigma_{xw}\Sigma_w^{-1}w_i$ and $\Pi_w^\perp x_i = x_i - \Sigma_{xw}\Sigma_w^{-1}w_i$.

Unlike IV methods, the surrogate-variable approach removes from x_i the component linearly explained by the surrogate controls. Under exact validity, the residualized regressors satisfy $\mathbb{E}[(\Pi_w^\perp x_i)\varsigma_i] = 0$, which, under the centered formulation and nonsingular Σ_w, is equivalent to $\mathbb{E}[(x_i - \Sigma_{xw}\Sigma_w^{-1}w_i)\varsigma_i] = 0$. This condition allows the original regressor to be endogenous, with $\mathbb{E}(x_i\varsigma_i) \neq 0$, and does not require w_i itself to be exogenous. Rather, w_i is used to capture the part of x_i associated with the structural error, so that after decomposing $x_i = \Pi_w x_i + \Pi_w^\perp x_i$, the remaining variation $\Pi_w^\perp x_i$ is orthogonal to ς_i. Equivalently, $\mathbb{E}(x_i\varsigma_i) = \Sigma_{xw}\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)$, meaning that the linear dependence between x_i and ς_i is fully absorbed by the component of x_i explained by w_i. Thus, unlike a valid instrument z_i, which directly satisfies $\mathbb{E}(z_i\varsigma_i) = 0$, the surrogate variable constructs an exogenous residualized regressor through partialling out. We allow $\delta_{\mathrm{SV}} := \mathbb{E}[(\Pi_w^\perp x_i)\varsigma_i] \neq 0$ in the subsequent analysis, where $\delta_{\mathrm{SV}} = 0$ is the exact-validity benchmark and $\delta_{\mathrm{SV}} \neq 0$ measures the remaining residual confounding and resulting bias relative to β.

We begin with the corresponding population identification result.

Proposition 3 (Statistical identification under surrogate-control residualization)**.** Under (2.1), let $\delta_{\mathrm{SV}} := \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\}$ and $Q_{\mathrm{SV}} := \mathbb{E}\{(\Pi_w^\perp x_i)x_i^\top\}$. If $Q_{\mathrm{SV}} \in \mathbb{S}_{++}^p$, the residualized population moment equation $\mathbb{E}[(\Pi_w^\perp x_i)(y_i - x_i^\top\theta)] = 0$ has the unique solution $\beta_{\mathrm{SV}} = \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$. In particular, $\delta_{\mathrm{SV}} = 0$ implies $\beta_{\mathrm{SV}} = \beta$.

Proposition 3 establishes the population target of the surrogate-control residualization moment. When the residualized moment bias vanishes, this target coincides with the structural coefficient β; otherwise, it is the partialling-out pseudo-target β_{SV}. In practice, estimation is typically carried out by a partialling-out procedure.

The partialling-out estimator can be viewed as a two-step procedure, mirroring the logic of 2SLS. Let $W = (w_1, \ldots, w_n)^\top \in \mathbb{R}^{n\times m}$ be the surrogate-control matrix. We require $\mathrm{rank}(W) = m$ and $\mathrm{rank}(P_W^\perp X) = p$, which necessarily implies $m + p \leq n$. Throughout the fixed-dimensional asymptotic analysis, m and p are treated as fixed as $n \to \infty$. At the sample level, partialling out W gives the residualized design $P_W^\perp X$ and residualized outcome $P_W^\perp Y$. Regressing the latter on the former gives

$$\widehat{\beta}_{\mathrm{SV}} := (X^\top P_W^\perp X)^{-1}X^\top P_W^\perp Y. \tag{2.4}$$

Remark 4 (Related methods)**.** Among existing works, the auxiliary variables framework of Miao et al. (2023, Section 3) is perhaps the closest in spirit to our setting. At a high level, both papers study identification under latent confounding by leveraging variables external to the primary outcome regression, rather than by consistently estimating the latent confounder itself. In particular, their framework treats an auxiliary variable as any variable satisfying an exclusion-type condition relative to the outcome given (X, U), a class broad enough to include instrumental variables as well as certain proxy-type variables. This makes their paper especially relevant conceptually, because it places IV-like and proxy-like devices under a common identification perspective.

The main difference is that Miao et al. (2023) develop a *nonparametric* identification theory for the potential outcome distribution, based on high-level equivalence and completeness conditions and on solving an integral equation, whereas our paper works with a linear structural parameter and focuses on the *estimation geometry* induced by different ancillary variables. For example, Theorem 1 of Miao et al. (2023) provides sufficient conditions to recover the joint distribution of three variables (Z, X, U), where Z is an auxiliary variable that satisfies the exclusion restriction condition $Z \perp\!\!\!\perp Y \mid (X, U)$. In our framework, instrumental variables and surrogate controls are not only two instances of auxiliary information, but also two distinct projected-moment mechanisms for the same linear structural coefficient. This leads to parallel statistical-target and inference results for the two single-source estimators, together with a unified GMM-style theory for combining their moment information under exact and locally misspecified restrictions. Thus, rather than pursuing the same nonparametric identification target as Miao et al. (2023), our contribution is to characterize the projection, rank, first-order, and weighting structure of these two deconfounding strategies within a linear working model.

The proximal inference framework (Miao et al., 2018, 2024) provides nonparametric identification by solving an integral equation under the existence of a bridge function linking negative control outcomes and negative control exposures.

As mentioned in Remark 2, estimating $\widehat{\beta}_{\mathrm{IV}}$ does not require a true first-stage linear structural equation. For identification and consistency of 2SLS, it is enough that the IV moment condition $\mathbb{E}(z_i\varsigma_i) = 0$ holds together with the rank condition on Σ_{zx}. Thus, when a linear model is imposed on x against z, it is best viewed as a linear projection or working representation, rather than as a primitive causal assumption.

For SV, recall that $\Pi_w^{\perp} x_i = x_i - \Sigma_{xw}\Sigma_w^{-1} w_i$ and $Q_{\mathrm{SV}} = \mathbb{E}[(\Pi_w^{\perp} x_i) x_i^{\top}]$. Define $e_{\mathrm{SV},i} := y_i - x_i^{\top}\beta_{\mathrm{SV}} - w_i^{\top}\Sigma_w^{-1}\mathbb{E}[w_i(y_i - x_i^{\top}\beta_{\mathrm{SV}})]$, so that $\mathbb{E}(w_i e_{\mathrm{SV},i}) = 0$. The first-order linearization term of the partialling-out normal equation can then be written as $\psi_{\mathrm{SV},i} := (\Pi_w^{\perp} x_i) e_{\mathrm{SV},i}$, and define $\Sigma_{\mathrm{SV}} := \mathrm{Var}(\psi_{\mathrm{SV},i})$.

Assumption 2 (Partialling-out conditions with observed controls)**.** Suppose the following conditions hold.

(i) Residualized relevance. Suppose $\Sigma_w \in \mathbb{S}_{++}^m$ and $Q_{\mathrm{SV}} \in \mathbb{S}_{++}^p$.

(ii) Second moments and nondegeneracy. Suppose $\mathbb{E}\|x_i\varsigma_i\|_2^2 < \infty$, $\mathbb{E}\|x_i x_i^{\top}\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|x_i w_i^{\top}\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|w_i e_{\mathrm{SV},i}\|_2^2 < \infty$, and $\Sigma_{\mathrm{SV}} \in \mathbb{S}_{++}^p$.

Theorem 4 (Statistical properties of partialling-out under residualized moment bias)**.** Under Assumption 2 (i), as $n \to \infty$, the following statements hold.

(1) Consistency: $\widehat{\beta}_{\mathrm{SV}} \xrightarrow{\mathrm{p}} \beta_{\mathrm{SV}}$.

(2) Asymptotic normality: If Assumption 2 (ii) also holds, then $\sqrt{n}(\widehat{\beta}_{\mathrm{SV}} - \beta_{\mathrm{SV}}) \xrightarrow{\mathrm{d}} \mathcal{N}(0, V_{\mathrm{SV}})$, where $V_{\mathrm{SV}} := Q_{\mathrm{SV}}^{-1}\Sigma_{\mathrm{SV}}Q_{\mathrm{SV}}^{-1}$.

2.4 Causal interpretation

The results in Proposition 1, Proposition 3, Theorem 2, and Theorem 4 are statistical target and estimation results. In the biased-moment formulation used above, they do not by themselves say that the feasible IV or SV estimator recovers the structural coefficient β. Rather, they show that the corresponding population targets $\beta_{\mathrm{IV}} = \beta + Q_{\mathrm{IV}}^{-1}\delta_{\mathrm{IV}}$ and $\beta_{\mathrm{SV}} = \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$ are well defined and consistently estimable under the stated relevance and moment conditions. The structural coefficient β is recovered as the target only in the no-bias subcases $\delta_{\mathrm{IV}} = 0$ and $\delta_{\mathrm{SV}} = 0$. Thus, the vectors δ_{IV} and δ_{SV} should be viewed as explicit measures of the remaining violation of the ideal deconfounding restrictions. A causal interpretation therefore requires additional structural assumptions explaining why these bias terms vanish, are negligible, or can otherwise be controlled.

For instrumental variables, the usual valid-IV model can be viewed as a restricted potential-outcome formulation (Wooldridge, 2010). Suppose the observed outcome satisfies consistency, $y_i = y_i(x_i)$, and the instrument is causally valid in the sense of exclusion, $y_i(x_i, z_i) = y_i(x_i)$, together with exogeneity with respect to the latent determinants of the response function. If one further imposes the homogeneous linear structural restriction $y_i(x) = \alpha_i + \beta^{\top} x$, so that the slope does not vary across units, then the observed-data equation becomes $y_i = \alpha_i + \beta^{\top} x_i$. After centering, the remaining disturbance is orthogonal to the valid instrument variation, and hence the projected IV bias satisfies $\delta_{\mathrm{IV}} = \mathbb{E}[(\Pi_z x_i)\varsigma_i] = 0$. In this classical no-bias case, the IV population target collapses to $\beta_{\mathrm{IV}} = \beta$, and β can be interpreted as a common causal effect, or as the average treatment effect when such a common-effect restriction is substantively appropriate.

This also clarifies the role of the IV moment condition. The classical condition $\mathbb{E}[z_i(y_i - x_i^\top \beta)] = 0$, or equivalently the projected condition $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top \beta)] = 0$, is not itself the primitive causal assumption. It is the statistical implication of exclusion, exogeneity, and the homogeneous linear response restriction. When the instrument remains associated with latent determinants of the outcome, the bias term δ_{IV} is generally nonzero. In that case, 2SLS still has a well-defined population target β_{IV}, but this target is shifted away from the structural coefficient β by $Q_{\text{IV}}^{-1}\delta_{\text{IV}}$. Hence, under invalid or only approximately valid instruments, the IV estimator should be interpreted as a biased or sensitivity-adjusted statistical target rather than as direct identification of the causal coefficient.

Without the homogeneous-slope restriction, even the no-bias IV estimand should not generally be read as a universal structural slope. Once treatment effects are heterogeneous or the response surface is nonlinear, linear IV no longer recovers a single common effect shared by all units. The canonical example is the binary-treatment setting, where under monotonicity IV identifies a local average treatment effect for compliers rather than a population-wide structural coefficient (Imbens and Angrist, 1994). In the present setting, the regressor may be continuous or may vary in intensity, so conventional linear IV and 2SLS are better understood as targeting a weighted-average causal response generated by the instrument-induced variation. If $\delta_{\text{IV}} = 0$, this weighted response may still have a causal interpretation, but it is not generally a universal slope; if $\delta_{\text{IV}} \neq 0$, it is additionally contaminated by the remaining association between the instrument-selected component of x_i and the structural disturbance.

For surrogate-control adjustment, the causal interpretation is different. The statistical restriction is now summarized by the residualized bias term $\delta_{\text{SV}} = \mathbb{E}[(\Pi_w^{\perp} x_i)\varsigma_i]$. When $\delta_{\text{SV}} = 0$, the variation in x_i left after partialling out the surrogate controls is orthogonal to the structural disturbance, and the partialling-out target satisfies $\beta_{\text{SV}} = \beta$. A causal reading of this no-bias case requires the surrogate variable w_i to capture the confounding-relevant component of x_i, so that the residualized regressor $\Pi_w^{\perp} x_i$ can be treated as exogenous in the outcome equation. Such a justification may arise, for example, when w_i is a valid pre-treatment proxy or surrogate for the latent confounding structure, although the precise conditions depend on the scientific setting and the measurement mechanism.

If the surrogate controls are imperfect, then δ_{SV} need not vanish. In that case, the partialling-out estimator still targets the well-defined population quantity $\beta_{\text{SV}} = \beta + Q_{\text{SV}}^{-1}\delta_{\text{SV}}$, but this quantity should be interpreted as a residualized statistical target rather than as the structural causal coefficient itself. The term $Q_{\text{SV}}^{-1}\delta_{\text{SV}}$ measures the remaining shift induced by residual confounding after surrogate adjustment. Thus, the SV formulation is useful not only in the ideal case where surrogate adjustment fully removes confounding, but also in approximate settings where the residual bias can be explicitly represented, bounded, or studied as a sensitivity parameter.

Overall, the IV and SV approaches can be viewed as two different routes from structural assumptions to statistical moments. In the IV setting, the identifying variation comes from the component of x_i induced by the instrument; in the SV setting, it comes from the residual variation in x_i left after surrogate-based deconfounding. In the no-bias subcases $\delta_{\text{IV}} = 0$ and $\delta_{\text{SV}} = 0$, these moment restrictions can support a causal interpretation of β, subject to the additional structural assumptions described above. In the more general biased formulation, the same moment structure remains useful, but the estimators target β_{IV} and β_{SV}, with the bias terms making explicit how far the statistical targets may be from the structural coefficient.

Remark 5 (Exogeneity, moment bias, and causal direction)**.** In the linear-in-variables IV framework, exogeneity is often written as the second-moment orthogonality condition $\mathbb{E}(z_i\varsigma_i) = 0$, or equivalently $\text{Cov}(z_i, \varsigma_i) = 0$ under a mean-zero normalization. In the projected formulation used in this paper, the corresponding no-bias condition is $\delta_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)\varsigma_i] = 0$. This covariance-based expression is a consequence of the linear moment structure, rather than an intrinsic requirement of IV or GMM more generally. From a purely statistical point of view, 2SLS is a method for solving sample analogues of moment equations; when the projected moment bias is allowed to be nonzero, the estimator remains centered around the shifted target $\beta_{\text{IV}} = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$, rather than around the structural coefficient β.

This distinction is useful because moment equations themselves do not encode a causal arrow between z_i and x_i. Formally, one may write down a moment restriction such as $\mathbb{E}(z_i\varsigma_i) = 0$, $\mathbb{E}(\varsigma_i \mid z_i) = 0$, or $\delta_{\text{IV}} = 0$ without specifying whether z_i causes x_i, whether z_i is only associated with x_i, or whether both arise from a common design mechanism. However, the credibility of such a restriction is ultimately judged through the causal structure. For example, if the data-generating process is closer to the chain $\varsigma_i \to x_i \to z_i$, so that the disturbance affects x_i and z_i is then constructed from or strongly driven by x_i, then z_i will generally inherit information about ς_i. In that case, the no-bias condition $\delta_{\text{IV}} = 0$ is difficult to defend, and the IV estimator should be interpreted as targeting the biased population quantity β_{IV}, with $Q_{\text{IV}}^{-1}\delta_{\text{IV}}$ describing the resulting shift.

Thus, the classical IV narrative $z_i \to x_i$ is not required by the algebra of 2SLS itself, but it is often the most plausible way to justify why the instrument-induced component of x_i is unrelated to the structural disturbance. In applications where this narrative is only approximately true, the biased formulation above makes the remaining violation explicit through δ_{IV}. This allows the IV moment to be used either as an ideal identifying restriction when $\delta_{\text{IV}} = 0$, or as a sensitivity and approximation device when $\delta_{\text{IV}} \neq 0$.

A closely related issue arises for surrogate-control residualization. The statistical no-bias condition is $\delta_{\text{SV}} := \mathbb{E}[(\Pi_w^{\perp} x_i)\varsigma_i] = 0$, which says that the part of x_i left after partialling out w_i is orthogonal to the structural disturbance. If w_i is a valid pre-treatment proxy for the confounding structure, this condition may be causally plausible. By contrast, if the data-generating process is better described by a chain such as $\varsigma_i \to x_i \to w_i$, then w_i may itself carry information transmitted through the endogenous regressor, and the residualized moment condition becomes harder to justify. In that case, the partialling-out estimator should be interpreted as targeting $\beta_{\text{SV}} = \beta + Q_{\text{SV}}^{-1}\delta_{\text{SV}}$, with $Q_{\text{SV}}^{-1}\delta_{\text{SV}}$ measuring the residual confounding left after surrogate adjustment.

Therefore, in both IV and SV approaches, the statistical procedures can be defined through projection and moment equations without imposing a particular causal direction as part of the estimator. The causal direction matters because it is what makes the no-bias restrictions $\delta_{\text{IV}} = 0$ and $\delta_{\text{SV}} = 0$ credible. When those restrictions are not fully credible, the biased-moment formulation keeps the estimation theory intact while making clear that the estimators target β_{IV} and β_{SV}, rather than necessarily the structural coefficient β.

3 Combining instruments and surrogate controls

The IV and SV approaches developed above provide two projected moment restrictions for the same structural parameter β. When both z_i and w_i are observed, it is therefore natural to combine these restrictions within a common GMM framework. Throughout this section, our primary estimator is the projected GMM estimator $\widehat{\beta}(M)$, indexed by a weighting matrix M.

The organization separates the common estimation framework from the specification regime and the subsequent choice of weighting matrix. We first establish the projected GMM formulation and its complementary identification structure. We then study the same estimator under two regimes: exact specification and $n^{-1/2}$-local misspecification. After establishing the corresponding first-order theories, we compare the optimal weighting principles across the two regimes. Under exact specification, optimal weighting is a variance-minimization problem, whereas under local misspecification it becomes a bias–variance tradeoff. Finally, we study a one-parameter weighted least-squares path as an interpretable restricted subfamily of the general projected GMM estimator.

3.1 Projected GMM formulation and complementary identification

Consider the linear structural model (2.2). We begin with the projected moment system that underlies both specification regimes studied below. Under exact projected validity, $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$ and $\mathbb{E}\{(\Pi_w^{\perp} x_i)\varsigma_i\} = 0_p$, or equivalently $\delta_{\text{IV}} = \delta_{\text{SV}} = 0_p$, so the IV and SV population targets coincide with the structural coefficient β whenever the separate targets are well defined.

Define the stacked projected regressor by $h_i := \{(\Pi_z x_i)^\top, (\Pi_w^{\perp} x_i)^\top\}^\top \in \mathbb{R}^{2p}$ and its Jacobian by $J := \mathbb{E}(h_i x_i^\top) = \{Q_{\text{IV}}^\top, Q_{\text{SV}}^\top\}^\top$. The combined population restriction and its identification condition are

$$\mathbb{E}\{h_i(y_i - x_i^\top \beta)\} = 0_{2p}, \tag{3.1}$$

$$\operatorname{rank}(J) = p \quad \Longleftrightarrow \quad \ker(Q_{\text{IV}}) \cap \ker(Q_{\text{SV}}) = \{0_p\}. \tag{3.2}$$

The second equivalence follows because J has full column rank if and only if there is no nonzero parameter direction annihilated by both projected moment blocks.

Thus, neither Q_{IV} nor Q_{SV} needs to have full rank on its own, provided that the two blocks do not fail in the same parameter direction. In particular, $\operatorname{rank}(Q_{\text{IV}}) = \operatorname{rank}(\Sigma_{zx}) \leq q$, so $q < p$ rules out IV-only identification of the full p-vector but does not rule out identification by the combined system. There is no analogous requirement $m \geq p$ for the surrogate dimension because SV identification is determined by $\operatorname{rank}(Q_{\text{SV}})$.

We now define the estimator used throughout the remainder of this section. The sample analogue of (3.1) is $\widehat{g}(\theta) := n^{-1}\{[X^\top P_Z(Y - X\theta)]^\top, [X^\top P_W^{\perp}(Y - X\theta)]^\top\}^\top$. For a fixed $M \in \mathbb{S}_{++}^{2p}$, define

$$\widehat{\beta}(M) := \operatorname*{argmin}_{\theta \in \mathbb{R}^p} \widehat{g}(\theta)^\top M \widehat{g}(\theta). \tag{3.3}$$

Because $\widehat{g}(\theta)$ is affine in θ, the estimator is obtained from a linear first-order condition. Whenever the required inverse exists,

$$\widehat{\beta}(M) = \left\{ \begin{pmatrix} X^\top P_Z X \\ X^\top P_W^\perp X \end{pmatrix}^\top M \begin{pmatrix} X^\top P_Z X \\ X^\top P_W^\perp X \end{pmatrix} \right\}^{-1} \begin{pmatrix} X^\top P_Z X \\ X^\top P_W^\perp X \end{pmatrix}^\top M \begin{pmatrix} X^\top P_Z Y \\ X^\top P_W^\perp Y \end{pmatrix}. \tag{3.4}$$

The common factor n^{-1} in the sample moments cancels from the first-order condition and therefore does not appear in (3.4).

The family (3.3) nests the two single-source projection estimators whenever the corresponding block identifies the full coefficient vector. Using only the IV block gives $(X^\top P_Z X)^{-1} X^\top P_Z Y = \widehat{\beta}_{\mathrm{IV}}$, whereas using only the SV block gives $(X^\top P_W^\perp X)^{-1} X^\top P_W^\perp Y = \widehat{\beta}_{\mathrm{SV}}$, provided that the relevant matrix is nonsingular. Within either exactly identified projected block, a positive definite weighting matrix does not affect the estimator.

The distinction between moment dimension and identifying rank remains important. The projected IV block is p-dimensional even when $q < p$, but its Jacobian then has rank at most $q < p$. Thus the IV block may fail to identify the full coefficient vector while still contributing identifying directions to the stacked system. Conversely, when $q > p$, the raw q-dimensional IV restriction may contain overidentifying information beyond that retained by its p-dimensional projection. We return to this qualification when discussing optimal weighting in Section 4.

3.2 Projected GMM under correct specification

We first study the GMM family (3.3) under exact projected validity. Because the feasible moments use estimated projection operators, their first-order behavior is not generally represented by $h_i \varsigma_i$ alone. Recall from Sections 2.2 and 2.3 that $\psi_{\mathrm{IV},i}$ and $\psi_{\mathrm{SV},i}$ denote the first-order scores of the feasible projected IV and SV moments. Under joint projected validity,

$$\begin{aligned} \psi_{\mathrm{IV},i} &= (\Pi_z x_i)\varsigma_i + \{x_i - \Pi_z x_i\} z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i \varsigma_i), \\ \psi_{\mathrm{SV},i} &= (\Pi_w^\perp x_i)\{\varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i \varsigma_i)\}. \end{aligned} \tag{3.5}$$

Projected IV validity does not generally imply $\mathbb{E}(z_i \varsigma_i) = 0_q$, and residualized SV validity does not generally imply $\mathbb{E}(w_i \varsigma_i) = 0_m$. Hence, the correction terms in (3.5) need not vanish. Under the stronger classical IV restriction $\mathbb{E}(z_i \varsigma_i) = 0_q$, however, $\psi_{\mathrm{IV},i} = (\Pi_z x_i)\varsigma_i$.

Stack the scores as $\psi_i := (\psi_{\mathrm{IV},i}^\top, \psi_{\mathrm{SV},i}^\top)^\top$, let $\Sigma := \mathrm{Var}(\psi_i)$, and for any admissible M define $K_M := (J^\top M J)^{-1} J^\top M$ and $V(M) := K_M \Sigma K_M^\top$.

Assumption 3 (Combined GMM conditions)**.** Suppose the following conditions hold.

(i) **Moment restriction and relevance.** Suppose $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$, $\mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = 0_p$, $\Sigma_z \in \mathbb{S}_{++}^q$, $\Sigma_w \in \mathbb{S}_{++}^m$, and $\mathrm{rank}(J) = p$. The weighting matrix $M \in \mathbb{S}_{++}^{2p}$ is fixed.

(ii) **Second moments and estimator nondegeneracy.** Suppose $\mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|z_i\{\varsigma_i - z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\}\|_2^2 < \infty$, $\mathbb{E}\|x_i \varsigma_i\|_2^2 < \infty$, $\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|w_i\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\}\|_2^2 < \infty$, and $V(M) \in \mathbb{S}_{++}^p$.

Remark 6. The combined system differs slightly from a classical GMM problem in which each moment function is evaluated directly from observed variables. Here, the population moments depend on Π_z and $\Pi_w^\perp$, whereas their feasible counterparts use $P_Z X$ and $P_W^\perp X$. Estimating these projections contributes to the first-order expansion, so the relevant score is (3.5), rather than simply $h_i \varsigma_i$.

Assumption 3(ii) ensures that the score components and empirical projection remainders have finite second moments. Consequently, Σ and its cross-covariance blocks are well defined. Positive definiteness of Σ is not required for asymptotic normality; the condition $V(M) \in \mathbb{S}_{++}^p$ instead allows redundant projected moment directions while ensuring nondegenerate inference for β. We impose $\Sigma \in \mathbb{S}_{++}^{2p}$ separately when its inverse is required for efficient weighting.

Theorem 5 (Consistency and asymptotic normality of combined GMM)**.** Under Assumption 3(i), as $n \to \infty$, the following statements hold.

(1) Consistency: $\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta$.

(2) Asymptotic normality: if Assumption 3(ii) also holds, then $\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\{0, V(M)\}$.

Theorem 5 establishes the exact-specification benchmark for the entire projected GMM family. Identification is governed by the stacked Jacobian J, while M determines how the information from the two projected blocks enters the first-order variance. Importantly, because the moments are exactly centered, M affects first-order precision but not asymptotic bias. We next contrast this benchmark with the same estimator under local misspecification.

3.3 Projected GMM under local misspecification

We now retain the same GMM estimator $\widehat{\beta}(M)$ but allow the projected moments to fail at the $n^{-1/2}$ scale. Following the approximate moment condition framework of Armstrong and Kolesár (2021), the misspecification is small enough to preserve consistency for β, but large enough to enter the first-order distribution as a local asymptotic bias.

For the stacked population moment, suppose

$$\mathbb{E}(h_i\varsigma_i) = \frac{\tau}{\sqrt{n}} = \frac{1}{\sqrt{n}}\begin{pmatrix}\tau_z \\ \tau_w\end{pmatrix}, \qquad \tau_z, \tau_w \in \mathbb{R}^p. \tag{3.6}$$

Thus, $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = \tau_z/\sqrt{n}$ and $\mathbb{E}\{(\Pi_w^{\perp} x_i)\varsigma_i\} = \tau_w/\sqrt{n}$.

For the local expansion around the structural coefficient β, define

$$\begin{aligned}\psi_{\mathrm{IV,miss},i} &:= (\Pi_z x_i)\varsigma_i + \{x_i - \Pi_z x_i\} z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i), \\ \psi_{\mathrm{SV,miss},i} &:= (\Pi_w^{\perp} x_i)\left\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\}.\end{aligned} \tag{3.7}$$

These scores differ from the single-source scores in Sections 2.2 and 2.3 because the present expansion is centered at the structural coefficient β, rather than at the single-source statistical targets.

Since $\mathbb{E}[\{x_i - \Pi_z x_i\} z_i^\top] = 0_{p\times q}$, the correction term in $\psi_{\mathrm{IV,miss},i}$ has mean zero, and therefore $\mathbb{E}(\psi_{\mathrm{IV,miss},i}) = \tau_z/\sqrt{n}$. Likewise, $\mathbb{E}\{(\Pi_w^{\perp} x_i) w_i^\top\} = 0_{p\times m}$, so $\mathbb{E}(\psi_{\mathrm{SV,miss},i}) = \tau_w/\sqrt{n}$.

Define $\psi_{\mathrm{miss},i} := (\psi_{\mathrm{IV,miss},i}^\top, \psi_{\mathrm{SV,miss},i}^\top)^\top \in \mathbb{R}^{2p}$. Then $\mathbb{E}(\psi_{\mathrm{miss},i}) = \tau/\sqrt{n}$. Hence, estimation of the projection operators affects the first-order covariance but does not alter the local misspecification vector τ.

Retain J and K_M from Section 3.1. Along the local sequence, let $\Sigma_{\mathrm{miss},n} := \mathrm{Var}(\psi_{\mathrm{miss},i})$ under the n-th data-generating law and suppose that $\Sigma_{\mathrm{miss},n}$ converges to Σ_{miss}. Let $\Sigma_{\mathrm{IV,miss}}$ and $\Sigma_{\mathrm{SV,miss}}$ denote its IV and SV diagonal blocks, and define $V_{\mathrm{miss}}(M) := K_M \Sigma_{\mathrm{miss}} K_M^\top$. The quantities $K_M\tau$ and $V_{\mathrm{miss}}(M)$ are therefore the local asymptotic bias and variance associated with M.

Assumption 4 (Combined GMM conditions under local misspecification)**.** Suppose the following conditions hold.

(i) **Local moment restriction and relevance.** Suppose (3.6) holds for a fixed $\tau \in \mathbb{R}^{2p}$, $\Sigma_z \in \mathbb{S}_{++}^q$, $\Sigma_w \in \mathbb{S}_{++}^m$, and the fixed symmetric weighting matrix $M \succeq 0$ satisfies $J^\top M J \in \mathbb{S}_{++}^p$.

(ii) **Second-moment conditions.** Suppose, uniformly along the local sequence, $\sup_n \mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$, $\sup_n \mathbb{E}\|z_i\{\varsigma_i - z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\}\|_2^2 < \infty$, $\sup_n \mathbb{E}\|x_i\varsigma_i\|_2^2 < \infty$, $\sup_n \mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty$, and $\sup_n \mathbb{E}\|w_i\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\}\|_2^2 < \infty$.

(iii) **Lindeberg condition, covariance convergence, and estimator nondegeneracy.** For every $\epsilon > 0$, suppose $\mathbb{E}[\|\psi_{\mathrm{miss},i} - \mathbb{E}(\psi_{\mathrm{miss},i})\|_2^2 \mathbf{1}\{\|\psi_{\mathrm{miss},i} - \mathbb{E}(\psi_{\mathrm{miss},i})\|_2 > \epsilon\sqrt{n}\}] \to 0$. Suppose further that, for some $\Sigma_{\mathrm{miss}} \in \mathbb{S}_+^{2p}$, $\|\Sigma_{\mathrm{miss},n} - \Sigma_{\mathrm{miss}}\|_{\mathrm{op}} \to 0$, and $V_{\mathrm{miss}}(M) = K_M \Sigma_{\mathrm{miss}} K_M^\top \in \mathbb{S}_{++}^p$.

Theorem 6 (Statistical properties of combined GMM under local misspecification)**.** Under Assumption 4 (i), as $n \to \infty$, the following statements hold.

(1) Consistency: $\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta$.

(2) Asymptotic normality: if Assumption 4 (ii)–(iii) also hold, then $\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\{K_M\tau, V_{\mathrm{miss}}(M)\}$.

The same conclusions apply to the IV-only and SV-only systems under the corresponding specializations of the projected moment vector, Jacobian, local-misspecification covariance matrix, local misspecification vector, and weighting matrix, provided that the corresponding identification conditions hold.

Theorem 6 differs from Theorem 5 only in the centering of the moments and the resulting triangular-array asymptotics. Under exact specification, the limiting distribution is centered at zero. Under local misspecification, the violation is of order $n^{-1/2}$, so $\widehat{\beta}(M)$ remains consistent for β, but its root-n distribution acquires the local bias $K_M\tau$. When $\tau = 0$, the local result reduces to the correctly specified benchmark.

Assumption 4 (iii) allows the data-generating law, and hence $\Sigma_{\text{miss},n}$, to vary along the local sequence. The Lindeberg condition rules out individual observations from dominating the root-n fluctuation, covariance convergence provides a stable limiting variance, and $V_{\text{miss}}(M) \in \mathbb{S}^p_{++}$ ensures nondegenerate inference.

The two preceding subsections therefore describe the same projected GMM estimator under two specification regimes. Under exact specification, M affects first-order variance only. Under local misspecification, M determines both the local sensitivity $K_M\tau$ and the variance $V_{\text{miss}}(M)$. This distinction determines the appropriate optimal weighting criterion.

4 Optimal weighting across specification regimes

The preceding results characterize the first-order behavior of $\widehat{\beta}(M)$ before imposing an optimal choice of M. We now compare the weighting problem across the two regimes. Under exact specification, the moments are centered and the natural criterion is variance minimization. Under local misspecification, the moments carry first-order bias, so optimal weighting instead balances potential bias against sampling variation.

4.1 Efficient weighting under correct specification

Under exact specification, the efficient GMM weight within the projected IV–SV system is $M^\star = \Sigma^{-1}$ whenever $\Sigma \in \mathbb{S}^{2p}_{++}$ (Hansen, 1982). The corresponding asymptotic variance is

$$V^\star_{\text{comb}} = (J^\top \Sigma^{-1} J)^{-1}. \tag{4.1}$$

Recall that $\Sigma_{\text{IV}} := \text{Var}(\psi_{\text{IV},i})$ and $\Sigma_{\text{SV}} := \text{Var}(\psi_{\text{SV},i})$, with $V_{\text{IV}} := Q_{\text{IV}}^{-1}\Sigma_{\text{IV}}Q_{\text{IV}}^{-1}$ and $V_{\text{SV}} := Q_{\text{SV}}^{-1}\Sigma_{\text{SV}}Q_{\text{SV}}^{-1}$ whenever the corresponding single-source systems are identified.

Proposition 7 (Optimal asymptotic variance and efficiency dominance)**.** Suppose the following conditions hold.

(i) Assumptions 1 and 2 hold, with $\delta_{\text{IV}} = \delta_{\text{SV}} = 0_p$.

(ii) The combined first-order covariance matrix satisfies $\Sigma \in \mathbb{S}^{2p}_{++}$.

Then the asymptotic variances of the IV-only and SV-only estimators are V_{IV} and V_{SV}, respectively, whereas the optimally weighted combined GMM estimator has asymptotic variance $V^\star_{\text{comb}}$. Moreover,

$$V^\star_{\text{comb}} \preceq V_{\text{IV}}, \qquad V^\star_{\text{comb}} \preceq V_{\text{SV}}. \tag{4.2}$$

Thus, when both single-source systems are themselves identified, optimally combining the two projected moment blocks is asymptotically at least as efficient as using either projected source alone.

Proposition 7 gives the natural efficiency benchmark within the projected system. The comparison requires the corresponding IV-only and SV-only estimators to be well defined, which is stronger than the combined rank condition (3.2). In particular, when $q < p$, the IV block cannot identify the full coefficient vector on its own, even though it may still supply directions that improve identification or precision in the stacked system. If both blocks are rank deficient but complementary, neither V_{IV} nor V_{SV} is an appropriate full-vector benchmark, whereas Theorem 5 continues to apply.

Remark 7 (Feasible two-step GMM)**.** The efficient weight $M^\star = \Sigma^{-1}$ is infeasible because Σ is unknown. Starting from any fixed $M_0 \in \mathbb{S}^{2p}_{++}$, let $\widehat{\beta}^{(1)}_{\text{comb}} := \widehat{\beta}(M_0)$, and construct the plug-in covariance estimator $\widehat{\Sigma}^{(1)}$ from the projected first-order scores as detailed in Appendix D. Proposition 14 gives $\widehat{\Sigma}^{(1)} \xrightarrow{\text{p}} \Sigma$. If $\Sigma \in \mathbb{S}^{2p}_{++}$, then $(\widehat{\Sigma}^{(1)})^{-1} \xrightarrow{\text{p}} \Sigma^{-1}$, and the feasible second-step estimator $\widehat{\beta}^{(2)}_{\text{comb}} := \widehat{\beta}\{(\widehat{\Sigma}^{(1)})^{-1}\}$ attains the variance in (4.1).

4.2 AMSE-optimal weighting under local misspecification

Under local misspecification, M determines both local sensitivity and precision. Let $\mathcal{C} \subseteq \mathbb{R}^{2p}$ be a researcher-specified set of plausible local misspecification values, as in Armstrong and Kolesár (2021). For example, one may take $\mathcal{C} = \{\tau = (\tau_z^\top, \tau_w^\top)^\top : \|\tau_z\|_2 \le \bar{c}_z,\ \|\tau_w\|_2 \le \bar{c}_w\}$.

For a fixed $\tau \in \mathcal{C}$, define the asymptotic mean squared error of $\widehat{\beta}(M)$ by $\mathrm{AMSE}_\tau(M) := \lim_{n\to\infty} n\,\mathbb{E}_\tau\|\widehat{\beta}(M) - \beta\|_2^2$, whenever the limit exists. Under exact specification, $\tau = 0$, so $\mathrm{AMSE}_0(M) = \mathrm{tr}\{V(M)\}$. Under the local sequence of Theorem 6, the limiting mean is $K_M\tau$ and the limiting covariance is $V_{\mathrm{miss}}(M)$, giving $\mathrm{AMSE}_\tau(M) = \|K_M\tau\|_2^2 + \mathrm{tr}\{V_{\mathrm{miss}}(M)\}$. Thus, unlike the exact-specification problem, the choice of M must now balance squared local asymptotic bias against variance.

Lemma 8 (Worst-case AMSE under local misspecification)**.** Let $\mathcal{C} \subseteq \mathbb{R}^{2p}$ be nonempty, and let $M \succeq 0$ be symmetric with $J^\top M J \in \mathbb{S}_{++}^p$. Under the conditions of Theorem 6, for each $\tau \in \mathcal{C}$, suppose that the corresponding squared root-n estimation errors are uniformly integrable along the local sequence. Then $\mathrm{AMSE}_\tau(M) = \|K_M\tau\|_2^2 + \mathrm{tr}\{V_{\mathrm{miss}}(M)\}$. Consequently,

$$\begin{aligned}\mathcal{R}_{\mathcal{C}}(M) &:= \sup_{\tau\in\mathcal{C}} \mathrm{AMSE}_\tau(M) \\ &= \sup_{\tau\in\mathcal{C}} \|K_M\tau\|_2^2 + \mathrm{tr}\{V_{\mathrm{miss}}(M)\}. \end{aligned} \tag{4.3}$$

Thus, $\mathcal{R}_{\mathcal{C}}(M)$ is the worst-case squared local asymptotic bias plus asymptotic variance. The local-risk-optimal combined GMM weight is any solution of

$$M_{\mathrm{opt}} \in \operatorname*{argmin}_{M\succeq 0:\ J^\top M J \in \mathbb{S}_{++}^p} \mathcal{R}_{\mathcal{C}}(M). \tag{4.4}$$

This weight need not coincide with $M^\star = \Sigma^{-1}$. The efficient GMM weight is optimal when $\tau = 0$, whereas M_{opt} allows for nonzero local violations and therefore downweights directions whose potential misspecification is large relative to their precision.

To compare the optimized combined procedure with its single-source counterparts, for an IV-only weight $M_z \succeq 0$ and an SV-only weight $M_w \succeq 0$, define $K_{M_z} := (Q_{\mathrm{IV}}^\top M_z Q_{\mathrm{IV}})^{-1} Q_{\mathrm{IV}}^\top M_z$ and $K_{M_w} := (Q_{\mathrm{SV}}^\top M_w Q_{\mathrm{SV}})^{-1} Q_{\mathrm{SV}}^\top M_w$, whenever the corresponding inverses exist. Let $\mathcal{C}_z := \{\tau_z : (\tau_z^\top, \tau_w^\top)^\top \in \mathcal{C} \text{ for some } \tau_w\}$ and $\mathcal{C}_w := \{\tau_w : (\tau_z^\top, \tau_w^\top)^\top \in \mathcal{C} \text{ for some } \tau_z\}$ denote the IV and SV coordinate projections of $\mathcal{C}$, respectively.

Define the single-source optimal local risks by

$$\mathcal{R}_{\mathrm{IV}}^\star := \inf_{M_z\succeq 0:\ Q_{\mathrm{IV}}^\top M_z Q_{\mathrm{IV}} \in \mathbb{S}_{++}^p} \left[\sup_{\tau_z\in\mathcal{C}_z} \|K_{M_z}\tau_z\|_2^2 + \mathrm{tr}\left\{K_{M_z}\Sigma_{\mathrm{IV,miss}}K_{M_z}^\top\right\}\right],$$

$$\mathcal{R}_{\mathrm{SV}}^\star := \inf_{M_w\succeq 0:\ Q_{\mathrm{SV}}^\top M_w Q_{\mathrm{SV}} \in \mathbb{S}_{++}^p} \left[\sup_{\tau_w\in\mathcal{C}_w} \|K_{M_w}\tau_w\|_2^2 + \mathrm{tr}\left\{K_{M_w}\Sigma_{\mathrm{SV,miss}}K_{M_w}^\top\right\}\right].$$

Similarly,

$$\mathcal{R}_{\mathrm{comb}}^\star := \inf_{M\succeq 0:\ J^\top M J \in \mathbb{S}_{++}^p} \left[\sup_{\tau\in\mathcal{C}} \|K_M\tau\|_2^2 + \mathrm{tr}\{V_{\mathrm{miss}}(M)\}\right].$$

Proposition 9 (Local-risk dominance of the optimally combined system)**.** Suppose the following conditions hold.

(i) The single-source Jacobians satisfy $Q_{\mathrm{IV}} \in \mathbb{S}_{++}^p$ and $Q_{\mathrm{SV}} \in \mathbb{S}_{++}^p$, the covariance matrices $\Sigma_{\mathrm{IV,miss}}$, $\Sigma_{\mathrm{SV,miss}}$, and Σ_{miss} are finite, and $\mathcal{C}_z$ and $\mathcal{C}_w$ are nonempty and bounded.

(ii) The local misspecification set satisfies $\mathcal{C} = \mathcal{C}_z \times \mathcal{C}_w$.

(iii) The combined weight class allows positive semidefinite block weights.

Then

$$\mathcal{R}_{\mathrm{comb}}^\star \le \mathcal{R}_{\mathrm{IV}}^\star, \qquad \mathcal{R}_{\mathrm{comb}}^\star \le \mathcal{R}_{\mathrm{SV}}^\star.$$

Thus, after optimizing the local bias–variance tradeoff, the combined IV–SV system has local risk no larger than either subsystem.

Proposition 9 is the local-misspecification analogue of Proposition 7. Under correct specification, the optimally combined system dominates in asymptotic variance; under local misspecification, it dominates in optimized local bias–variance risk. This does not imply that every fixed combined weight is preferable. A poorly chosen weight may place excessive mass on a more misspecified block and increase risk. The result instead reflects the flexibility of the combined weight class: after optimization, it can reproduce either single-source procedure by assigning zero weight to the other block, while retaining the possibility of improvement when the additional moments contribute enough precision relative to their allowed misspecification.

Remark 8 (Feasible estimation of the local-risk-optimal weight)**.** The local-risk-optimal weight M_{opt} is infeasible because J and Σ_{miss} are unknown. We therefore use a two-step plug-in procedure analogous to the feasible efficient GMM construction in Remark 7. Starting from any fixed $M_0 \in \mathbb{S}^{2p}_{++}$, let $\widehat{\beta}^{(1)}_{\mathrm{comb}} := \widehat{\beta}(M_0)$. Using this first-step estimator, construct $\widehat{\psi}^{(1)}_{\mathrm{IV,miss},i}$ and $\widehat{\psi}^{(1)}_{\mathrm{SV,miss},i}$, stack them as $\widehat{\psi}^{(1)}_{\mathrm{miss},i} := \{(\widehat{\psi}^{(1)}_{\mathrm{IV,miss},i})^\top, (\widehat{\psi}^{(1)}_{\mathrm{SV,miss},i})^\top\}^\top$, and let $\overline{\widehat{\psi}}^{(1)}_{\mathrm{miss}} := n^{-1}\sum_{i=1}^n \widehat{\psi}^{(1)}_{\mathrm{miss},i}$.

Estimate Σ_{miss} by

$$\widehat{\Sigma}^{(1)}_{\mathrm{miss}} := \frac{1}{n}\sum_{i=1}^n \left\{\widehat{\psi}^{(1)}_{\mathrm{miss},i} - \overline{\widehat{\psi}}^{(1)}_{\mathrm{miss}}\right\}\left\{\widehat{\psi}^{(1)}_{\mathrm{miss},i} - \overline{\widehat{\psi}}^{(1)}_{\mathrm{miss}}\right\}^\top.$$

By Theorem 6, $\widehat{\beta}^{(1)}_{\mathrm{comb}} \xrightarrow{\mathrm{p}} \beta$. Under Assumption 4 (ii) and the uniform quadratic-moment conditions in (D.1), the same score-by-score plug-in argument used in the proof of Proposition 14 gives

$$\frac{1}{n}\sum_{i=1}^n \left\|\widehat{\psi}^{(1)}_{\mathrm{miss},i} - \psi_{\mathrm{miss},i}\right\|_2^2 \xrightarrow{\mathrm{p}} 0.$$

Hence the plug-in covariance differs by $o_p(1)$ from the centered sample covariance based on $\psi_{\mathrm{miss},i}$. Since $\mathrm{Var}(\psi_{\mathrm{miss},i}) = \Sigma_{\mathrm{miss},n}$, the triangular-array law of large numbers and Assumption 4 (iii) imply $\widehat{\Sigma}^{(1)}_{\mathrm{miss}} \xrightarrow{\mathrm{p}} \Sigma_{\mathrm{miss}}$.

Estimate the stacked Jacobian by $\widehat{J} := n^{-1}\{(X^\top P_Z X)^\top, (X^\top P_W^\perp X)^\top\}^\top$. For any admissible $M \succeq 0$, define $\widehat{K}_M := (\widehat{J}^\top M \widehat{J})^{-1}\widehat{J}^\top M$ and $\widehat{V}_{\mathrm{miss}}(M) := \widehat{K}_M \widehat{\Sigma}^{(1)}_{\mathrm{miss}} \widehat{K}_M^\top$. The plug-in worst-case local risk is $\widehat{\mathcal{R}}_{\mathcal{C}}(M) := \sup_{\tau\in\mathcal{C}} \|\widehat{K}_M\tau\|_2^2 + \mathrm{tr}\{\widehat{V}_{\mathrm{miss}}(M)\}$, and the feasible local-risk-optimal weight is

$$\widehat{M}_{\mathrm{opt}} \in \operatorname*{argmin}_{M\succeq 0:\, \widehat{J}^\top M \widehat{J} \in \mathbb{S}^p_{++}} \widehat{\mathcal{R}}_{\mathcal{C}}(M).$$

Since multiplying M by a positive scalar does not change $\widehat{K}_M$, the optimization can be carried out over a normalized admissible class, for example by imposing $\mathrm{tr}(M) = 1$.

The corresponding second-step estimator remains in the same GMM family: $\widehat{\beta}^{(2)}_{\mathrm{comb}} := \widehat{\beta}(\widehat{M}_{\mathrm{opt}})$. Thus, conventional two-step GMM and local-risk-optimal GMM differ only in their second-step weighting rule. The former estimates the variance-optimal weight, whereas the latter minimizes estimated worst-case squared local bias plus variance. Consistency of the feasible risk criterion and its minimizer is established in Proposition 15.

Remark 9 (Misspecification-robust confidence intervals)**.** Following Armstrong and Kolesár (2021), Theorem 6 also yields misspecification-robust inference for scalar functionals of β. For a prespecified nonzero $a \in \mathbb{R}^p$, the statistic $\sqrt{n}\{a^\top\widehat{\beta}(M) - a^\top\beta\}$ has local asymptotic bias $a^\top K_M\tau$ and variance $\sigma^2_M(a) := a^\top V_{\mathrm{miss}}(M)a$. Let $B_{\mathcal{C}}(M;a) := \sup_{\tau\in\mathcal{C}} |a^\top K_M\tau|$, $\widehat{B}_{\mathcal{C}}(M;a) := \sup_{\tau\in\mathcal{C}} |a^\top \widehat{K}_M\tau|$, and $\widehat{\sigma}_M(a) := \{a^\top \widehat{V}_{\mathrm{miss}}(M)a\}^{1/2}$.

Provided that $\widehat{\sigma}_M(a) \xrightarrow{\mathrm{p}} \sigma_M(a)$, a $100(1-\alpha)\%$ misspecification-robust confidence interval for $a^\top\beta$ is

$$a^\top\widehat{\beta}(M) \pm \mathrm{cv}_\alpha\{\widehat{B}_{\mathcal{C}}(M;a)/\widehat{\sigma}_M(a)\}\,\widehat{\sigma}_M(a)/\sqrt{n},$$

where $\mathrm{cv}_\alpha(t)$ denotes the $1-\alpha$ quantile of $|Z|$ for $Z \sim \mathcal{N}(t,1)$. When $\mathcal{C} = \{0\}$, the worst-case bias is zero and $\mathrm{cv}_\alpha(0) = z_{1-\alpha/2}$, so the interval reduces to the usual Wald confidence interval. For inference on an individual coefficient β_j, one takes $a = e_j$. The weighting matrix that minimizes robust confidence-interval length need not coincide with the AMSE-optimal weight in (4.4).

The comparison between the two regimes can therefore be summarized directly. Under exact specification, $M^\star = \Sigma^{-1}$ minimizes first-order variance. Under local misspecification, M_{opt} instead minimizes the worst-case sum of squared local bias and variance in (4.3). Both procedures belong to the same projected GMM family; what changes across regimes is the criterion used to select its weighting matrix.

4.3 Restricted weighting: a one-parameter WLS path

The general projected GMM estimator allows a $2p \times 2p$ weighting matrix and remains the benchmark procedure. For interpretation, however, it is useful to consider a simpler one-parameter path that moves continuously between the IV and SV projection criteria. The resulting weighted least-squares estimator is not a separate estimation framework; it is a restricted subfamily of $\widehat{\beta}(M)$.

This restriction is deliberately narrower than the general stacked-GMM theory. To retain both endpoints $\lambda = 0$ and $\lambda = 1$, we work in the regime where the projected IV and SV blocks are individually nonsingular. Consequently, the full path considered below does not cover complementary-rank settings such as $q < p$, in which Theorem 5 may still identify β through the stacked system.

Recall that $\widehat{\beta}_{\mathrm{IV}}$ and $\widehat{\beta}_{\mathrm{SV}}$ minimize $\|P_Z(Y - Xb)\|_2^2$ and $\|P_W^{\perp}(Y - Xb)\|_2^2$, respectively. For $\lambda \in [0,1]$, define $\widehat{\beta}_\lambda := \operatorname{argmin}_b\{(1-\lambda)\|P_Z(Y - Xb)\|_2^2 + \lambda\|P_W^{\perp}(Y - Xb)\|_2^2\}$. Its first-order condition gives

$$\widehat{\beta}_\lambda = \{(1-\lambda)X^\top P_Z X + \lambda X^\top P_W^{\perp} X\}^{-1}\{(1-\lambda)X^\top P_Z Y + \lambda X^\top P_W^{\perp} Y\}. \tag{4.5}$$

In particular, $\widehat{\beta}_0 = \widehat{\beta}_{\mathrm{IV}}$ and $\widehat{\beta}_1 = \widehat{\beta}_{\mathrm{SV}}$.

The next result makes explicit that this path is contained in the projected GMM family.

Proposition 10 (Restricted-GMM interpretation)**.** Suppose that $n^{-1}X^\top P_Z X \in \mathbb{S}_{++}^p$ and $n^{-1}X^\top P_W^{\perp} X \in \mathbb{S}_{++}^p$. For $\lambda \in (0,1)$, define

$$M_\lambda := \begin{pmatrix} (1-\lambda)\big(n^{-1}X^\top P_Z X\big)^{-1} & 0 \\ 0 & \lambda\big(n^{-1}X^\top P_W^{\perp} X\big)^{-1} \end{pmatrix}.$$

Then $M_\lambda \in \mathbb{S}_{++}^{2p}$, and $\widehat{\beta}_\lambda = \widehat{\beta}(M_\lambda)$. Thus, for $\lambda \in (0,1)$, the WLS estimator is a one-parameter restricted subfamily of the joint GMM estimator. At $\lambda = 0$ and $\lambda = 1$, the corresponding positive-semidefinite block weights recover $\widehat{\beta}_{\mathrm{IV}}$ and $\widehat{\beta}_{\mathrm{SV}}$, respectively.

Proposition 10 clarifies the role of the WLS path. The unrestricted GMM estimator can use general weighting, including off-diagonal blocks that account for dependence between the IV and SV scores. By contrast, M_λ restricts attention to a one-dimensional block-diagonal family. The WLS path is therefore useful for interpreting how the estimator moves between the two projected criteria, but it is not the general efficiency or local-risk benchmark.

Having established this restricted-GMM interpretation, we next consider how λ trades off the two projected criteria within this one-dimensional family. This differs from the preceding analysis, which studies the unrestricted projected GMM estimator through unconditional first-order asymptotic criteria. To obtain a transparent finite-sample characterization of the restricted path, we condition on the realized design matrices (X, Z, W). This treats the information matrices along the path as fixed and isolates the contribution of the structural disturbance to estimation risk. The conditioning is only an analytical device and does not impose conditional validity of either projected moment restriction.

When β is vector-valued, there is generally no scalar λ that minimizes the variance matrix in the Loewner order. A scalar optimum therefore requires a scalar loss criterion. We use conditional trace risk.

For this analysis, define $S_Z := X^\top P_Z X$, $S_W := X^\top P_W^{\perp} X$, and $S_\lambda := (1-\lambda)S_Z + \lambda S_W$. Let $\mathcal{D} := \sigma(X, Z, W)$, $U_Z := X^\top P_Z \varsigma$, $U_W := X^\top P_W^{\perp}\varsigma$, $c_Z := \mathbb{E}(\|U_Z\|_2^2 \mid \mathcal{D})$, $c_{ZW} := \mathbb{E}(U_Z^\top U_W \mid \mathcal{D})$, and $c_W := \mathbb{E}(\|U_W\|_2^2 \mid \mathcal{D})$.

Proposition 11 (Conditional trace risk and a tractable surrogate)**.** Work conditionally on the realized design matrices (X, Z, W), treated as fixed throughout. Suppose the following conditions hold.

(i) $S_Z \in \mathbb{S}_{++}^p$ and $S_W \in \mathbb{S}_{++}^p$, or equivalently $S_\lambda \in \mathbb{S}_{++}^p$ for every $\lambda \in [0,1]$.

(ii) $c_Z < \infty$ and $c_W < \infty$.

Then $\widehat{\beta}_\lambda - \beta = S_\lambda^{-1}\{(1-\lambda)U_Z + \lambda U_W\}$. Let $U_\lambda := (1-\lambda)U_Z + \lambda U_W$. The conditional trace risk is

$$\begin{aligned} \mathcal{R}_{\mathrm{tr}}(\lambda) &:= \mathbb{E}(\|\widehat{\beta}_\lambda - \beta\|_2^2 \mid \mathcal{D}) \\ &= \operatorname{tr}\left[S_\lambda^{-1}\mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D})S_\lambda^{-1}\right]. \end{aligned} \tag{4.6}$$

Equivalently, $\mathcal{R}_{\mathrm{tr}}(\lambda) = \|S_\lambda^{-1}\mathbb{E}(U_\lambda \mid \mathcal{D})\|_2^2 + \operatorname{tr}\{S_\lambda^{-1}\operatorname{Var}(U_\lambda \mid \mathcal{D})S_\lambda^{-1}\}$.

Moreover, $\operatorname{tr}\{\mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D})\} = (1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W$, and therefore

$$\begin{aligned}\frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\lambda_{\max}(S_\lambda)^2} &\le \mathcal{R}_{\mathrm{tr}}(\lambda) \\ &\le \frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\lambda_{\min}(S_\lambda)^2}. \end{aligned} \tag{4.7}$$

If, in addition, $\lambda_{\max}(S_Z) \lesssim \lambda_{\min}(S_Z)$ and $\lambda_{\max}(S_W) \lesssim \lambda_{\min}(S_W)$, then uniformly over $\lambda \in [0,1]$,

$$\begin{aligned}\mathcal{R}_{\mathrm{tr}}(\lambda) &\asymp \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \\ &:= \frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\{(1-\lambda)\lambda_{\min}(S_Z) + \lambda\lambda_{\min}(S_W)\}^2}. \end{aligned} \tag{4.8}$$

If, moreover, $(c_W - c_{ZW})\lambda_{\min}(S_Z) + (c_Z - c_{ZW})\lambda_{\min}(S_W) \neq 0$, then $\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$ has at most one interior stationary point, given by

$$\lambda^\star_{\mathrm{sur}} = \frac{c_Z \lambda_{\min}(S_W) - c_{ZW}\lambda_{\min}(S_Z)}{(c_W - c_{ZW})\lambda_{\min}(S_Z) + (c_Z - c_{ZW})\lambda_{\min}(S_W)}. \tag{4.9}$$

Consequently, the minimizer of $\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$ over $[0,1]$ is obtained by comparing its values at $\lambda = 0$, $\lambda = 1$, and $\lambda = \lambda^\star_{\mathrm{sur}}$ whenever $\lambda^\star_{\mathrm{sur}} \in (0,1)$.

Proposition 11 is a finite-sample conditional risk characterization and does not require the projected score blocks to be conditionally centered. In particular, the unconditional IV and SV restrictions do not generally imply $\mathbb{E}(U_Z \mid \mathcal{D}) = 0$ or $\mathbb{E}(U_W \mid \mathcal{D}) = 0$. Indeed,

$$\begin{aligned} c_Z &= \operatorname{tr}\{X^\top P_Z \operatorname{Var}(\varsigma \mid \mathcal{D}) P_Z X\} + \|\mathbb{E}(U_Z \mid \mathcal{D})\|_2^2, \\ c_{ZW} &= \operatorname{tr}\{X^\top P_Z \operatorname{Var}(\varsigma \mid \mathcal{D}) P_W^\perp X\} + \mathbb{E}(U_Z \mid \mathcal{D})^\top \mathbb{E}(U_W \mid \mathcal{D}), \\ c_W &= \operatorname{tr}\{X^\top P_W^\perp \operatorname{Var}(\varsigma \mid \mathcal{D}) P_W^\perp X\} + \|\mathbb{E}(U_W \mid \mathcal{D})\|_2^2. \end{aligned}$$

Thus, c_Z, c_{ZW}, and c_W measure conditional second moments and incorporate both conditional bias and conditional variance. Under conditional centering, they reduce to covariance-based expressions.

The information matrix S_λ changes linearly with λ, whereas the conditional score second moment changes quadratically through c_Z, c_{ZW}, and c_W. The bounds in (4.7) separate these two components and motivate the tractable surrogate criterion in (4.8). The candidate $\lambda^\star_{\mathrm{sur}}$ therefore provides an interpretable scalar balance between the information strength and total conditional second moments of the two projected channels.

The unrestricted projected GMM estimator remains the benchmark procedure. Under exact specification its optimal weight is $M^\star = \Sigma^{-1}$, whereas under local misspecification its weight is chosen according to (4.4). The WLS path instead provides a transparent one-dimensional representation of the same IV–SV combination problem.

5 Simulation

5.1 Efficiency gains under correct specification

Our first experiment studies the finite-sample efficiency gains from combining two valid and individually identifying deconfounding channels. We consider the scalar design

$$z_i = s_i + \eta_i, \qquad w_i = (u_i, s_i)^\top, \qquad x_i = u_i + a s_i + v_i, \qquad y_i = \beta x_i + u_i + \varepsilon_i, \tag{5.1}$$

where $u_i, s_i, \eta_i, v_i, \varepsilon_i$ are mutually independent standard normal random variables. The parameter $a > 0$ controls the information carried by the projected IV channel while leaving the surrogate-control channel unchanged.

Under (5.1), both projected moment restrictions are correctly specified. In particular, $\Pi_z x_i = (a/2) z_i$, whereas partialling out $w_i = (u_i, s_i)^\top$ from x_i leaves $\Pi_w^\perp x_i = v_i$. Hence $Q_{\mathrm{IV}} = a^2/2$ and $Q_{\mathrm{SV}} = 1$, and both channels identify β whenever $a > 0$. The corresponding single-source asymptotic variances are $V_{\mathrm{IV}} = 4/a^2$ and $V_{\mathrm{SV}} = 2$, while the optimally weighted combined projected-GMM estimator has

$$V^\star_{\mathrm{comb}} = \frac{4}{a^2 + 2}.$$

The balanced regime therefore occurs at $a = \sqrt{2}$. We consider $a \in \{1, \sqrt{2}, 2\}$ as representative SV-dominant, balanced, and IV-dominant cases, respectively, and use a finer grid of a when plotting the full efficiency curves. Sample size varies over $n \in \{250, 500, 1000, 2000\}$, and each design point is evaluated using 1,000 Monte Carlo replications.

We compare the IV-only estimator, the SV-only estimator, an equally weighted projected-GMM estimator, the feasible efficient two-step GMM estimator from Remark 7, and the restricted WLS estimator $\widehat{\beta}_\lambda$. The oracle efficient variance is included as a population benchmark. We report empirical variance and mean squared error as functions of both n and a, together with average confidence-interval length. The main objective is not to numerically verify asymptotic normality, but to examine whether the feasible combination realizes the efficiency gains predicted by Proposition 7 across different information regimes and sample sizes.

The resulting comparison has a direct information-adaptation interpretation. When the SV channel dominates, the efficient combined estimator should behave similarly to the SV-only estimator; when the IV channel dominates, it should move toward the IV-only estimator. Around the balanced regime, both channels contain substantial nonredundant information, so the gain from efficient combination should be most visible relative to either single-source procedure.

We additionally examine inference under correct specification. For size, we set $\beta = 0$ and report the rejection probability of the two-sided 5% Wald test of $H_0 : \beta = 0$. To translate variance reduction into an inferential consequence, we also consider local alternatives $\beta = c/\sqrt{n}$ for several values of c and compare rejection probabilities across the competing procedures. This provides a finite-sample illustration of how efficiency gains affect local power rather than merely checking the validity of a central limit approximation.

Finally, we study the restricted WLS path over $\lambda \in [0, 1]$. For $a \in \{1, \sqrt{2}, 2\}$, we plot the empirical scaled risk $n\,\mathbb{E}\{(\widehat{\beta}_\lambda - \beta)^2\}$ against λ and compare its empirical minimizer with the surrogate choice $\lambda^\star_{\mathrm{sur}}$ in (4.9). This examines how closely the one-dimensional surrogate criterion in Proposition 11 tracks the risk-minimizing region of the restricted WLS family across different information regimes.

5.2 Robust weighting under local misspecification

Our second experiment studies the local bias–variance tradeoff when the two projected moment restrictions are only approximately valid. To isolate the weighting problem from differences in baseline information, we retain the design in (5.1) and fix the balanced case $a = \sqrt{2}$, but generate the outcome according to the triangular array

$$y_{i,n} = \beta x_i + u_i + \varepsilon_i + \frac{\tau_z}{a\sqrt{n}} z_i + \frac{\tau_w}{\sqrt{n}} v_i. \tag{5.2}$$

Since $\Pi_z x_i = (a/2) z_i$ and $\Pi_w^\perp x_i = v_i$, this construction gives

$$\mathbb{E}\{(\Pi_z x_i)(y_{i,n} - \beta x_i)\} = \frac{\tau_z}{\sqrt{n}}, \qquad \mathbb{E}\{(\Pi_w^\perp x_i)(y_{i,n} - \beta x_i)\} = \frac{\tau_w}{\sqrt{n}},$$

so that the local misspecification vector is controlled directly.

We consider three regimes: $(\tau_z, \tau_w) = (\kappa, 0)$, $(0, \kappa)$, and (κ, κ), corresponding respectively to IV-only, SV-only, and joint local misspecification. The sensitivity magnitude is varied over $\kappa \in \{0, 0.5, 1, 2\}$, while the sample size again varies over $n \in \{250, 500, 1000, 2000\}$. Each design point is evaluated using 1,000 Monte Carlo replications. For the local-risk procedure, κ is treated as a prespecified sensitivity radius rather than as an estimable bias parameter; the corresponding uncertainty set is chosen to contain the local-misspecification direction used in each design.

We compare the IV-only and SV-only procedures, conventional efficient GMM based on variance-optimal weighting, the feasible local-risk-optimal GMM procedure in Remark 8, and the restricted WLS family. The primary criterion is $n\,\mathbb{E}\|\widehat{\beta} - \beta\|_2^2$, together with its empirical squared-bias and variance components. These quantities directly parallel the AMSE decomposition in (4.3) and allow the effect of local misspecification to be separated from ordinary sampling variation.

The central comparison is between variance-optimal and local-risk-optimal weighting. At $\kappa = 0$, the efficient GMM weight is designed purely to minimize variance. As κ increases, however, the local-risk-optimal procedure should reduce its sensitivity to the channel exposed to misspecification, potentially accepting a modest variance increase in exchange for a larger reduction in squared local bias. The three regimes therefore show whether the proposed weighting rule adapts in the direction predicted by the local AMSE theory rather than simply reproducing the exact-specification efficient weight.

Finally, we compare conventional Wald intervals with the misspecification-robust intervals in Remark 9. We report empirical coverage and average interval length over n and κ. Conventional intervals ignore the

local bias term and may therefore lose coverage as the allowed violation increases, whereas the robust intervals incorporate the prespecified local-misspecification set. This final comparison translates the local risk analysis into an inferential consequence of approximate moment validity.

6 Real data

In the single-cell CRISPR application, let $Z \in \mathbb{R}^{n \times q}$ denote the one-hot perturbation-condition matrix, $X \in \mathbb{R}^{n \times p}$ the expression matrix for target genes, $Y \in \mathbb{R}^{n \times g}$ the expression matrix for outcome genes, and $W \in \mathbb{R}^{n \times m}$ the surrogate-control matrix constructed from negative-control or non-differentially expressed genes. Although this notation accommodates a joint multi-perturbation model $Y \sim X$, our primary analysis considers one perturbation target at a time. For target k, we use the scalar instrument $Z_{\cdot k}$, the target expression $X_{\cdot k}$, and the multivariate outcome matrix Y. Thus, the real-data analysis reflects the common regime in which each target has a low-dimensional perturbation instrument, while the surrogate-control matrix W provides richer information about latent cell-state and technical variation.

7 Discussion

- nonlinear models: approximated by linear
- GMM with measurement error in variables Evdokimov and Zeleneev (2023)

References

Adams, J. and Hansen, N. R. (2025). Substitute adjustment via recovery of latent variables. *Journal of Machine Learning Research*, 26(274):1–44.

Anderson, T. W. and Rubin, H. (1949). Estimation of the parameters of a single equation in a complete system of stochastic equations. *The Annals of mathematical statistics*, 20(1):46–63.

Armstrong, T. B. and Kolesár, M. (2021). Sensitivity analysis using approximate moment condition models. *Quantitative Economics*, 12(1):77–108.

Du, J.-H., Roeder, K., and Wasserman, L. (2026). Assumption-lean post-integrated inference with surrogate-control outcomes. *Biometrika*, page asag004.

Dukes, O., Richardson, D. B., Shahn, Z., Robins, J. M., and Tchetgen Tchetgen, E. J. (2025). Using negative controls to identify causal effects with invalid instrumental variables. *Biometrika*, 112(1):asae064.

Evdokimov, K. S. and Zeleneev, A. (2023). Simple estimation of semiparametric models with measurement errors. *arXiv preprint arXiv:2306.14311*.

Hansen, L. P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica: Journal of the econometric society*, pages 1029–1054.

Imbens, G., Kallus, N., Mao, X., and Wang, Y. (2025). Long-term causal inference under persistent confounding via data combination. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(2):362–388.

Imbens, G. W. and Angrist, J. D. (1994). Identification and estimation of local average treatment effects. *Econometrica*, 62(2):467–475.

Kallus, N. and Mao, X. (2025). On the role of surrogates in the efficient estimation of treatment effects with limited outcome data. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(2):480–509.

Miao, W., Geng, Z., and Tchetgen Tchetgen, E. J. (2018). Identifying causal effects with proxy variables of an unmeasured confounder. *Biometrika*, 105(4):987–993.

Miao, W., Hu, W., Ogburn, E. L., and Zhou, X.-H. (2023). Identifying effects of multiple treatments in the presence of unmeasured confounding. *Journal of the American Statistical Association*, 118(543):1953–1967.

Miao, W., Shi, X., Li, Y., and Tchetgen Tchetgen, E. J. (2024). A confounding bridge approach for double negative control inference on causal effects. *Statistical Theory and Related Fields*, pages 1–12.

Sargan, J. D. (1958). The estimation of economic relationships using instrumental variables. *Econometrica: Journal of the econometric society*, pages 393–415.

Shi, X., Miao, W., Nelson, J. C., and Tchetgen Tchetgen, E. J. (2020). Multiply robust causal inference with double-negative control adjustment for categorical unmeasured confounding. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 82(2):521–540.

Stock, J. H. and Yogo, M. (2002). Testing for weak instruments in linear iv regression.

Tchetgen Tchetgen, E. J., Ying, A., Cui, Y., Shi, X., and Miao, W. (2024). An introduction to proximal causal inference. *Statistical Science*, 39(3):375–390.

Theil, H. (1953). Repeated least squares applied to complete equation systems. *The Hague: central planning bureau*.

Wang, Y. and Blei, D. M. (2019). The blessings of multiple causes. *Journal of the American Statistical Association*, 114(528):1574–1596.

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data*. MIT press.

A Proof

A.1 Proof of Proposition 1

Proof of Proposition 1. Recall that the projected IV population moment condition is $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top \theta)] = 0_p$, where $\theta \in \mathbb{R}^p$ is a candidate coefficient vector. Using $y_i = x_i^\top \beta + \varsigma_i$, we have

$$\mathbb{E}\big[(\Pi_z x_i)\big(y_i - x_i^\top \theta\big)\big] = \mathbb{E}\big[(\Pi_z x_i) x_i^\top\big]\,(\beta - \theta) + \mathbb{E}[(\Pi_z x_i)\varsigma_i]\,.$$

By the definitions $Q_{\text{IV}} := \mathbb{E}[(\Pi_z x_i) x_i^\top]$ and $\delta_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)\varsigma_i]$, the population moment condition is therefore equivalent to

$$Q_{\text{IV}}(\theta - \beta) = \delta_{\text{IV}}. \tag{A.1}$$

By assumption, $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, and hence Q_{IV} is nonsingular. Left-multiplying (A.1) by Q_{IV}^{-1} gives

$$\theta = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}.$$

Thus a solution exists.

To see that the solution is unique, suppose that both θ_1 and θ_2 satisfy (A.1). Then $Q_{\text{IV}}(\theta_1 - \theta_2) = 0_p$. Since Q_{IV} is nonsingular, its null space contains only 0_p, so $\theta_1 - \theta_2 = 0_p$, and hence $\theta_1 = \theta_2$. Therefore the projected IV population moment uniquely identifies

$$\beta_{\text{IV}} = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}.$$

Finally, if $\delta_{\text{IV}} = 0_p$, then $Q_{\text{IV}}(\beta_{\text{IV}} - \beta) = 0_p$. Nonsingularity of Q_{IV} implies $\beta_{\text{IV}} = \beta$. Hence, when the projected IV moment bias vanishes, the projected IV population moment identifies the structural coefficient β. □

A.2 Proof of Theorem 2

Proof of Theorem 2. Recall that $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$, $Q_{\text{IV}} = \mathbb{E}[(\Pi_z x_i) x_i^\top] = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$, $\delta_{\text{IV}} = \mathbb{E}[(\Pi_z x_i)\varsigma_i]$, and $\beta_{\text{IV}} = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$.

Define $r_{\text{IV},i} := y_i - x_i^\top \beta_{\text{IV}} = \varsigma_i - x_i^\top Q_{\text{IV}}^{-1}\delta_{\text{IV}}$, and write $r_{\text{IV}} := (r_{\text{IV},1}, \ldots, r_{\text{IV},n})^\top = Y - X\beta_{\text{IV}}$. Then

$$\mathbb{E}[(\Pi_z x_i) r_{\text{IV},i}] = \delta_{\text{IV}} - Q_{\text{IV}} Q_{\text{IV}}^{-1}\delta_{\text{IV}} = 0_p.$$

Hence, β_{IV} satisfies the population projected normal equation

$$\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i}) = 0_p. \tag{A.2}$$

Importantly, (A.2) does not generally imply $\mathbb{E}(z_i r_{\text{IV},i}) = 0_q$.

Using $Y = X\beta_{\text{IV}} + r_{\text{IV}}$, the 2SLS estimator satisfies the exact decomposition

$$\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}} = \left(\frac{1}{n} X^\top P_Z X\right)^{-1}\left(\frac{1}{n} X^\top P_Z r_{\text{IV}}\right). \tag{A.3}$$

Consistency. Suppose Assumption 1 (i) holds. We first consider the denominator in (A.3).

Under the maintained second-moment conditions, the Cauchy–Schwarz inequality gives $\mathbb{E}\|x_i z_i^\top\|_{\text{F}} \le \{\mathbb{E}\|x_i\|_2^2\}^{1/2}\{\mathbb{E}\|z_i\|_2^2\}^{1/2} < \infty$, while $\mathbb{E}\|z_i z_i^\top\|_{\text{F}} = \mathbb{E}\|z_i\|_2^2 < \infty$. Therefore, the weak law of large numbers applied entrywise gives $\frac{1}{n}X^\top Z \xrightarrow{\text{p}} \Sigma_{xz}$, $\frac{1}{n}Z^\top Z \xrightarrow{\text{p}} \Sigma_z$, and $\frac{1}{n}Z^\top X \xrightarrow{\text{p}} \Sigma_{zx}$. Since $\Sigma_z \in \mathbb{S}^q_{++}$, the continuous mapping theorem gives $(\frac{1}{n}Z^\top Z)^{-1} \xrightarrow{\text{p}} \Sigma_z^{-1}$. Thus,

$$\frac{1}{n} X^\top P_Z X = \left(\frac{1}{n} X^\top Z\right)\left(\frac{1}{n} Z^\top Z\right)^{-1}\left(\frac{1}{n} Z^\top X\right) \xrightarrow{\text{p}} Q_{\text{IV}}. \tag{A.4}$$

Because $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, $(\frac{1}{n}X^\top P_Z X)^{-1} \xrightarrow{\text{p}} Q_{\text{IV}}^{-1}$.

We next consider the numerator. Since $r_{\text{IV},i} = \varsigma_i - x_i^\top Q_{\text{IV}}^{-1}\delta_{\text{IV}}$, we have $\mathbb{E}(r_{\text{IV},i}^2) \le 2\mathbb{E}(\varsigma_i^2) + 2\|Q_{\text{IV}}^{-1}\delta_{\text{IV}}\|_2^2\mathbb{E}\|x_i\|_2^2 < \infty$. It follows from Cauchy–Schwarz that $\mathbb{E}\|z_i r_{\text{IV},i}\|_2 < \infty$, and hence $\frac{1}{n}\sum_{i=1}^n z_i r_{\text{IV},i} \xrightarrow{\text{p}} \mathbb{E}(z_i r_{\text{IV},i})$. Therefore,

$$\frac{1}{n} X^\top P_Z r_{\text{IV}} = \left(\frac{1}{n} X^\top Z\right)\left(\frac{1}{n} Z^\top Z\right)^{-1}\frac{1}{n}\sum_{i=1}^n z_i r_{\text{IV},i} \xrightarrow{\text{p}} \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i}) = 0_p,$$

where the last equality follows from (A.2).

Combining this result with (A.4) and (A.3) gives $\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}} \xrightarrow{\text{p}} 0_p$. Hence, $\widehat{\beta}_{\text{IV}} \xrightarrow{\text{p}} \beta_{\text{IV}}$.

Asymptotic normality. Suppose further that Assumption 1 (ii) holds. Multiplying (A.3) by $\sqrt{n}$ gives

$$\sqrt{n}(\widehat{\beta}_{\mathrm{IV}} - \beta_{\mathrm{IV}}) = \left(\frac{1}{n}X^\top P_Z X\right)^{-1}\left(\frac{1}{\sqrt{n}}X^\top P_Z r_{\mathrm{IV}}\right). \tag{A.5}$$

By (A.4), the first factor converges in probability to Q_{IV}^{-1}. It therefore remains to derive the limiting distribution of the normalized numerator.

Directly expanding the numerator gives

$$\frac{1}{\sqrt{n}}X^\top P_Z r_{\mathrm{IV}} = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i r_{\mathrm{IV},i}. \tag{A.6}$$

The projected normal equation (A.2) only requires $\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i}) = 0_p$, while $\mathbb{E}(z_i r_{\mathrm{IV},i})$ itself need not vanish. Consequently, $n^{-1/2}\sum_{i=1}^n z_i r_{\mathrm{IV},i}$ may contain a mean component of order $\sqrt{n}$, so this sum is not generally $O_p(1)$. Thus, unlike a directly centered CLT term, the random projection coefficient in (A.6) cannot immediately be replaced by its probability limit.

To obtain a centered term, define

$$e_{\mathrm{IV},i} := r_{\mathrm{IV},i} - z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i}).$$

The consistency argument gives $\mathbb{E}\|z_i r_{\mathrm{IV},i}\|_2 < \infty$, so $\mathbb{E}(z_i r_{\mathrm{IV},i})$ is finite. By construction,

$$\mathbb{E}(z_i e_{\mathrm{IV},i}) = \mathbb{E}(z_i r_{\mathrm{IV},i}) - \Sigma_z\Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i}) = 0_q.$$

Assumption 1 (ii) further gives $\mathbb{E}\|z_i e_{\mathrm{IV},i}\|_2^2 < \infty$. Hence, the multivariate central limit theorem implies $n^{-1/2}\sum_{i=1}^n z_i e_{\mathrm{IV},i} = O_p(1)$.

Writing $e_{\mathrm{IV}} := (e_{\mathrm{IV},1}, \ldots, e_{\mathrm{IV},n})^\top$, the preceding definition gives $r_{\mathrm{IV}} = e_{\mathrm{IV}} + Z\Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i})$. At this point, the IV argument differs from the corresponding SV argument. For SV, the residual-maker eliminates the ancillary-variable component because $P_W^\perp W = 0$. For IV, however, $P_Z Z = Z$, so the component in the span of Z is retained. Substituting the decomposition of r_{IV} into (A.6) therefore gives the exact decomposition

$$\begin{aligned}\frac{1}{\sqrt{n}}X^\top P_Z r_{\mathrm{IV}} = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i e_{\mathrm{IV},i} \\ + \frac{1}{\sqrt{n}}\sum_{i=1}^n x_i z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i}).\end{aligned} \tag{A.7}$$

Thus, in contrast to SV, recentering does not remove the ancillary-variable component but instead produces an additional first-order correction term.

For the first term in (A.7), the consistency argument gives

$$\left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1} = \Sigma_{xz}\Sigma_z^{-1} + o_p(1).$$

Since $n^{-1/2}\sum_{i=1}^n z_i e_{\mathrm{IV},i} = O_p(1)$, the estimation error in this coefficient contributes only $o_p(1)$. Hence,

$$\left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i e_{\mathrm{IV},i} = \frac{1}{\sqrt{n}}\sum_{i=1}^n (\Pi_z x_i) e_{\mathrm{IV},i} + o_p(1).$$

Substituting this expression into (A.7) motivates defining

$$\psi_{\mathrm{IV},i} := (\Pi_z x_i) e_{\mathrm{IV},i} + x_i z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i}). \tag{A.8}$$

The normalized numerator then admits the first-order representation

$$\frac{1}{\sqrt{n}}X^\top P_Z r_{\mathrm{IV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{IV},i} + o_p(1). \tag{A.9}$$

It remains to verify that the summands in (A.9) are centered and have finite second moments. Using $\mathbb{E}(z_i e_{\mathrm{IV},i}) = 0_q$,

$$\mathbb{E}(\psi_{\mathrm{IV},i}) = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i e_{\mathrm{IV},i}) + \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i}) = 0_p,$$

where the last equality follows from (A.2).

For the second moment, $\mathbb{E}\|(\Pi_z x_i)e_{\mathrm{IV},i}\|_2^2 \le \|\Sigma_{xz}\Sigma_z^{-1}\|_{\mathrm{op}}^2 \mathbb{E}\|z_i e_{\mathrm{IV},i}\|_2^2 < \infty$. Moreover,

$$\mathbb{E}\left\|x_i z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i})\right\|_2^2 \le \left\|\Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i})\right\|_2^2 \mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty.$$

Therefore, $\mathbb{E}\|\psi_{\mathrm{IV},i}\|_2^2 < \infty$. The second bound is precisely where the stronger condition $\mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$ in Assumption 1(ii) is needed. Under valid IV, $\mathbb{E}(z_i r_{\mathrm{IV},i}) = 0_q$, so the correction term in (A.8) vanishes and this additional bound is unnecessary for that term.

Because the observations are i.i.d., the $\psi_{\mathrm{IV},i}$ are i.i.d., centered, and have finite second moments. Therefore, the multivariate central limit theorem gives

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{IV},i} \xrightarrow{\mathrm{d}} \mathcal{N}(0,\Sigma_{\mathrm{IV}}), \qquad \Sigma_{\mathrm{IV}} := \mathrm{Var}(\psi_{\mathrm{IV},i}). \tag{A.10}$$

Combining (A.9) and (A.10) yields $\frac{1}{\sqrt{n}}X^\top P_Z r_{\mathrm{IV}} \xrightarrow{\mathrm{d}} \mathcal{N}(0,\Sigma_{\mathrm{IV}})$.

Finally, combining this result with (A.5), (A.4), and Slutsky's theorem gives

$$\sqrt{n}(\widehat{\beta}_{\mathrm{IV}} - \beta_{\mathrm{IV}}) \xrightarrow{\mathrm{d}} \mathcal{N}\left(0, Q_{\mathrm{IV}}^{-1}\Sigma_{\mathrm{IV}}Q_{\mathrm{IV}}^{-1}\right) = \mathcal{N}(0,V_{\mathrm{IV}}), \tag{A.11}$$

where $V_{\mathrm{IV}} := Q_{\mathrm{IV}}^{-1}\Sigma_{\mathrm{IV}}Q_{\mathrm{IV}}^{-1}$. This completes the proof.

□

A.3 Proof of Proposition 3

Proof of Proposition 3. Recall that the residualized population moment condition is $\mathbb{E}[(\Pi_w^\perp x_i)(y_i - x_i^\top\theta)] = 0_p$, where $\theta \in \mathbb{R}^p$ is a candidate coefficient vector. Using $y_i = x_i^\top\beta + \varsigma_i$, we have

$$\mathbb{E}\left[(\Pi_w^\perp x_i)\left(y_i - x_i^\top\theta\right)\right] = \mathbb{E}\left[(\Pi_w^\perp x_i)x_i^\top\right](\beta-\theta) + \mathbb{E}\left[(\Pi_w^\perp x_i)\varsigma_i\right].$$

By the definitions $Q_{\mathrm{SV}} := \mathbb{E}[(\Pi_w^\perp x_i)x_i^\top]$ and $\delta_{\mathrm{SV}} := \mathbb{E}[(\Pi_w^\perp x_i)\varsigma_i]$, the population moment condition is therefore equivalent to

$$Q_{\mathrm{SV}}(\theta-\beta) = \delta_{\mathrm{SV}}. \tag{A.12}$$

By assumption, $Q_{\mathrm{SV}} \in \mathbb{S}_{++}^p$, and hence Q_{SV} is nonsingular. Left-multiplying (A.12) by Q_{SV}^{-1} gives

$$\theta = \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}.$$

Thus a solution exists.

To see that the solution is unique, suppose that both θ_1 and θ_2 satisfy (A.12). Then $Q_{\mathrm{SV}}(\theta_1-\theta_2) = 0_p$. Since Q_{SV} is nonsingular, its null space contains only 0_p, so $\theta_1 - \theta_2 = 0_p$, and hence $\theta_1 = \theta_2$. Therefore the residualized population moment uniquely identifies

$$\beta_{\mathrm{SV}} = \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}.$$

Finally, if $\delta_{\mathrm{SV}} = 0_p$, then $Q_{\mathrm{SV}}(\beta_{\mathrm{SV}} - \beta) = 0_p$. Nonsingularity of Q_{SV} implies $\beta_{\mathrm{SV}} = \beta$. Hence, when the residualized moment bias vanishes, the surrogate-control population moment identifies the structural coefficient β. □

A.4 Proof of Theorem 4

Proof of Theorem 4. Recall that $\Pi_w^\perp x_i = x_i - \Sigma_{xw}\Sigma_w^{-1}w_i$, $Q_{\mathrm{SV}} = \mathbb{E}\{(\Pi_w^\perp x_i)x_i^\top\}$, $\delta_{\mathrm{SV}} = \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\}$, and $\beta_{\mathrm{SV}} = \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$.

Within this proof, define $\Gamma_w := \Sigma_w^{-1}\Sigma_{wx}$, $\widetilde{x}_i := x_i - \Gamma_w^\top w_i = \Pi_w^\perp x_i$, and let $\widetilde{X}$ denote the matrix whose i-th row is $\widetilde{x}_i^\top$. Then

$$x_i = \Gamma_w^\top w_i + \widetilde{x}_i, \qquad X = W\Gamma_w + \widetilde{X}. \tag{A.13}$$

By the definition of the population linear projection,

$$\mathbb{E}(\widetilde{x}_i w_i^\top) = \Sigma_{xw} - \Gamma_w^\top\Sigma_w = 0_{p\times m}. \tag{A.14}$$

Consequently,

$$Q_{\mathrm{SV}} = \mathbb{E}(\widetilde{x}_i x_i^\top) = \mathbb{E}(\widetilde{x}_i \widetilde{x}_i^\top) + \mathbb{E}(\widetilde{x}_i w_i^\top)\Gamma_w = \mathbb{E}(\widetilde{x}_i \widetilde{x}_i^\top).$$

Hence Q_{SV} is symmetric, and it is positive definite by Assumption 2(i).

Define $r_{\mathrm{SV},i} := y_i - x_i^\top \beta_{\mathrm{SV}} = \varsigma_i - x_i^\top Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$, and write $r_{\mathrm{SV}} := (r_{\mathrm{SV},1}, \ldots, r_{\mathrm{SV},n})^\top$. Using $\widetilde{x}_i = \Pi_w^\perp x_i$, we obtain

$$\begin{aligned}\mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) &= \mathbb{E}(\widetilde{x}_i \varsigma_i) - \mathbb{E}(\widetilde{x}_i x_i^\top) Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}} \\ &= \delta_{\mathrm{SV}} - Q_{\mathrm{SV}} Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}} \\ &= 0_p.\end{aligned} \tag{A.15}$$

Thus, β_{SV} satisfies the population residualized normal equation.

Since $P_W^\perp W = 0$, decomposition (A.13) gives $P_W^\perp X = P_W^\perp \widetilde{X}$. Because $P_W^\perp$ is symmetric, $X^\top P_W^\perp X = \widetilde{X}^\top P_W^\perp \widetilde{X}$ and $X^\top P_W^\perp r_{\mathrm{SV}} = \widetilde{X}^\top P_W^\perp r_{\mathrm{SV}}$. The estimator therefore satisfies the exact decomposition

$$\widehat{\beta}_{\mathrm{SV}} - \beta_{\mathrm{SV}} = \left(\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X}\right)^{-1} \left(\frac{1}{n}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}}\right). \tag{A.16}$$

Consistency. Suppose Assumption 2 (i) holds. The maintained second-moment conditions and the weak law of large numbers give $\frac{1}{n}\widetilde{X}^\top \widetilde{X} \xrightarrow{\mathrm{p}} \mathbb{E}(\widetilde{x}_i \widetilde{x}_i^\top) = Q_{\mathrm{SV}}$, $\frac{1}{n}\widetilde{X}^\top W \xrightarrow{\mathrm{p}} \mathbb{E}(\widetilde{x}_i w_i^\top) = 0_{p\times m}$, and $\frac{1}{n}W^\top W \xrightarrow{\mathrm{p}} \Sigma_w$. Since $\Sigma_w \in \mathbb{S}_{++}^m$,

$$\begin{aligned}\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X} &= \frac{1}{n}\widetilde{X}^\top \widetilde{X} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\left(\frac{1}{n}W^\top \widetilde{X}\right) \\ &\xrightarrow{\mathrm{p}} Q_{\mathrm{SV}}.\end{aligned}$$

It follows that $\{\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X}\}^{-1} \xrightarrow{\mathrm{p}} Q_{\mathrm{SV}}^{-1}$. Similarly, the weak law of large numbers gives $\frac{1}{n}\sum_{i=1}^n \widetilde{x}_i r_{\mathrm{SV},i} \xrightarrow{\mathrm{p}} \mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) = 0_p$ by (A.15), and $\frac{1}{n}\sum_{i=1}^n w_i r_{\mathrm{SV},i} \xrightarrow{\mathrm{p}} \mathbb{E}(w_i r_{\mathrm{SV},i})$. Therefore,

$$\begin{aligned}\frac{1}{n}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}} &= \frac{1}{n}\sum_{i=1}^n \widetilde{x}_i r_{\mathrm{SV},i} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^n w_i r_{\mathrm{SV},i}\right) \\ &\xrightarrow{\mathrm{p}} 0_p.\end{aligned}$$

Combining these limits with (A.16) yields

$$\widehat{\beta}_{\mathrm{SV}} \xrightarrow{\mathrm{p}} \beta_{\mathrm{SV}}.$$

Asymptotic normality. Suppose further that Assumption 2 (ii) holds. Multiplying (A.16) by $\sqrt{n}$ gives

$$\sqrt{n}(\widehat{\beta}_{\mathrm{SV}} - \beta_{\mathrm{SV}}) = \left(\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X}\right)^{-1} \frac{1}{\sqrt{n}}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}}.$$

The first factor converges in probability to Q_{SV}^{-1}, so it remains to derive the limiting distribution of the normalized numerator.

A direct expansion gives

$$\frac{1}{\sqrt{n}}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \widetilde{x}_i r_{\mathrm{SV},i} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n w_i r_{\mathrm{SV},i}. \tag{A.17}$$

Although $\mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) = 0_p$, this does not imply $\mathbb{E}(w_i r_{\mathrm{SV},i}) = 0_m$. Hence the last sum in (A.17) may contain a nonzero mean component of order $\sqrt{n}$, so the second term cannot yet be treated as negligible.

To center this sum, define

$$c_{\mathrm{SV}} := \Sigma_w^{-1}\mathbb{E}(w_i r_{\mathrm{SV},i}), \qquad e_{\mathrm{SV},i} := r_{\mathrm{SV},i} - w_i^\top c_{\mathrm{SV}}. \tag{A.18}$$

Then $\mathbb{E}(w_i e_{\mathrm{SV},i}) = \mathbb{E}(w_i r_{\mathrm{SV},i}) - \Sigma_w c_{\mathrm{SV}} = 0_m$. Writing $e_{\mathrm{SV}} := (e_{\mathrm{SV},1}, \ldots, e_{\mathrm{SV},n})^\top$, equation (A.18) gives $r_{\mathrm{SV}} = e_{\mathrm{SV}} + W c_{\mathrm{SV}}$. Since $P_W^\perp W = 0$, we have the exact identity

$$P_W^\perp r_{\mathrm{SV}} = P_W^\perp e_{\mathrm{SV}}.$$

The corrected numerator therefore satisfies

$$\frac{1}{\sqrt{n}}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \widetilde{x}_i e_{\mathrm{SV},i} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n w_i e_{\mathrm{SV},i}.$$

Assumption 2(ii) and $\mathbb{E}(w_i e_{\mathrm{SV},i}) = 0_m$ imply $n^{-1/2}\sum_{i=1}^n w_i e_{\mathrm{SV},i} = O_p(1)$ by the multivariate central limit theorem. Moreover, $\frac{1}{n}\widetilde{X}^\top W = o_p(1)$ and $\{\frac{1}{n}W^\top W\}^{-1} = O_p(1)$. Thus, the second term is $o_p(1)$.

Define $\psi_{\mathrm{SV},i} := \widetilde{x}_i e_{\mathrm{SV},i} = (\Pi_w^\perp x_i) e_{\mathrm{SV},i}$. The preceding argument gives the first-order representation

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{SV},i} + o_p(1). \tag{A.19}$$

The score is centered because, by (A.14) and (A.15),

$$\mathbb{E}(\psi_{\mathrm{SV},i}) = \mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) - \mathbb{E}(\widetilde{x}_i w_i^\top)c_{\mathrm{SV}} = 0_p.$$

It remains to verify that $\mathbb{E}\|\psi_{\mathrm{SV},i}\|_2^2 < \infty$. Since $r_{\mathrm{SV},i} = \varsigma_i - x_i^\top Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$,

$$\mathbb{E}\|x_i r_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i\varsigma_i\|_2^2 + 2\|Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}\|_2^2\mathbb{E}\|x_i x_i^\top\|_{\mathrm{op}}^2 < \infty.$$

The consistency argument and the nonsingularity of Σ_w imply that c_{SV} in (A.18) is finite. Hence,

$$\mathbb{E}\|x_i e_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i r_{\mathrm{SV},i}\|_2^2 + 2\|c_{\mathrm{SV}}\|_2^2\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty.$$

Finally, $\psi_{\mathrm{SV},i} = x_i e_{\mathrm{SV},i} - \Gamma_w^\top w_i e_{\mathrm{SV},i}$, so

$$\mathbb{E}\|\psi_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i e_{\mathrm{SV},i}\|_2^2 + 2\|\Gamma_w^\top\|_{\mathrm{op}}^2\mathbb{E}\|w_i e_{\mathrm{SV},i}\|_2^2 < \infty.$$

Because the observations are i.i.d., the random vectors $\psi_{\mathrm{SV},i}$ are i.i.d., centered, and have finite second moments. The multivariate central limit theorem therefore gives

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{SV},i} \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma_{\mathrm{SV}}), \qquad \Sigma_{\mathrm{SV}} := \mathrm{Var}(\psi_{\mathrm{SV},i}).$$

Combining this result with (A.19) yields

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp r_{\mathrm{SV}} \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma_{\mathrm{SV}}).$$

Finally, Slutsky's theorem gives

$$\sqrt{n}(\widehat{\beta}_{\mathrm{SV}} - \beta_{\mathrm{SV}}) \xrightarrow{\mathrm{d}} \mathcal{N}\left(0, Q_{\mathrm{SV}}^{-1}\Sigma_{\mathrm{SV}}Q_{\mathrm{SV}}^{-1}\right).$$

Thus,

$$V_{\mathrm{SV}} = Q_{\mathrm{SV}}^{-1}\Sigma_{\mathrm{SV}}Q_{\mathrm{SV}}^{-1}.$$

This completes the proof.

□

A.5 Proof of Theorem 5

Proof of Theorem 5. Recall that $\varsigma_i = y_i - x_i^\top\beta$ and

$$\widehat{g}(\theta) = \frac{1}{n}\begin{pmatrix} X^\top P_Z(Y - X\theta) \\ X^\top P_W^\perp(Y - X\theta)\end{pmatrix}.$$

Let

$$\widehat{J}_n := \frac{1}{n}\begin{pmatrix} X^\top P_Z X \\ X^\top P_W^\perp X\end{pmatrix}.$$

Since the sample moment is linear in θ,

$$\widehat{g}(\theta) = \widehat{g}(\beta) - \widehat{J}_n(\theta - \beta). \tag{A.20}$$

Assumption 3 (i) gives $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$ and $\mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = 0_p$. Thus, both projected moment blocks are correctly specified at β.

The first-order condition for $\widehat{\beta}(M)$ is $\widehat{J}_n^\top M\widehat{g}\{\widehat{\beta}(M)\} = 0_p$. Using (A.20), we obtain the exact estimation-error decomposition

$$\widehat{\beta}(M) - \beta = \left(\widehat{J}_n^\top M\widehat{J}_n\right)^{-1}\widehat{J}_n^\top M\widehat{g}(\beta). \tag{A.21}$$

Consistency. Suppose Assumption 3 (i) holds. We first establish the probability limit of $\widehat{J}_n$.

For the IV block, the maintained second-moment conditions and the weak law of large numbers give $\frac{1}{n}X^\top Z \xrightarrow{\mathrm{p}} \Sigma_{xz}$, $\frac{1}{n}Z^\top Z \xrightarrow{\mathrm{p}} \Sigma_z$, and $\frac{1}{n}Z^\top X \xrightarrow{\mathrm{p}} \Sigma_{zx}$. Since $\Sigma_z \in \mathbb{S}^q_{++}$, $(\frac{1}{n}Z^\top Z)^{-1} \xrightarrow{\mathrm{p}} \Sigma_z^{-1}$. Hence,

$$\frac{1}{n}X^\top P_Z X = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}Z^\top X\right) \xrightarrow{\mathrm{p}} Q_{\mathrm{IV}}.$$

For the SV block, the weak law of large numbers gives $\frac{1}{n}X^\top X \xrightarrow{\mathrm{p}} \Sigma_x$, $\frac{1}{n}X^\top W \xrightarrow{\mathrm{p}} \Sigma_{xw}$, $\frac{1}{n}W^\top W \xrightarrow{\mathrm{p}} \Sigma_w$, and $\frac{1}{n}W^\top X \xrightarrow{\mathrm{p}} \Sigma_{wx}$. Since $\Sigma_w \in \mathbb{S}^m_{++}$, $(\frac{1}{n}W^\top W)^{-1} \xrightarrow{\mathrm{p}} \Sigma_w^{-1}$. Therefore,

$$\frac{1}{n}X^\top P_W^\perp X \xrightarrow{\mathrm{p}} \Sigma_x - \Sigma_{xw}\Sigma_w^{-1}\Sigma_{wx} = Q_{\mathrm{SV}}.$$

Combining the two blocks gives

$$\widehat{J}_n \xrightarrow{\mathrm{p}} \begin{pmatrix} Q_{\mathrm{IV}} \\ Q_{\mathrm{SV}} \end{pmatrix} = J. \tag{A.22}$$

We next establish the probability limit of $\widehat{g}(\beta)$. For the IV block,

$$\frac{1}{n}X^\top P_Z \varsigma = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^n z_i\varsigma_i\right) \xrightarrow{\mathrm{p}} \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) = 0_p,$$

where the final equality follows from $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$. In particular, this does not require the stronger raw IV restriction $\mathbb{E}(z_i\varsigma_i) = 0_q$.

For the SV block,

$$\frac{1}{n}X^\top P_W^\perp \varsigma \xrightarrow{\mathrm{p}} \mathbb{E}(x_i\varsigma_i) - \Sigma_{xw}\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i) = \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = 0_p.$$

Hence,

$$\widehat{g}(\beta) \xrightarrow{\mathrm{p}} 0_{2p}. \tag{A.23}$$

Since $M \in \mathbb{S}^{2p}_{++}$ and J has full column rank p, $J^\top M J \in \mathbb{S}^p_{++}$. Indeed, for every nonzero $c \in \mathbb{R}^p$, $c^\top J^\top M J c = (Jc)^\top M (Jc) > 0$. It follows from (A.22) and the continuous mapping theorem that

$$\left(\widehat{J}_n^\top M \widehat{J}_n\right)^{-1}\widehat{J}_n^\top M \xrightarrow{\mathrm{p}} (J^\top M J)^{-1} J^\top M = K_M.$$

Combining this result with (A.21) and (A.23) gives $\widehat{\beta}(M) - \beta \xrightarrow{\mathrm{p}} 0_p$. Therefore, $\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta$.

Asymptotic normality. Suppose further that Assumption 3 (ii) holds. Multiplying (A.21) by $\sqrt{n}$ gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} = \left(\widehat{J}_n^\top M \widehat{J}_n\right)^{-1}\widehat{J}_n^\top M \sqrt{n}\,\widehat{g}(\beta). \tag{A.24}$$

By the consistency argument, the matrix multiplying $\sqrt{n}\,\widehat{g}(\beta)$ converges to K_M. It therefore remains to derive a central limit theorem for the two projected moment blocks in $\sqrt{n}\,\widehat{g}(\beta)$.

For notational convenience within this proof, define $c_z := \Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)$ and $c_w := \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)$, and let $e_{z,i} := \varsigma_i - z_i^\top c_z$ and $e_{w,i} := \varsigma_i - w_i^\top c_w$. The standing second-moment conditions imply that $\mathbb{E}(z_i\varsigma_i)$ and $\mathbb{E}(w_i\varsigma_i)$ are finite, so c_z and c_w are finite nonrandom vectors. By construction, $\mathbb{E}(z_i e_{z,i}) = 0_q$ and $\mathbb{E}(w_i e_{w,i}) = 0_m$.

We first consider the projected IV block. A direct expansion in terms of ς_i cannot generally be treated by replacing the sample projection coefficient by its probability limit, because the projected restriction $\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) = 0_p$ does not imply $\mathbb{E}(z_i\varsigma_i) = 0_q$. Thus, $n^{-1/2}\sum_{i=1}^n z_i\varsigma_i$ may contain a mean component of order $\sqrt{n}$.

Using $\varsigma_i = e_{z,i} + z_i^\top c_z$, the IV block admits the exact decomposition

$$\frac{1}{\sqrt{n}}X^\top P_Z \varsigma = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i e_{z,i}$$

$$+\frac{1}{\sqrt{n}}\sum_{i=1}^{n} x_i z_i^\top c_z. \tag{A.25}$$

The second term remains because $P_Z Z = Z$; hence, the component of the structural error lying in the span of Z is retained by the IV projection.

Assumption 3 (ii) gives $\mathbb{E}\|z_i e_{z,i}\|_2^2 < \infty$, so $n^{-1/2}\sum_{i=1}^n z_i e_{z,i} = O_p(1)$. Moreover,

$$\left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1} = \Sigma_{xz}\Sigma_z^{-1} + o_p(1).$$

Therefore, the first term in (A.25) equals $n^{-1/2}\sum_{i=1}^n(\Pi_z x_i)e_{z,i} + o_p(1)$, and hence

$$\frac{1}{\sqrt{n}}X^\top P_Z \varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\left\{(\Pi_z x_i)e_{z,i} + x_i z_i^\top c_z\right\} + o_p(1). \tag{A.26}$$

We next consider the projected SV block. Here the same recentering has a different consequence. Since $\varsigma_i = e_{w,i} + w_i^\top c_w$ and $P_W^\perp W = 0$, the component lying in the span of W is removed exactly:

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp \varsigma = \frac{1}{\sqrt{n}}X^\top P_W^\perp e_w,$$

where $e_w = (e_{w,1}, \ldots, e_{w,n})^\top$.

Let $\Pi_w^\perp X$ denote the matrix whose i-th row is $(\Pi_w^\perp x_i)^\top$. Since $X = \Pi_w^\perp X + W\Sigma_w^{-1}\Sigma_{wx}$, the identity $P_W^\perp W = 0$ implies $P_W^\perp X = P_W^\perp(\Pi_w^\perp X)$. Expanding the sample residual-maker therefore gives

$$\begin{aligned}\frac{1}{\sqrt{n}}X^\top P_W^\perp \varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}(\Pi_w^\perp x_i)e_{w,i} \\ -\left\{\frac{1}{n}\sum_{i=1}^{n}(\Pi_w^\perp x_i)w_i^\top\right\}\left(\frac{1}{n}\sum_{i=1}^{n} w_i w_i^\top\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} w_i e_{w,i}.\end{aligned} \tag{A.27}$$

By the definition of the population residualization, $\mathbb{E}\{(\Pi_w^\perp x_i)w_i^\top\} = 0_{p\times m}$. The weak law of large numbers therefore gives $n^{-1}\sum_{i=1}^n(\Pi_w^\perp x_i)w_i^\top = o_p(1)$. Also, $(n^{-1}\sum_{i=1}^n w_i w_i^\top)^{-1} \xrightarrow{p} \Sigma_w^{-1}$, while $\mathbb{E}(w_i e_{w,i}) = 0_m$ and $\mathbb{E}\|w_i e_{w,i}\|_2^2 < \infty$ imply $n^{-1/2}\sum_{i=1}^n w_i e_{w,i} = O_p(1)$. Hence, the second term in (A.27) is $o_p(1)$, so

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp \varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}(\Pi_w^\perp x_i)e_{w,i} + o_p(1). \tag{A.28}$$

Equations (A.26) and (A.28) collect the first-order variation of the two feasible projected moment blocks. Recall from Section 3.1 that $\psi_{\mathrm{IV},i}$ and $\psi_{\mathrm{SV},i}$ denote the corresponding first-order scores. Under the notation introduced above, these previously defined scores admit the equivalent representations

$$\psi_{\mathrm{IV},i} = (\Pi_z x_i)e_{z,i} + x_i z_i^\top c_z, \qquad \psi_{\mathrm{SV},i} = (\Pi_w^\perp x_i)e_{w,i}.$$

Recall also that $\psi_i = (\psi_{\mathrm{IV},i}^\top, \psi_{\mathrm{SV},i}^\top)^\top$ and $\Sigma = \mathrm{Var}(\psi_i)$. Stacking (A.26) and (A.28) gives the first-order representation

$$\sqrt{n}\,\widehat{g}(\beta) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_i + o_p(1). \tag{A.29}$$

It remains to verify that the summands in (A.29) are centered and have finite second moments.

For the IV component, using $\mathbb{E}(z_i e_{z,i}) = 0_q$,

$$\mathbb{E}(\psi_{\mathrm{IV},i}) = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i e_{z,i}) + \Sigma_{xz}c_z = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) = 0_p,$$

where the last equality follows from the projected IV restriction. For the SV component,

$$\mathbb{E}(\psi_{\mathrm{SV},i}) = \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} - \mathbb{E}\{(\Pi_w^\perp x_i)w_i^\top\}c_w = 0_p.$$

Hence,

$$\mathbb{E}(\psi_i) = \begin{pmatrix} \mathbb{E}(\psi_{\mathrm{IV},i}) \\ \mathbb{E}(\psi_{\mathrm{SV},i}) \end{pmatrix} = \begin{pmatrix} 0_p \\ 0_p \end{pmatrix} = 0_{2p}.$$

For the IV second moment, $\mathbb{E}\|(\Pi_z x_i)e_{z,i}\|_2^2 \le \|\Sigma_{xz}\Sigma_z^{-1}\|_{\mathrm{op}}^2 \mathbb{E}\|z_i e_{z,i}\|_2^2 < \infty$. Moreover, $\mathbb{E}\|x_i z_i^\top c_z\|_2^2 \le \|c_z\|_2^2 \mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$. Thus, $\mathbb{E}\|\psi_{\mathrm{IV},i}\|_2^2 < \infty$.

For the SV component, first note that $x_i e_{w,i} = x_i \varsigma_i - x_i w_i^\top c_w$. Therefore,

$$\mathbb{E}\|x_i e_{w,i}\|_2^2 \le 2\mathbb{E}\|x_i \varsigma_i\|_2^2 + 2\|c_w\|_2^2 \mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty.$$

Since $(\Pi_w^\perp x_i)e_{w,i} = x_i e_{w,i} - \Sigma_{xw}\Sigma_w^{-1} w_i e_{w,i}$, we further have

$$\mathbb{E}\|\psi_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i e_{w,i}\|_2^2 + 2\|\Sigma_{xw}\Sigma_w^{-1}\|_{\mathrm{op}}^2 \mathbb{E}\|w_i e_{w,i}\|_2^2 < \infty.$$

Consequently, $\mathbb{E}\|\psi_i\|_2^2 < \infty$, so $\Sigma = \mathrm{Var}(\psi_i)$ is well defined.

Because the observations are i.i.d., the ψ_i are i.i.d., centered, and have finite second moments. The multivariate central limit theorem therefore gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_i \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma). \tag{A.30}$$

Combining (A.29) and (A.30) yields

$$\sqrt{n}\, \widehat{g}(\beta) \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma).$$

Finally, by (A.22),

$$\left(\widehat{J}_n^\top M \widehat{J}_n\right)^{-1} \widehat{J}_n^\top M \xrightarrow{\mathrm{p}} (J^\top M J)^{-1} J^\top M = K_M.$$

Combining this result with (A.24) and Slutsky's theorem gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\left(0,\, K_M \Sigma K_M^\top\right) = \mathcal{N}\{0, V(M)\}, \tag{A.31}$$

where $V(M) = K_M \Sigma K_M^\top$. By Assumption 3 (ii), $V(M) \in \mathbb{S}_{++}^p$, so the limiting distribution is nondegenerate. This completes the proof.

□

A.6 Proof of Theorem 6

Proof of Theorem 6. Recall that $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$, $Q_{\mathrm{IV}} = \mathbb{E}[(\Pi_z x_i)x_i^\top]$, $\Pi_w^\perp x_i = x_i - \Sigma_{xw}\Sigma_w^{-1} w_i$, and $Q_{\mathrm{SV}} = \mathbb{E}[(\Pi_w^\perp x_i)x_i^\top]$. Define

$$\widehat{J}_n := \frac{1}{n}\begin{pmatrix} X^\top P_Z X \\ X^\top P_W^\perp X \end{pmatrix}.$$

Since $\widehat{g}(\theta) = n^{-1}\left(X^\top P_Z (Y - X\theta)^\top,\, X^\top P_W^\perp (Y - X\theta)^\top\right)^\top$, the sample moment is linear in θ, so

$$\widehat{g}(\theta) = \widehat{g}(\beta) - \widehat{J}_n(\theta - \beta). \tag{A.32}$$

We first establish the probability limit of the sample Jacobian. For the IV block,

$$\frac{1}{n} X^\top P_Z X = \left(\frac{1}{n} X^\top Z\right)\left(\frac{1}{n} Z^\top Z\right)^{-1}\left(\frac{1}{n} Z^\top X\right).$$

By the weak law of large numbers, $\frac{1}{n} X^\top Z \xrightarrow{\mathrm{p}} \Sigma_{xz}$, $\frac{1}{n} Z^\top Z \xrightarrow{\mathrm{p}} \Sigma_z$, and $\frac{1}{n} Z^\top X \xrightarrow{\mathrm{p}} \Sigma_{zx}$. Since $\Sigma_z \in \mathbb{S}_{++}^q$, the continuous mapping theorem gives

$$\frac{1}{n} X^\top P_Z X \xrightarrow{\mathrm{p}} \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx} = Q_{\mathrm{IV}}.$$

For the SV block,

$$\frac{1}{n} X^\top P_W^\perp X = \frac{1}{n} X^\top X - \left(\frac{1}{n} X^\top W\right)\left(\frac{1}{n} W^\top W\right)^{-1}\left(\frac{1}{n} W^\top X\right).$$

The weak law of large numbers gives $\frac{1}{n}X^\top X \xrightarrow{\text{p}} \mathbb{E}(x_i x_i^\top)$, $\frac{1}{n}X^\top W \xrightarrow{\text{p}} \Sigma_{xw}$, $\frac{1}{n}W^\top X \xrightarrow{\text{p}} \Sigma_{wx}$, and $\frac{1}{n}W^\top W \xrightarrow{\text{p}} \Sigma_w$. Since $\Sigma_w \in \mathbb{S}^m_{++}$,

$$\frac{1}{n}X^\top P_W^\perp X \xrightarrow{\text{p}} \mathbb{E}(x_i x_i^\top) - \Sigma_{xw}\Sigma_w^{-1}\Sigma_{wx} = Q_{\text{SV}}.$$

Hence,

$$\widehat{J}_n \xrightarrow{\text{p}} \begin{pmatrix} Q_{\text{IV}} \\ Q_{\text{SV}} \end{pmatrix} = J. \tag{A.33}$$

By Assumption 4 (i), $J^\top M J \in \mathbb{S}^p_{++}$. Thus, by (A.33) and the continuous mapping theorem,

$$\left(\widehat{J}_n^\top M \widehat{J}_n\right)^{-1} \widehat{J}_n^\top M \xrightarrow{\text{p}} (J^\top M J)^{-1} J^\top M = K_M. \tag{A.34}$$

The first-order condition for minimizing $\widehat{g}(\theta)^\top M \widehat{g}(\theta)$ is $\widehat{J}_n^\top M \widehat{g}\{\widehat{\beta}(M)\} = 0$. Substituting (A.32) therefore gives, with probability approaching one,

$$\widehat{\beta}(M) - \beta = \left(\widehat{J}_n^\top M \widehat{J}_n\right)^{-1} \widehat{J}_n^\top M \widehat{g}(\beta). \tag{A.35}$$

Consistency. Under the local misspecification sequence,

$$\mathbb{E}(h_i \varsigma_i) = \frac{\tau}{\sqrt{n}} = \frac{1}{\sqrt{n}} \begin{pmatrix} \tau_z \\ \tau_w \end{pmatrix}.$$

Thus, $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = \tau_z/\sqrt{n} \to 0_p$ and $\mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = \tau_w/\sqrt{n} \to 0_p$.

For the IV block, since $Y - X\beta = \varsigma$,

$$\frac{1}{n}X^\top P_Z \varsigma = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^n z_i \varsigma_i\right).$$

The weak law of large numbers along the local sequence and the continuous mapping theorem give

$$\frac{1}{n}X^\top P_Z \varsigma - \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) \xrightarrow{\text{p}} 0_p.$$

Since $\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) = \mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = \tau_z/\sqrt{n} \to 0_p$, it follows that $\frac{1}{n}X^\top P_Z(Y - X\beta) \xrightarrow{\text{p}} 0_p$.

For the SV block,

$$\frac{1}{n}X^\top P_W^\perp \varsigma = \frac{1}{n}X^\top \varsigma - \left(\frac{1}{n}X^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\left(\frac{1}{n}W^\top \varsigma\right).$$

Again by the weak law of large numbers and the continuous mapping theorem,

$$\frac{1}{n}X^\top P_W^\perp \varsigma - \left\{\mathbb{E}(x_i\varsigma_i) - \Sigma_{xw}\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\} \xrightarrow{\text{p}} 0_p.$$

The term in braces equals $\mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = \tau_w/\sqrt{n} \to 0_p$. Therefore, $\frac{1}{n}X^\top P_W^\perp(Y - X\beta) \xrightarrow{\text{p}} 0_p$.

Combining the two blocks gives

$$\widehat{g}(\beta) \xrightarrow{\text{p}} 0_{2p}. \tag{A.36}$$

Equations (A.35), (A.34), and (A.36) then imply

$$\widehat{\beta}(M) \xrightarrow{\text{p}} \beta.$$

Asymptotic normality. Suppose further that Assumption 4 (ii)–(iii) hold. Multiplying (A.35) by $\sqrt{n}$ gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} = \left(\widehat{J}_n^\top M \widehat{J}_n\right)^{-1} \widehat{J}_n^\top M \sqrt{n}\,\widehat{g}(\beta). \tag{A.37}$$

By (A.34), the matrix multiplying $\sqrt{n}\,\widehat{g}(\beta)$ converges in probability to K_M. It therefore remains to derive the limiting distribution of the normalized stacked sample moment.

We first obtain a first-order linear representation for its two projected moment blocks. For the IV block,

$$\frac{1}{\sqrt{n}}X^\top P_Z\varsigma = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i\varsigma_i.$$

Because $\mathbb{E}(z_i\varsigma_i)$ need not vanish, write $\varsigma_i = \{\varsigma_i - z_i^\top\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\} + z_i^\top\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)$. Using this decomposition together with the equivalent representation of $\psi_{\mathrm{IV,miss},i}$ in (3.7), a direct rearrangement gives

$$\begin{aligned}&\frac{1}{\sqrt{n}}X^\top P_Z\varsigma - \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{IV,miss},i}\\ &= \left\{\left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1} - \Sigma_{xz}\Sigma_z^{-1}\right\}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i\left\{\varsigma_i - z_i^\top\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\right\}.\end{aligned}$$

The first factor is $o_p(1)$ by the same law-of-large-numbers and continuous-mapping argument used above. The summands in the second factor are centered because $\mathbb{E}[z_i\{\varsigma_i - z_i^\top\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\}] = 0_q$. Moreover, Assumption 4 (ii) gives a uniformly bounded second moment for these summands. Hence their normalized sum is $O_p(1)$, and therefore

$$\frac{1}{\sqrt{n}}X^\top P_Z\varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{IV,miss},i} + o_p(1). \tag{A.38}$$

For the SV block, the same argument applies after removing the component of ς_i linearly explained by w_i. Since $P_W^\perp W = 0$,

$$P_W^\perp\varsigma = P_W^\perp\left\{\varsigma - W\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\}.$$

Expanding $P_W^\perp$ and comparing the resulting expression with $\psi_{\mathrm{SV,miss},i}$ gives

$$\begin{aligned}&\frac{1}{\sqrt{n}}X^\top P_W^\perp\varsigma - \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{SV,miss},i}\\ &= -\left\{\left(\frac{1}{n}X^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1} - \Sigma_{xw}\Sigma_w^{-1}\right\}\frac{1}{\sqrt{n}}\sum_{i=1}^n w_i\left\{\varsigma_i - w_i^\top\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\}.\end{aligned}$$

Again, the first factor is $o_p(1)$. The summands in the second factor are centered because $\mathbb{E}[w_i\{\varsigma_i - w_i^\top\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\}] = 0_m$, and their second moments are uniformly bounded by Assumption 4 (ii). Thus, the normalized sum is $O_p(1)$, so

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp\varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{SV,miss},i} + o_p(1). \tag{A.39}$$

Stacking (A.38) and (A.39) yields the key first-order representation

$$\sqrt{n}\,\widehat{g}(\beta) = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{miss},i} + o_p(1). \tag{A.40}$$

We next verify the mean and second-moment properties of the score. By the population IV projection property, $\mathbb{E}[\{x_i - \Pi_z x_i\}z_i^\top] = 0_{p\times q}$, and hence $\mathbb{E}(\psi_{\mathrm{IV,miss},i}) = \mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = \tau_z/\sqrt{n}$. Similarly, $\mathbb{E}\{(\Pi_w^\perp x_i)w_i^\top\} = 0_{p\times m}$, so $\mathbb{E}(\psi_{\mathrm{SV,miss},i}) = \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = \tau_w/\sqrt{n}$. Therefore,

$$\mathbb{E}(\psi_{\mathrm{miss},i}) = \frac{\tau}{\sqrt{n}}. \tag{A.41}$$

It remains to verify that $\Sigma_{\mathrm{miss},n} = \mathrm{Var}(\psi_{\mathrm{miss},i})$ is well defined. For the IV block, the equivalent score representation gives

$$\mathbb{E}\|\psi_{\mathrm{IV,miss},i}\|_2^2 \le 2\|\Sigma_{xz}\Sigma_z^{-1}\|_{\mathrm{op}}^2\mathbb{E}\left\|z_i\left\{\varsigma_i - z_i^\top\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\right\}\right\|_2^2 + 2\left\|\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\right\|_2^2\mathbb{E}\|x_iz_i^\top\|_{\mathrm{op}}^2.$$

The first term is finite by Assumption 4 (ii). For the second term, Cauchy–Schwarz gives $\mathbb{E}\|z_i\varsigma_i\|_2 \le \{\mathbb{E}\|z_i\|_2^2\}^{1/2}\{\mathbb{E}(\varsigma_i^2)\}^{1/2} < \infty$. Since $\Sigma_z \in \mathbb{S}_{++}^q$, $\|\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\|_2 < \infty$, while $\mathbb{E}\|x_iz_i^\top\|_{\mathrm{op}}^2 < \infty$ by Assumption 4 (ii). Consequently, $\mathbb{E}\|\psi_{\mathrm{IV,miss},i}\|_2^2 < \infty$.

For the SV block, first note that

$$\mathbb{E}\left\|x_i\left\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\}\right\|_2^2$$
$$\leq 2\mathbb{E}\|x_i\varsigma_i\|_2^2 + 2\left\|\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\|_2^2 \mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2.$$

The first term is finite by Assumption 4 (ii). Moreover, Cauchy–Schwarz gives $\mathbb{E}\|w_i\varsigma_i\|_2 \leq \{\mathbb{E}\|w_i\|_2^2\}^{1/2}\{\mathbb{E}(\varsigma_i^2)\}^{1/2} < \infty$. Since $\Sigma_w \in \mathbb{S}_{++}^m$, $\|\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\|_2 < \infty$, and $\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty$ by Assumption 4 (ii). Hence the preceding expectation is finite.

Using

$$\psi_{\mathrm{SV,miss},i} = x_i\left\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\}$$
$$- \Sigma_{xw}\Sigma_w^{-1} w_i\left\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\},$$

we therefore obtain

$$\mathbb{E}\|\psi_{\mathrm{SV,miss},i}\|_2^2 \leq 2\mathbb{E}\left\|x_i\left\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\}\right\|_2^2$$
$$+ 2\|\Sigma_{xw}\Sigma_w^{-1}\|_{\mathrm{op}}^2\mathbb{E}\left\|w_i\left\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\}\right\|_2^2 < \infty,$$

where the last term is finite by Assumption 4 (ii). Thus, $\mathbb{E}\|\psi_{\mathrm{miss},i}\|_2^2 < \infty$, and $\Sigma_{\mathrm{miss},n} = \mathrm{Var}(\psi_{\mathrm{miss},i})$ is well defined along the local sequence.

We can now derive the limiting distribution of the normalized stacked score. By (A.41),

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{miss},i} = \frac{1}{\sqrt{n}}\sum_{i=1}^n\{\psi_{\mathrm{miss},i} - \mathbb{E}(\psi_{\mathrm{miss},i})\} + \tau. \tag{A.42}$$

Thus, it is enough to establish a central limit theorem for the centered normalized sum.

Fix any $a \in \mathbb{R}^{2p}$. Along the local sequence, the scalar summands $a^\top\{\psi_{\mathrm{miss},i} - \mathbb{E}(\psi_{\mathrm{miss},i})\}$ are row-wise independent and centered. Moreover,

$$\mathrm{Var}\left[a^\top\frac{1}{\sqrt{n}}\sum_{i=1}^n\{\psi_{\mathrm{miss},i} - \mathbb{E}(\psi_{\mathrm{miss},i})\}\right] = a^\top\Sigma_{\mathrm{miss},n}a$$
$$\longrightarrow a^\top\Sigma_{\mathrm{miss}}a,$$

where the convergence follows from Assumption 4 (iii).

The same assumption also imposes the vector Lindeberg condition. Since $|a^\top v| \leq \|a\|_2\|v\|_2$ for every $v \in \mathbb{R}^{2p}$, it implies the corresponding scalar Lindeberg condition for $a^\top\{\psi_{\mathrm{miss},i} - \mathbb{E}(\psi_{\mathrm{miss},i})\}$. The scalar Lindeberg–Feller central limit theorem therefore gives

$$a^\top\frac{1}{\sqrt{n}}\sum_{i=1}^n\{\psi_{\mathrm{miss},i} - \mathbb{E}(\psi_{\mathrm{miss},i})\} \xrightarrow{\mathrm{d}} \mathcal{N}\left(0,\, a^\top\Sigma_{\mathrm{miss}}a\right).$$

Because this holds for every fixed $a \in \mathbb{R}^{2p}$, the Cramér–Wold device yields

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n\{\psi_{\mathrm{miss},i} - \mathbb{E}(\psi_{\mathrm{miss},i})\} \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma_{\mathrm{miss}}). \tag{A.43}$$

Combining (A.43) with (A.42), and recalling that τ is fixed along the local sequence, gives

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{miss},i} \xrightarrow{\mathrm{d}} \mathcal{N}(\tau, \Sigma_{\mathrm{miss}}). \tag{A.44}$$

The mean shift from 0 to τ is precisely the first-order effect of the $n^{-1/2}$-local moment violation.

The first-order representation (A.40) gives $\sqrt{n}\,\widehat{g}(\beta) = n^{-1/2}\sum_{i=1}^n \psi_{\mathrm{miss},i} + o_p(1)$. Hence, combining this representation with (A.44) and applying Slutsky's theorem yields

$$\sqrt{n}\,\widehat{g}(\beta) \xrightarrow{\mathrm{d}} \mathcal{N}(\tau, \Sigma_{\mathrm{miss}}). \tag{A.45}$$

Finally, recall from (A.37) that

$$\sqrt{n}\{\widehat{\beta}(M)-\beta\}=\left(\widehat{J}_n^\top M\widehat{J}_n\right)^{-1}\widehat{J}_n^\top M\sqrt{n}\,\widehat{g}(\beta).$$

By (A.34), $(\widehat{J}_n^\top M\widehat{J}_n)^{-1}\widehat{J}_n^\top M \xrightarrow{\mathrm{p}} K_M$, while (A.45) gives $\sqrt{n}\,\widehat{g}(\beta)\xrightarrow{\mathrm{d}}\mathcal{N}(\tau,\Sigma_{\mathrm{miss}})$. Another application of Slutsky's theorem therefore gives

$$\sqrt{n}\{\widehat{\beta}(M)-\beta\}\xrightarrow{\mathrm{d}}\mathcal{N}\left(K_M\tau,\,K_M\Sigma_{\mathrm{miss}}K_M^\top\right)=\mathcal{N}\{K_M\tau,V_{\mathrm{miss}}(M)\}. \tag{A.46}$$

Assumption 4 (iii) ensures that $V_{\mathrm{miss}}(M)\in\mathbb{S}^p_{++}$, so the limiting distribution is nondegenerate.

For the IV-only projected moment system, the same argument applies with $J=Q_{\mathrm{IV}}$, $\Sigma_{\mathrm{miss}}=\Sigma_{\mathrm{IV,miss}}$, $\tau=\tau_z$, and $M=M_z$. Likewise, for the SV-only system, it applies with $J=Q_{\mathrm{SV}}$, $\Sigma_{\mathrm{miss}}=\Sigma_{\mathrm{SV,miss}}$, $\tau=\tau_w$, and $M=M_w$. Whenever the corresponding single-source Jacobian is nonsingular, $K_{M_z}=Q_{\mathrm{IV}}^{-1}$ for the IV-only system and $K_{M_w}=Q_{\mathrm{SV}}^{-1}$ for the SV-only system. Hence, in the exactly identified single-source systems, the local asymptotic bias and variance do not depend on the particular admissible weighting matrix.

This completes the proof.

□

A.7 Proof of Proposition 7

Proof of Proposition 7. Recall that the first-order score of the combined system is

$$\psi_i=\begin{pmatrix}\psi_{\mathrm{IV},i}\\ \psi_{\mathrm{SV},i}\end{pmatrix},\qquad \Sigma=\mathrm{Var}(\psi_i),$$

where, under the correctly specified benchmark $\beta_{\mathrm{IV}}=\beta_{\mathrm{SV}}=\beta$,

$$\begin{aligned}\psi_{\mathrm{IV},i}&=(\Pi_z x_i)\varsigma_i+\{x_i-\Pi_z x_i\}z_i^\top\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i),\\ \psi_{\mathrm{SV},i}&=(\Pi_w^\perp x_i)\left\{\varsigma_i-w_i^\top\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\}.\end{aligned}$$

Recall also that $\Sigma_{\mathrm{IV}}=\mathrm{Var}(\psi_{\mathrm{IV},i})$ and $\Sigma_{\mathrm{SV}}=\mathrm{Var}(\psi_{\mathrm{SV},i})$. For the proof, write

$$\Sigma_{\mathrm{IV,SV}}:=\mathrm{Cov}(\psi_{\mathrm{IV},i},\psi_{\mathrm{SV},i}),$$

so that

$$\Sigma=\begin{pmatrix}\Sigma_{\mathrm{IV}} & \Sigma_{\mathrm{IV,SV}}\\ \Sigma_{\mathrm{IV,SV}}^\top & \Sigma_{\mathrm{SV}}\end{pmatrix}.$$

The Jacobian of the stacked projected moment system is

$$J=\begin{pmatrix}Q_{\mathrm{IV}}\\ Q_{\mathrm{SV}}\end{pmatrix}.$$

We first derive the optimal asymptotic variance of the combined system. For any admissible weighting matrix M, Theorem 5 gives

$$\sqrt{n}\{\widehat{\beta}(M)-\beta\}\xrightarrow{\mathrm{d}}\mathcal{N}\{0,V(M)\},$$

where

$$V(M)=K_M\Sigma K_M^\top,\qquad K_M=(J^\top MJ)^{-1}J^\top M.$$

Notice that

$$K_MJ=(J^\top MJ)^{-1}J^\top MJ=I_p.$$

Consider the weight

$$M^\star=\Sigma^{-1}.$$

Since $\Sigma\in\mathbb{S}^{2p}_{++}$, this weight is well defined and positive definite. The corresponding coefficient matrix is

$$K^\star:=(J^\top\Sigma^{-1}J)^{-1}J^\top\Sigma^{-1},$$

and its asymptotic variance is

$$\begin{aligned} V^\star_{\text{comb}} &:= K^\star \Sigma (K^\star)^\top \\ &= (J^\top \Sigma^{-1} J)^{-1} J^\top \Sigma^{-1} \Sigma \Sigma^{-1} J (J^\top \Sigma^{-1} J)^{-1} \\ &= (J^\top \Sigma^{-1} J)^{-1}. \end{aligned}$$

Hence,

$$V^\star_{\text{comb}} = \left\{ \begin{pmatrix} Q_{\text{IV}} \\ Q_{\text{SV}} \end{pmatrix}^\top \Sigma^{-1} \begin{pmatrix} Q_{\text{IV}} \\ Q_{\text{SV}} \end{pmatrix} \right\}^{-1}.$$

To verify that this weight is optimal, let M be any other admissible weight and write

$$D_M := K_M - K^\star.$$

Since both $K_M J = I_p$ and $K^\star J = I_p$,

$$D_M J = 0.$$

Moreover,

$$K^\star \Sigma = (J^\top \Sigma^{-1} J)^{-1} J^\top.$$

It follows that

$$\begin{aligned} D_M \Sigma (K^\star)^\top &= D_M J (J^\top \Sigma^{-1} J)^{-1} \\ &= 0, \end{aligned}$$

and similarly

$$K^\star \Sigma D_M^\top = 0.$$

Therefore,

$$\begin{aligned} V(M) &= K_M \Sigma K_M^\top \\ &= (K^\star + D_M) \Sigma (K^\star + D_M)^\top \\ &= K^\star \Sigma (K^\star)^\top + D_M \Sigma D_M^\top \\ &= V^\star_{\text{comb}} + D_M \Sigma D_M^\top. \end{aligned}$$

Since $\Sigma \succ 0$,

$$D_M \Sigma D_M^\top \succeq 0.$$

Thus,

$$V(M) \succeq V^\star_{\text{comb}}$$

for every admissible M, which establishes the optimality of $M^\star = \Sigma^{-1}$.

We next consider the two single-source systems. Under the conditions of the proposition, Q_{IV} and Q_{SV} are nonsingular. The IV-only projected moment system contains p moments for p parameters, so it is exactly identified. Its asymptotic variance, as established in Theorem 2, is

$$V_{\text{IV}} = Q_{\text{IV}}^{-1} \Sigma_{\text{IV}} Q_{\text{IV}}^{-1}.$$

Equivalently,

$$V_{\text{IV}} = \left(Q_{\text{IV}}^\top \Sigma_{\text{IV}}^{-1} Q_{\text{IV}} \right)^{-1}.$$

Indeed,

$$\begin{aligned} \left(Q_{\text{IV}}^\top \Sigma_{\text{IV}}^{-1} Q_{\text{IV}} \right)^{-1} &= Q_{\text{IV}}^{-1} \Sigma_{\text{IV}} (Q_{\text{IV}}^\top)^{-1} \\ &= Q_{\text{IV}}^{-1} \Sigma_{\text{IV}} Q_{\text{IV}}^{-1}, \end{aligned}$$

where the last equality uses the symmetry of Q_{IV}.

Similarly, the SV-only projected moment system is exactly identified, and Theorem 4 gives

$$V_{\text{SV}} = Q_{\text{SV}}^{-1} \Sigma_{\text{SV}} Q_{\text{SV}}^{-1} = \left(Q_{\text{SV}}^\top \Sigma_{\text{SV}}^{-1} Q_{\text{SV}} \right)^{-1}.$$

We now compare the optimally combined system with the IV-only system. Since $\Sigma \succ 0$, its principal blocks Σ_{IV} and Σ_{SV} are positive definite. Moreover, the Schur complement

$$\Sigma_{\text{SV}} - \Sigma_{\text{IV,SV}}^\top \Sigma_{\text{IV}}^{-1} \Sigma_{\text{IV,SV}}$$

is positive definite. For notational convenience within this proof, write

$$S_{\mathrm{SV}|\mathrm{IV}} := \Sigma_{\mathrm{SV}} - \Sigma_{\mathrm{IV},\mathrm{SV}}^\top \Sigma_{\mathrm{IV}}^{-1} \Sigma_{\mathrm{IV},\mathrm{SV}}.$$

Then $S_{\mathrm{SV}|\mathrm{IV}} \succ 0$.

The block-inverse formula gives

$$\Sigma^{-1} = \begin{pmatrix} \Sigma_{\mathrm{IV}}^{-1} + \Sigma_{\mathrm{IV}}^{-1}\Sigma_{\mathrm{IV},\mathrm{SV}} S_{\mathrm{SV}|\mathrm{IV}}^{-1} \Sigma_{\mathrm{IV},\mathrm{SV}}^\top \Sigma_{\mathrm{IV}}^{-1} & -\Sigma_{\mathrm{IV}}^{-1}\Sigma_{\mathrm{IV},\mathrm{SV}} S_{\mathrm{SV}|\mathrm{IV}}^{-1} \\ -S_{\mathrm{SV}|\mathrm{IV}}^{-1} \Sigma_{\mathrm{IV},\mathrm{SV}}^\top \Sigma_{\mathrm{IV}}^{-1} & S_{\mathrm{SV}|\mathrm{IV}}^{-1} \end{pmatrix}.$$

Therefore,

$$\begin{aligned} (V_{\mathrm{comb}}^\star)^{-1} &= J^\top \Sigma^{-1} J \\ &= Q_{\mathrm{IV}}^\top \left\{ \Sigma_{\mathrm{IV}}^{-1} + \Sigma_{\mathrm{IV}}^{-1}\Sigma_{\mathrm{IV},\mathrm{SV}} S_{\mathrm{SV}|\mathrm{IV}}^{-1} \Sigma_{\mathrm{IV},\mathrm{SV}}^\top \Sigma_{\mathrm{IV}}^{-1} \right\} Q_{\mathrm{IV}} \\ &\quad - Q_{\mathrm{IV}}^\top \Sigma_{\mathrm{IV}}^{-1}\Sigma_{\mathrm{IV},\mathrm{SV}} S_{\mathrm{SV}|\mathrm{IV}}^{-1} Q_{\mathrm{SV}} \\ &\quad - Q_{\mathrm{SV}}^\top S_{\mathrm{SV}|\mathrm{IV}}^{-1} \Sigma_{\mathrm{IV},\mathrm{SV}}^\top \Sigma_{\mathrm{IV}}^{-1} Q_{\mathrm{IV}} \\ &\quad + Q_{\mathrm{SV}}^\top S_{\mathrm{SV}|\mathrm{IV}}^{-1} Q_{\mathrm{SV}}. \end{aligned}$$

Collecting the last four terms into a quadratic form yields

$$\begin{aligned} (V_{\mathrm{comb}}^\star)^{-1} &= Q_{\mathrm{IV}}^\top \Sigma_{\mathrm{IV}}^{-1} Q_{\mathrm{IV}} \\ &\quad + \left(Q_{\mathrm{SV}} - \Sigma_{\mathrm{IV},\mathrm{SV}}^\top \Sigma_{\mathrm{IV}}^{-1} Q_{\mathrm{IV}}\right)^\top S_{\mathrm{SV}|\mathrm{IV}}^{-1} \left(Q_{\mathrm{SV}} - \Sigma_{\mathrm{IV},\mathrm{SV}}^\top \Sigma_{\mathrm{IV}}^{-1} Q_{\mathrm{IV}}\right). \end{aligned}$$

Since

$$Q_{\mathrm{IV}}^\top \Sigma_{\mathrm{IV}}^{-1} Q_{\mathrm{IV}} = V_{\mathrm{IV}}^{-1},$$

we obtain

$$\begin{aligned} (V_{\mathrm{comb}}^\star)^{-1} &= V_{\mathrm{IV}}^{-1} \\ &\quad + \left(Q_{\mathrm{SV}} - \Sigma_{\mathrm{IV},\mathrm{SV}}^\top \Sigma_{\mathrm{IV}}^{-1} Q_{\mathrm{IV}}\right)^\top S_{\mathrm{SV}|\mathrm{IV}}^{-1} \left(Q_{\mathrm{SV}} - \Sigma_{\mathrm{IV},\mathrm{SV}}^\top \Sigma_{\mathrm{IV}}^{-1} Q_{\mathrm{IV}}\right). \end{aligned}$$

Because $S_{\mathrm{SV}|\mathrm{IV}}^{-1} \succ 0$, the second term is positive semidefinite. Hence,

$$(V_{\mathrm{comb}}^\star)^{-1} \succeq V_{\mathrm{IV}}^{-1}.$$

Both matrices are positive definite. Since matrix inversion reverses the Loewner order,

$$V_{\mathrm{comb}}^\star \preceq V_{\mathrm{IV}}.$$

The comparison with the SV-only system follows symmetrically. Since $\Sigma \succ 0$, the other Schur complement

$$S_{\mathrm{IV}|\mathrm{SV}} := \Sigma_{\mathrm{IV}} - \Sigma_{\mathrm{IV},\mathrm{SV}} \Sigma_{\mathrm{SV}}^{-1} \Sigma_{\mathrm{IV},\mathrm{SV}}^\top$$

is also positive definite. Expanding Σ^{-1} around its $(2,2)$ block gives

$$\begin{aligned} (V_{\mathrm{comb}}^\star)^{-1} &= Q_{\mathrm{SV}}^\top \Sigma_{\mathrm{SV}}^{-1} Q_{\mathrm{SV}} \\ &\quad + \left(Q_{\mathrm{IV}} - \Sigma_{\mathrm{IV},\mathrm{SV}} \Sigma_{\mathrm{SV}}^{-1} Q_{\mathrm{SV}}\right)^\top S_{\mathrm{IV}|\mathrm{SV}}^{-1} \left(Q_{\mathrm{IV}} - \Sigma_{\mathrm{IV},\mathrm{SV}} \Sigma_{\mathrm{SV}}^{-1} Q_{\mathrm{SV}}\right). \end{aligned}$$

Since

$$Q_{\mathrm{SV}}^\top \Sigma_{\mathrm{SV}}^{-1} Q_{\mathrm{SV}} = V_{\mathrm{SV}}^{-1},$$

we obtain

$$\begin{aligned} (V_{\mathrm{comb}}^\star)^{-1} &= V_{\mathrm{SV}}^{-1} \\ &\quad + \left(Q_{\mathrm{IV}} - \Sigma_{\mathrm{IV},\mathrm{SV}} \Sigma_{\mathrm{SV}}^{-1} Q_{\mathrm{SV}}\right)^\top S_{\mathrm{IV}|\mathrm{SV}}^{-1} \left(Q_{\mathrm{IV}} - \Sigma_{\mathrm{IV},\mathrm{SV}} \Sigma_{\mathrm{SV}}^{-1} Q_{\mathrm{SV}}\right). \end{aligned}$$

The second term is positive semidefinite because $S_{\mathrm{IV}|\mathrm{SV}}^{-1} \succ 0$. Therefore,

$$(V_{\mathrm{comb}}^\star)^{-1} \succeq V_{\mathrm{SV}}^{-1},$$

and inversion gives

$$V^{\star}_{\mathrm{comb}} \preceq V_{\mathrm{SV}}.$$

Finally, consider the special case in which the two first-order score blocks are uncorrelated,

$$\mathrm{Cov}(\psi_{\mathrm{IV},i}, \psi_{\mathrm{SV},i}) = \Sigma_{\mathrm{IV,SV}} = 0_{p\times p}.$$

Then

$$\Sigma = \begin{pmatrix} \Sigma_{\mathrm{IV}} & 0 \\ 0 & \Sigma_{\mathrm{SV}} \end{pmatrix}, \qquad \Sigma^{-1} = \begin{pmatrix} \Sigma_{\mathrm{IV}}^{-1} & 0 \\ 0 & \Sigma_{\mathrm{SV}}^{-1} \end{pmatrix}.$$

Consequently,

$$\begin{aligned}
(V^{\star}_{\mathrm{comb}})^{-1} &= J^{\top}\Sigma^{-1}J \\
&= \begin{pmatrix} Q_{\mathrm{IV}} \\ Q_{\mathrm{SV}} \end{pmatrix}^{\top} \begin{pmatrix} \Sigma_{\mathrm{IV}}^{-1} & 0 \\ 0 & \Sigma_{\mathrm{SV}}^{-1} \end{pmatrix} \begin{pmatrix} Q_{\mathrm{IV}} \\ Q_{\mathrm{SV}} \end{pmatrix} \\
&= Q_{\mathrm{IV}}^{\top}\Sigma_{\mathrm{IV}}^{-1}Q_{\mathrm{IV}} + Q_{\mathrm{SV}}^{\top}\Sigma_{\mathrm{SV}}^{-1}Q_{\mathrm{SV}} \\
&= V_{\mathrm{IV}}^{-1} + V_{\mathrm{SV}}^{-1}.
\end{aligned}$$

Thus, when the two first-order score blocks are uncorrelated, their information contributions are additive. This completes the proof.

□

A.8 Proof of Lemma 8

Proof of Lemma 8. For any fixed $\tau \in \mathcal{C}$, Theorem 6 gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} Z_{\tau}, \qquad Z_{\tau} \sim \mathcal{N}\{K_M\tau, V_{\mathrm{miss}}(M)\}.$$

By the assumed uniform integrability of the squared root-n estimation errors, convergence in distribution implies convergence of the corresponding second moments. Therefore,

$$\begin{aligned}
\mathrm{AMSE}_{\tau}(M) &:= \lim_{n\to\infty} n\,\mathbb{E}_{\tau}\left[\|\widehat{\beta}(M) - \beta\|_2^2\right] \\
&= \lim_{n\to\infty} \mathbb{E}_{\tau}\left[\left\|\sqrt{n}\{\widehat{\beta}(M) - \beta\}\right\|_2^2\right] \\
&= \mathbb{E}\left(\|Z_{\tau}\|_2^2\right).
\end{aligned}$$

For any random vector Z with finite second moment,

$$\begin{aligned}
\mathbb{E}\|Z\|_2^2 &= \mathbb{E}\left[\|Z - \mathbb{E}(Z) + \mathbb{E}(Z)\|_2^2\right] \\
&= \mathbb{E}\|Z - \mathbb{E}(Z)\|_2^2 + \|\mathbb{E}(Z)\|_2^2,
\end{aligned}$$

because the cross term has expectation zero. Moreover,

$$\begin{aligned}
\mathbb{E}\|Z - \mathbb{E}(Z)\|_2^2 &= \mathbb{E}\left[\mathrm{tr}\left\{(Z - \mathbb{E}Z)(Z - \mathbb{E}Z)^{\top}\right\}\right] \\
&= \mathrm{tr}\{\mathrm{Var}(Z)\}.
\end{aligned}$$

Applying this identity to $Z_{\tau} \sim \mathcal{N}\{K_M\tau, V_{\mathrm{miss}}(M)\}$ gives

$$\begin{aligned}
\mathbb{E}\|Z_{\tau}\|_2^2 &= \|\mathbb{E}(Z_{\tau})\|_2^2 + \mathrm{tr}\{\mathrm{Var}(Z_{\tau})\} \\
&= \|K_M\tau\|_2^2 + \mathrm{tr}\{V_{\mathrm{miss}}(M)\}.
\end{aligned}$$

Hence,

$$\mathrm{AMSE}_{\tau}(M) = \|K_M\tau\|_2^2 + \mathrm{tr}\{V_{\mathrm{miss}}(M)\}.$$

Taking the supremum over $\tau \in \mathcal{C}$ gives

$$\begin{aligned}
\mathcal{R}_{\mathcal{C}}(M) &:= \sup_{\tau\in\mathcal{C}} \mathrm{AMSE}_{\tau}(M) \\
&= \sup_{\tau\in\mathcal{C}} \left[\|K_M\tau\|_2^2 + \mathrm{tr}\{V_{\mathrm{miss}}(M)\}\right].
\end{aligned}$$

Since $V_{\text{miss}}(M)$ does not depend on the particular $\tau \in \mathcal{C}$, its trace can be taken outside the supremum. Thus,

$$\mathcal{R}_{\mathcal{C}}(M) = \sup_{\tau \in \mathcal{C}} \|K_M \tau\|_2^2 + \operatorname{tr}\{V_{\text{miss}}(M)\}.$$

This proves the stated result.

□

A.9 Proof of Proposition 9

Proof of Proposition 9. Recall that the Jacobian, local misspecification vector, and first-order score covariance matrix of the combined IV–SV moment system can be partitioned as

$$J = \begin{pmatrix} Q_{\text{IV}} \\ Q_{\text{SV}} \end{pmatrix}, \qquad \tau = \begin{pmatrix} \tau_z \\ \tau_w \end{pmatrix}, \qquad \Sigma_{\text{miss}} = \begin{pmatrix} \Sigma_{\text{IV,miss}} & \Sigma_{\text{IV,SV,miss}} \\ \Sigma_{\text{SV,IV,miss}} & \Sigma_{\text{SV,miss}} \end{pmatrix},$$

where

$$\Sigma_{\text{IV,SV,miss}} := \operatorname{Cov}(\psi_{\text{IV,miss},i}, \psi_{\text{SV,miss},i}), \qquad \Sigma_{\text{SV,IV,miss}} := \Sigma_{\text{IV,SV,miss}}^{\top}.$$

We first establish dominance relative to the IV-only system. Let $M_z \succeq 0$ be any feasible IV-only weighting matrix, so that

$$Q_{\text{IV}}^{\top} M_z Q_{\text{IV}} \in \mathbb{S}_{++}^{p}.$$

Embed M_z into the combined system by defining

$$\widetilde{M}_z := \begin{pmatrix} M_z & 0 \\ 0 & 0 \end{pmatrix}.$$

To verify that this is a feasible combined weight, first note that, for any conformable vectors a_z and a_w,

$$\begin{pmatrix} a_z^{\top} & a_w^{\top} \end{pmatrix} \widetilde{M}_z \begin{pmatrix} a_z \\ a_w \end{pmatrix} = a_z^{\top} M_z a_z \geq 0.$$

Therefore, $\widetilde{M}_z \succeq 0$. Although $\widetilde{M}_z$ is generally singular because it assigns zero weight to the SV block, positive definiteness of the weighting matrix itself is not required. It is sufficient that the weighted Jacobian remain nonsingular. In the present case,

$$\begin{aligned} J^{\top} \widetilde{M}_z J &= \begin{pmatrix} Q_{\text{IV}}^{\top} & Q_{\text{SV}}^{\top} \end{pmatrix} \begin{pmatrix} M_z & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} Q_{\text{IV}} \\ Q_{\text{SV}} \end{pmatrix} \\ &= \begin{pmatrix} Q_{\text{IV}}^{\top} M_z & 0 \end{pmatrix} \begin{pmatrix} Q_{\text{IV}} \\ Q_{\text{SV}} \end{pmatrix} \\ &= Q_{\text{IV}}^{\top} M_z Q_{\text{IV}}. \end{aligned}$$

The right-hand side belongs to $\mathbb{S}_{++}^{p}$ by the feasibility of M_z. Hence, $J^{\top} \widetilde{M}_z J$ is positive definite, and $\widetilde{M}_z$ is a feasible weighting matrix for the combined system.

The corresponding combined-system sensitivity matrix is

$$\begin{aligned} K_{\widetilde{M}_z} &= (J^{\top} \widetilde{M}_z J)^{-1} J^{\top} \widetilde{M}_z \\ &= (Q_{\text{IV}}^{\top} M_z Q_{\text{IV}})^{-1} \begin{pmatrix} Q_{\text{IV}}^{\top} & Q_{\text{SV}}^{\top} \end{pmatrix} \begin{pmatrix} M_z & 0 \\ 0 & 0 \end{pmatrix} \\ &= (Q_{\text{IV}}^{\top} M_z Q_{\text{IV}})^{-1} \begin{pmatrix} Q_{\text{IV}}^{\top} M_z & 0 \end{pmatrix} \\ &= \begin{pmatrix} K_{M_z} & 0 \end{pmatrix}, \end{aligned}$$

where

$$K_{M_z} := (Q_{\text{IV}}^{\top} M_z Q_{\text{IV}})^{-1} Q_{\text{IV}}^{\top} M_z.$$

The zero block in $K_{\widetilde{M}_z}$ shows that the embedded estimator is insensitive to the SV moment block. In particular, for any

$$\tau = \begin{pmatrix} \tau_z \\ \tau_w \end{pmatrix} \in \mathcal{C},$$

its local asymptotic bias is

$$\begin{aligned} K_{\widetilde{M}_z}\tau &= \begin{pmatrix} K_{M_z} & 0 \end{pmatrix} \begin{pmatrix} \tau_z \\ \tau_w \end{pmatrix} \\ &= K_{M_z}\tau_z. \end{aligned}$$

Thus, the misspecification component τ_w does not affect the embedded estimator. Because $\mathcal{C} = \mathcal{C}_z \times \mathcal{C}_w$, it follows that

$$\begin{aligned} \sup_{\tau\in\mathcal{C}} \|K_{\widetilde{M}_z}\tau\|_2^2 &= \sup_{\substack{\tau_z\in\mathcal{C}_z \\ \tau_w\in\mathcal{C}_w}} \|K_{M_z}\tau_z\|_2^2 \\ &= \sup_{\tau_z\in\mathcal{C}_z} \|K_{M_z}\tau_z\|_2^2. \end{aligned}$$

Hence, the worst-case squared local bias of the embedded combined estimator is exactly the worst-case squared local bias of the IV-only estimator evaluated at M_z.

Next, using the block decomposition of Σ_{miss}, the asymptotic covariance matrix of the embedded estimator is

$$\begin{aligned} K_{\widetilde{M}_z}\Sigma_{\text{miss}}K_{\widetilde{M}_z}^\top &= \begin{pmatrix} K_{M_z} & 0 \end{pmatrix} \begin{pmatrix} \Sigma_{\text{IV,miss}} & \Sigma_{\text{IV,SV,miss}} \\ \Sigma_{\text{SV,IV,miss}} & \Sigma_{\text{SV,miss}} \end{pmatrix} \begin{pmatrix} K_{M_z}^\top \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} K_{M_z}\Sigma_{\text{IV,miss}} & K_{M_z}\Sigma_{\text{IV,SV,miss}} \end{pmatrix} \begin{pmatrix} K_{M_z}^\top \\ 0 \end{pmatrix} \\ &= K_{M_z}\Sigma_{\text{IV,miss}}K_{M_z}^\top. \end{aligned}$$

In particular, the cross-covariance blocks $\Sigma_{\text{IV,SV,miss}}$ and $\Sigma_{\text{SV,IV,miss}}$ disappear because the estimator assigns zero sensitivity to the SV moment block. Consequently,

$$\begin{aligned} \mathcal{R}_{\mathcal{C}}(\widetilde{M}_z) = \sup_{\tau_z\in\mathcal{C}_z} &\|K_{M_z}\tau_z\|_2^2 \\ &+ \operatorname{tr}\left\{K_{M_z}\Sigma_{\text{IV,miss}}K_{M_z}^\top\right\}. \end{aligned}$$

This is precisely the IV-only local risk evaluated at M_z. Therefore, the embedding preserves both the worst-case local bias and the asymptotic variance.

Since $\widetilde{M}_z$ belongs to the feasible class of combined weighting matrices, the definition of $\mathcal{R}^\star_{\text{comb}}$ implies that, for every feasible M_z,

$$\begin{aligned} \mathcal{R}^\star_{\text{comb}} &= \inf_{M\succeq 0:\ J^\top MJ\in\mathbb{S}^p_{++}} \mathcal{R}_{\mathcal{C}}(M) \\ &\le \mathcal{R}_{\mathcal{C}}(\widetilde{M}_z) \\ &= \sup_{\tau_z\in\mathcal{C}_z} \|K_{M_z}\tau_z\|_2^2 + \operatorname{tr}\left\{K_{M_z}\Sigma_{\text{IV,miss}}K_{M_z}^\top\right\}. \end{aligned}$$

Because this inequality holds for every feasible IV-only weighting matrix, taking the infimum over all such M_z gives

$$\begin{aligned} \mathcal{R}^\star_{\text{comb}} &\le \inf_{M_z\succeq 0:\ Q_{\text{IV}}^\top M_z Q_{\text{IV}}\in\mathbb{S}^p_{++}} \left[\sup_{\tau_z\in\mathcal{C}_z} \|K_{M_z}\tau_z\|_2^2 + \operatorname{tr}\left\{K_{M_z}\Sigma_{\text{IV,miss}}K_{M_z}^\top\right\}\right] \\ &= \mathcal{R}^\star_{\text{IV}}. \end{aligned}$$

Notice that this argument uses an infimum and therefore does not require the IV-only optimal weight to be attained. Equivalently, one may apply the argument to an arbitrarily close-to-optimal IV-only weight and then let the approximation error converge to zero.

We next establish the corresponding result for the SV-only system. Let $M_w \succeq 0$ be any feasible SV-only weighting matrix, so that

$$Q_{\text{SV}}^\top M_w Q_{\text{SV}} \in \mathbb{S}^p_{++}.$$

Embed M_w into the combined system by defining

$$\widetilde{M}_w := \begin{pmatrix} 0 & 0 \\ 0 & M_w \end{pmatrix}.$$

For any conformable vectors a_z and a_w,

$$\begin{pmatrix} a_z^\top & a_w^\top \end{pmatrix} \widetilde{M}_w \begin{pmatrix} a_z \\ a_w \end{pmatrix} = a_w^\top M_w a_w \ge 0,$$

and hence $\widetilde{M}_w \succeq 0$. Moreover,

$$\begin{aligned} J^\top \widetilde{M}_w J &= \begin{pmatrix} Q_{\mathrm{IV}}^\top & Q_{\mathrm{SV}}^\top \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & M_w \end{pmatrix} \begin{pmatrix} Q_{\mathrm{IV}} \\ Q_{\mathrm{SV}} \end{pmatrix} \\ &= \begin{pmatrix} 0 & Q_{\mathrm{SV}}^\top M_w \end{pmatrix} \begin{pmatrix} Q_{\mathrm{IV}} \\ Q_{\mathrm{SV}} \end{pmatrix} \\ &= Q_{\mathrm{SV}}^\top M_w Q_{\mathrm{SV}} \in \mathbb{S}_{++}^p. \end{aligned}$$

Thus, $\widetilde{M}_w$ is feasible for the combined system.

Its sensitivity matrix satisfies

$$\begin{aligned} K_{\widetilde{M}_w} &= (J^\top \widetilde{M}_w J)^{-1} J^\top \widetilde{M}_w \\ &= (Q_{\mathrm{SV}}^\top M_w Q_{\mathrm{SV}})^{-1} \begin{pmatrix} Q_{\mathrm{IV}}^\top & Q_{\mathrm{SV}}^\top \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & M_w \end{pmatrix} \\ &= (Q_{\mathrm{SV}}^\top M_w Q_{\mathrm{SV}})^{-1} \begin{pmatrix} 0 & Q_{\mathrm{SV}}^\top M_w \end{pmatrix} \\ &= \begin{pmatrix} 0 & K_{M_w} \end{pmatrix}, \end{aligned}$$

where

$$K_{M_w} := (Q_{\mathrm{SV}}^\top M_w Q_{\mathrm{SV}})^{-1} Q_{\mathrm{SV}}^\top M_w.$$

Thus, the embedded estimator is insensitive to the IV moment block. In particular,

$$\begin{aligned} K_{\widetilde{M}_w} \tau &= \begin{pmatrix} 0 & K_{M_w} \end{pmatrix} \begin{pmatrix} \tau_z \\ \tau_w \end{pmatrix} \\ &= K_{M_w} \tau_w. \end{aligned}$$

Using $\mathcal{C} = \mathcal{C}_z \times \mathcal{C}_w$, we therefore obtain

$$\begin{aligned} \sup_{\tau \in \mathcal{C}} \|K_{\widetilde{M}_w} \tau\|_2^2 &= \sup_{\substack{\tau_z \in \mathcal{C}_z \\ \tau_w \in \mathcal{C}_w}} \|K_{M_w} \tau_w\|_2^2 \\ &= \sup_{\tau_w \in \mathcal{C}_w} \|K_{M_w} \tau_w\|_2^2. \end{aligned}$$

Hence, the worst-case squared local bias coincides with that of the SV-only estimator evaluated at M_w.

Similarly,

$$\begin{aligned} K_{\widetilde{M}_w} \Sigma_{\mathrm{miss}} K_{\widetilde{M}_w}^\top &= \begin{pmatrix} 0 & K_{M_w} \end{pmatrix} \begin{pmatrix} \Sigma_{\mathrm{IV,miss}} & \Sigma_{\mathrm{IV,SV,miss}} \\ \Sigma_{\mathrm{SV,IV,miss}} & \Sigma_{\mathrm{SV,miss}} \end{pmatrix} \begin{pmatrix} 0 \\ K_{M_w}^\top \end{pmatrix} \\ &= \begin{pmatrix} K_{M_w} \Sigma_{\mathrm{SV,IV,miss}} & K_{M_w} \Sigma_{\mathrm{SV,miss}} \end{pmatrix} \begin{pmatrix} 0 \\ K_{M_w}^\top \end{pmatrix} \\ &= K_{M_w} \Sigma_{\mathrm{SV,miss}} K_{M_w}^\top. \end{aligned}$$

Again, the cross-covariance blocks disappear because the IV block receives zero sensitivity. It follows that

$$\begin{aligned} \mathcal{R}_{\mathcal{C}}(\widetilde{M}_w) = &\sup_{\tau_w \in \mathcal{C}_w} \|K_{M_w} \tau_w\|_2^2 \\ &+ \operatorname{tr}\left\{ K_{M_w} \Sigma_{\mathrm{SV,miss}} K_{M_w}^\top \right\}, \end{aligned}$$

which is exactly the SV-only local risk evaluated at M_w. Since $\widetilde{M}_w$ is feasible for the combined problem, for every feasible M_w,

$$\mathcal{R}_{\mathrm{comb}}^\star \le \mathcal{R}_{\mathcal{C}}(\widetilde{M}_w).$$

Taking the infimum over all feasible SV-only weighting matrices gives

$$\begin{aligned} \mathcal{R}_{\mathrm{comb}}^\star &\le \inf_{M_w \succeq 0:\, Q_{\mathrm{SV}}^\top M_w Q_{\mathrm{SV}} \in \mathbb{S}_{++}^p} \left[\sup_{\tau_w \in \mathcal{C}_w} \|K_{M_w} \tau_w\|_2^2 + \operatorname{tr}\left\{ K_{M_w} \Sigma_{\mathrm{SV,miss}} K_{M_w}^\top \right\} \right] \\ &= \mathcal{R}_{\mathrm{SV}}^\star. \end{aligned}$$

The two block-diagonal embeddings show that the feasible combined weighting class contains exact copies of both the IV-only and SV-only weighting classes, with their corresponding local risks preserved. Therefore,

$$\mathcal{R}_{\mathrm{comb}}^\star \le \mathcal{R}_{\mathrm{IV}}^\star, \qquad \mathcal{R}_{\mathrm{comb}}^\star \le \mathcal{R}_{\mathrm{SV}}^\star.$$

The result concerns the optimally weighted combined system; it does not imply that every fixed combined weighting matrix has lower local risk than the two subsystem estimators. □

A.10 Proof of Proposition 10

Proof of Proposition 10. Let $r(\beta) := Y - X\beta$. Recall that the stacked sample moment is

$$\widehat{g}(\beta) = \frac{1}{n}\begin{pmatrix} X^\top P_Z r(\beta) \\ X^\top P_W^\perp r(\beta) \end{pmatrix}.$$

Under the block-diagonal weighting matrix M_λ, the restricted GMM criterion can be written as

$$\widehat{g}(\beta)^\top M_\lambda \widehat{g}(\beta) = \mathcal{J}_{Z,\lambda}(\beta) + \mathcal{J}_{W,\lambda}(\beta),$$

where

$$\mathcal{J}_{Z,\lambda}(\beta) := \frac{1-\lambda}{n} r(\beta)^\top P_Z X (X^\top P_Z X)^{-1} X^\top P_Z r(\beta),$$
$$\mathcal{J}_{W,\lambda}(\beta) := \frac{\lambda}{n} r(\beta)^\top P_W^\perp X (X^\top P_W^\perp X)^{-1} X^\top P_W^\perp r(\beta).$$

Indeed, for the projected IV block,

$$\left\{\frac{1}{n}X^\top P_Z r(\beta)\right\}^\top (1-\lambda)\left(\frac{1}{n}X^\top P_Z X\right)^{-1}\left\{\frac{1}{n}X^\top P_Z r(\beta)\right\}$$
$$= \frac{1-\lambda}{n} r(\beta)^\top P_Z X (X^\top P_Z X)^{-1} X^\top P_Z r(\beta),$$

and the same calculation gives the SV component.

Since $\partial r(\beta)/\partial \beta^\top = -X$, and $P_Z X (X^\top P_Z X)^{-1} X^\top P_Z$ is symmetric, differentiation of the IV component gives

$$\nabla_\beta \mathcal{J}_{Z,\lambda}(\beta) = -\frac{2(1-\lambda)}{n} X^\top P_Z X (X^\top P_Z X)^{-1} X^\top P_Z r(\beta)$$
$$= -\frac{2(1-\lambda)}{n} X^\top P_Z r(\beta).$$

Similarly,

$$\nabla_\beta \mathcal{J}_{W,\lambda}(\beta) = -\frac{2\lambda}{n} X^\top P_W^\perp X (X^\top P_W^\perp X)^{-1} X^\top P_W^\perp r(\beta)$$
$$= -\frac{2\lambda}{n} X^\top P_W^\perp r(\beta).$$

Therefore, the first-order condition for minimizing $\widehat{g}(\beta)^\top M_\lambda \widehat{g}(\beta)$ is

$$(1-\lambda) X^\top P_Z r(\beta) + \lambda X^\top P_W^\perp r(\beta) = 0.$$

Substituting $r(\beta) = Y - X\beta$ gives

$$(1-\lambda) X^\top P_Z Y + \lambda X^\top P_W^\perp Y$$
$$- \left\{(1-\lambda) X^\top P_Z X + \lambda X^\top P_W^\perp X\right\}\beta = 0.$$

Hence, whenever the weighted Gram matrix is nonsingular, the minimizer is

$$\widehat{\beta}_\lambda = \left\{(1-\lambda) X^\top P_Z X + \lambda X^\top P_W^\perp X\right\}^{-1}\left\{(1-\lambda) X^\top P_Z Y + \lambda X^\top P_W^\perp Y\right\},$$

which is precisely the weighted least-squares estimator defined in Section 4.3.

For $\lambda \in (0,1)$, both diagonal blocks of M_λ are positive definite, so $M_\lambda \in \mathbb{S}^{2p}_{++}$ and the preceding calculation gives the stated restricted-GMM representation within the weighting class of Section 3.1. At $\lambda = 0$ and $\lambda = 1$, M_λ is positive semidefinite rather than positive definite, but the same quadratic criterion remains well defined and the preceding first-order calculation yields the IV-only and SV-only endpoint estimators, respectively. □

A.11 Proof of Proposition 11

Proof of Proposition 11. Let

$$\mathcal{D} := \sigma(X, Z, W)$$

denote the sigma-field generated by the realized design matrices. Throughout the proof, all conditional expectations, variances, and covariances are taken with respect to $\mathcal{D}$.

For notational convenience, define the two score blocks

$$U_Z := X^\top P_Z \varsigma, \qquad U_W := X^\top P_W^\perp \varsigma,$$

and their weighted combination

$$U_\lambda := (1-\lambda)U_Z + \lambda U_W.$$

Using $Y = X\beta + \varsigma$ in (4.5), we obtain

$$\begin{aligned}
\widehat{\beta}_\lambda &= S_\lambda^{-1}\left\{(1-\lambda)X^\top P_Z Y + \lambda X^\top P_W^\perp Y\right\} \\
&= S_\lambda^{-1}\left[(1-\lambda)X^\top P_Z(X\beta+\varsigma) + \lambda X^\top P_W^\perp(X\beta+\varsigma)\right] \\
&= S_\lambda^{-1}\left[\left\{(1-\lambda)X^\top P_Z X + \lambda X^\top P_W^\perp X\right\}\beta + U_\lambda\right].
\end{aligned}$$

By the definition

$$S_\lambda = (1-\lambda)S_Z + \lambda S_W = (1-\lambda)X^\top P_Z X + \lambda X^\top P_W^\perp X,$$

the preceding display reduces to

$$\widehat{\beta}_\lambda - \beta = S_\lambda^{-1}U_\lambda = S_\lambda^{-1}\left\{(1-\lambda)X^\top P_Z\varsigma + \lambda X^\top P_W^\perp \varsigma\right\}.$$

Define

$$\mu_Z := \mathbb{E}(U_Z \mid \mathcal{D}), \qquad \mu_W := \mathbb{E}(U_W \mid \mathcal{D}),$$

and hence

$$\mu_\lambda := \mathbb{E}(U_\lambda \mid \mathcal{D}) = (1-\lambda)\mu_Z + \lambda\mu_W.$$

Because S_λ is measurable with respect to $\mathcal{D}$,

$$\mathbb{E}(\widehat{\beta}_\lambda - \beta \mid \mathcal{D}) = S_\lambda^{-1}\mu_\lambda.$$

Thus, the estimator need not be conditionally centered even when the population moment restrictions are valid.

Because S_λ is symmetric positive definite, S_λ^{-1} is also symmetric. Therefore,

$$\begin{aligned}
\mathcal{R}_{\mathrm{tr}}(\lambda) &= \mathbb{E}\Big(\|\widehat{\beta}_\lambda - \beta\|_2^2 \mid \mathcal{D}\Big) \\
&= \mathrm{tr}\left[S_\lambda^{-1}\mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D})S_\lambda^{-1}\right].
\end{aligned}$$

Using

$$\mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D}) = \mathrm{Var}(U_\lambda \mid \mathcal{D}) + \mu_\lambda\mu_\lambda^\top,$$

we equivalently obtain

$$\mathcal{R}_{\mathrm{tr}}(\lambda) = \mathrm{tr}\left[S_\lambda^{-1}\,\mathrm{Var}(U_\lambda \mid \mathcal{D})S_\lambda^{-1}\right] + \|S_\lambda^{-1}\mu_\lambda\|_2^2.$$

This proves the conditional bias–variance decomposition in (4.6).

Define

$$\Omega_\lambda := \mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D}).$$

Then $\Omega_\lambda \succeq 0$. By linearity,

$$\begin{aligned}
\Omega_\lambda = {} & (1-\lambda)^2\mathbb{E}(U_Z U_Z^\top \mid \mathcal{D}) \\
& + \lambda(1-\lambda)\mathbb{E}(U_Z U_W^\top \mid \mathcal{D}) \\
& + \lambda(1-\lambda)\mathbb{E}(U_W U_Z^\top \mid \mathcal{D}) \\
& + \lambda^2\mathbb{E}(U_W U_W^\top \mid \mathcal{D}).
\end{aligned}$$

Taking traces and using

$$\mathrm{tr}(U_Z U_W^\top) = U_Z^\top U_W$$

gives

$$\operatorname{tr}(\Omega_\lambda) = (1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W.$$

For later use, write

$$N(\lambda) := (1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W.$$

The cyclic property of the trace gives

$$\mathcal{R}_{\mathrm{tr}}(\lambda) = \operatorname{tr}\left(\Omega_\lambda S_\lambda^{-2}\right).$$

We use the following elementary trace inequality. For any two positive semidefinite matrices A and B of the same dimension,

$$\lambda_{\min}(B)\operatorname{tr}(A) \le \operatorname{tr}(AB) \le \lambda_{\max}(B)\operatorname{tr}(A).$$

Indeed,

$$\lambda_{\min}(B) I \preceq B \preceq \lambda_{\max}(B) I.$$

Premultiplying and postmultiplying by $A^{1/2}$ gives

$$\lambda_{\min}(B) A \preceq A^{1/2} B A^{1/2} \preceq \lambda_{\max}(B) A.$$

Taking traces and using $\operatorname{tr}(A^{1/2}BA^{1/2}) = \operatorname{tr}(AB)$ establishes the claimed inequality.

Applying this inequality with

$$A = \Omega_\lambda, \qquad B = S_\lambda^{-2},$$

yields

$$\lambda_{\min}(S_\lambda^{-2})\operatorname{tr}(\Omega_\lambda) \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \lambda_{\max}(S_\lambda^{-2})\operatorname{tr}(\Omega_\lambda).$$

If $s_1, \dots, s_p$ are the eigenvalues of S_λ, then the eigenvalues of S_λ^{-2} are $s_1^{-2}, \dots, s_p^{-2}$. Therefore,

$$\lambda_{\min}(S_\lambda^{-2}) = \lambda_{\max}(S_\lambda)^{-2}, \qquad \lambda_{\max}(S_\lambda^{-2}) = \lambda_{\min}(S_\lambda)^{-2}.$$

Since $\operatorname{tr}(\Omega_\lambda) = N(\lambda)$, we obtain

$$\frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\lambda_{\max}(S_\lambda)^2} \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\lambda_{\min}(S_\lambda)^2}.$$

We now establish the uniform comparison with the surrogate risk. Define

$$a_Z := \lambda_{\min}(S_Z), \qquad a_W := \lambda_{\min}(S_W),$$

and

$$d_\lambda := (1-\lambda)a_Z + \lambda a_W.$$

Because $S_\lambda \succ 0$ for every $\lambda \in [0,1]$, in particular $S_Z = S_0 \succ 0$ and $S_W = S_1 \succ 0$. Hence

$$a_Z > 0, \qquad a_W > 0, \qquad d_\lambda > 0.$$

For every unit vector $v \in \mathbb{R}^p$,

$$\begin{aligned} v^\top S_\lambda v &= (1-\lambda)v^\top S_Z v + \lambda v^\top S_W v \\ &\ge (1-\lambda)\lambda_{\min}(S_Z) + \lambda\lambda_{\min}(S_W) \\ &= d_\lambda. \end{aligned}$$

Taking the minimum over all unit vectors gives

$$\lambda_{\min}(S_\lambda) \ge d_\lambda.$$

Similarly,

$$v^\top S_\lambda v \le (1-\lambda)\lambda_{\max}(S_Z) + \lambda\lambda_{\max}(S_W),$$

and hence

$$\lambda_{\max}(S_\lambda) \le (1-\lambda)\lambda_{\max}(S_Z) + \lambda\lambda_{\max}(S_W).$$

Under

$$\lambda_{\max}(S_Z) \lesssim \lambda_{\min}(S_Z), \qquad \lambda_{\max}(S_W) \lesssim \lambda_{\min}(S_W),$$

there exists a constant $C \geq 1$, independent of $\lambda \in [0,1]$, such that

$$\lambda_{\max}(S_Z) \leq C a_Z, \qquad \lambda_{\max}(S_W) \leq C a_W.$$

It follows that

$$\lambda_{\max}(S_\lambda) \leq C\{(1-\lambda)a_Z + \lambda a_W\} = C d_\lambda.$$

Combining the preceding bounds gives

$$d_\lambda \leq \lambda_{\min}(S_\lambda) \leq \lambda_{\max}(S_\lambda) \leq C d_\lambda$$

uniformly over $\lambda \in [0,1]$.

Substituting these inequalities into the trace-risk bounds yields

$$\frac{N(\lambda)}{C^2 d_\lambda^2} \leq \mathcal{R}_{\mathrm{tr}}(\lambda) \leq \frac{N(\lambda)}{d_\lambda^2}.$$

By definition,

$$\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) = \frac{N(\lambda)}{d_\lambda^2} = \frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\{(1-\lambda)a_Z + \lambda a_W\}^2}.$$

Therefore,

$$C^{-2}\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \leq \mathcal{R}_{\mathrm{tr}}(\lambda) \leq \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$$

uniformly over $\lambda \in [0,1]$, which proves

$$\mathcal{R}_{\mathrm{tr}}(\lambda) \asymp \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \qquad \text{uniformly over } \lambda \in [0,1].$$

It remains to characterize the stationary points of the surrogate risk. Retain the definition of $N(\lambda)$ above and let

$$D(\lambda) := (1-\lambda)a_Z + \lambda a_W.$$

Then

$$\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) = \frac{N(\lambda)}{D(\lambda)^2}.$$

Expanding $N(\lambda)$ gives

$$N(\lambda) = c_Z + 2(c_{ZW} - c_Z)\lambda + (c_Z - 2c_{ZW} + c_W)\lambda^2,$$

so

$$N'(\lambda) = 2(c_{ZW} - c_Z) + 2(c_Z - 2c_{ZW} + c_W)\lambda.$$

Moreover,

$$D(\lambda) = a_Z + (a_W - a_Z)\lambda, \qquad D'(\lambda) = a_W - a_Z.$$

Since $D(\lambda) > 0$ on $[0,1]$, differentiation gives

$$\widetilde{\mathcal{R}}'_{\mathrm{tr}}(\lambda) = \frac{N'(\lambda)D(\lambda) - 2N(\lambda)D'(\lambda)}{D(\lambda)^3}.$$

Thus, every interior stationary point must satisfy

$$N'(\lambda)D(\lambda) - 2N(\lambda)D'(\lambda) = 0.$$

A direct expansion shows that the quadratic terms cancel and that

$$\frac{1}{2}\left\{N'(\lambda)D(\lambda) - 2N(\lambda)D'(\lambda)\right\} = \Big\{(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W\Big\}\lambda - \left(c_Z a_W - c_{ZW} a_Z\right).$$

Therefore, provided

$$(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W \neq 0,$$

the stationary-point equation is linear and has the unique solution

$$\lambda^{\star}_{\mathrm{sur}} = \frac{c_Z a_W - c_{ZW} a_Z}{(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W}.$$

Consequently, there is at most one interior stationary point. Substituting back

$$a_Z = \lambda_{\min}(S_Z), \qquad a_W = \lambda_{\min}(S_W)$$

gives the expression stated in the proposition.

Finally, $\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$ is continuous on the compact interval $[0,1]$, because $D(\lambda) > 0$ throughout this interval. Hence it attains a global minimum. Any minimizer must occur either at a boundary point or at an interior stationary point. It is therefore sufficient to compare

$$\widetilde{\mathcal{R}}_{\mathrm{tr}}(0), \qquad \widetilde{\mathcal{R}}_{\mathrm{tr}}(1),$$

and, whenever $\lambda^{\star}_{\mathrm{sur}} \in (0,1)$, the additional candidate

$$\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda^{\star}_{\mathrm{sur}}).$$

This completes the proof. □

B Projection geometry under low-rank factor structure

Let $U \in \mathbb{R}^{n\times r}$ and $\Gamma \in \mathbb{R}^{q\times r}$, and write $B := \Gamma^\top$ so that the (rank-r) signal is

$$S := U\Gamma^\top = UB \in \mathbb{R}^{n\times q}.$$

For any full-column-rank matrix A, define the orthogonal projector onto its column space by

$$P_A := A(A^\top A)^{-1}A^\top \in \mathbb{R}^{n\times n}.$$

Noiseless case: $W = U\Gamma^\top$. Since each column of $W = UB$ is a linear combination of columns of U, we have $\mathcal{C}(W) \subseteq \mathcal{C}(U)$, and therefore the corresponding orthogonal projections satisfy

$$P_U P_W = P_W, \qquad P_W P_U = P_W, \qquad \text{and} \qquad P_U - P_W \succeq 0.$$

Moreover, $P_W = P_U$ holds if and only if $\mathcal{C}(W) = \mathcal{C}(U)$, equivalently $\operatorname{rank}(UB) = \operatorname{rank}(U) = r$; in particular, if U and Γ are both full column rank r, then $\mathcal{C}(W) = \mathcal{C}(U)$ and $P_W = P_U$.

Noisy case: $W = U\Gamma^\top + E$. Exact nesting is retained only when the noise does not inject components outside $\mathcal{C}(U)$:

$$\mathcal{C}(W) \subseteq \mathcal{C}(U) \quad \Longleftrightarrow \quad (I - P_U)W = 0 \quad \Longleftrightarrow \quad (I - P_U)E = 0.$$

Hence, even when U and Γ are full column rank, generic noise E destroys the identities $P_U P_W = P_W$ and $P_W = P_U$.

A stable analogue is obtained by projecting onto the leading rank-r left singular subspace of W. Let $\widehat{U}_r \in \mathbb{R}^{n\times r}$ collect the top r left singular vectors of W, and define $\widehat{P}_r := \widehat{U}_r\widehat{U}_r^\top$. If $\|E\|_2 < \sigma_r(S)$ (where $\sigma_r(S)$ is the smallest nonzero singular value of S), then standard subspace perturbation bounds yield

$$\|\widehat{P}_r - P_U\|_2 = \|\widehat{P}_r - P_S\|_2 \;\le\; \frac{2\|E\|_2}{\sigma_r(S) - \|E\|_2} \;\lesssim\; \frac{\|E\|_2}{\sigma_r(S)}.$$

In addition, $\sigma_r(S) \ge \sigma_r(U)\sigma_r(\Gamma)$ provides a convenient lower bound, so subspace (and hence projector) consistency holds when $\|E\|_2 \ll \sigma_r(U)\sigma_r(\Gamma)$.

C Raw versus projected IV information

The projected representation also places the IV and SV restrictions on the same p-dimensional scale. The next result shows that this compression does not sacrifice any IV identifying direction.

Proposition 12 (Projection preserves identifying directions)**.** Under (2.1), suppose the following conditions hold.

(i) The instrument covariance matrix satisfies $\Sigma_z \in \mathbb{S}^q_{++}$.

(ii) The projected IV and residualized SV moments are well defined, with $Q_{\text{IV}} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$ and $Q_{\text{SV}} = \mathbb{E}\{(\Pi_w^\perp x_i)x_i^\top\}$.

Then $\ker(Q_{\text{IV}}) = \ker(\Sigma_{zx})$. Consequently, for $J = \{Q_{\text{IV}}^\top, Q_{\text{SV}}^\top\}^\top$ and $J_{\text{raw}} := \{\Sigma_{zx}^\top, Q_{\text{SV}}^\top\}^\top$, we have $\ker(J) = \ker(J_{\text{raw}})$ and $\operatorname{rank}(J) = \operatorname{rank}(J_{\text{raw}})$. Hence, the projected-IV-plus-residualized-SV system and the corresponding raw-IV-plus-residualized-SV system preserve exactly the same identifying directions for β, and $\operatorname{rank}(J) = p$ if and only if $\operatorname{rank}(J_{\text{raw}}) = p$.

Moreover, if $\operatorname{rank}(\Sigma_{zx}) = p$, any linear compression of the q-dimensional IV moment that preserves all IV identifying directions must have dimension at least p. The projected instrument $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1}z_i$ has dimension p and therefore attains this minimum.

The main analysis adopts the projected IV moment because it places the IV and SV channels on a common p-dimensional scale while preserving all IV identifying directions. Projection, however, need not preserve all efficiency information contained in the original q-dimensional IV moment. This section isolates that distinction by comparing the raw and projected IV systems directly, without involving the SV block.

Recall that the projected IV restriction is $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$, whereas the classical raw-IV restriction is $\mathbb{E}(z_i\varsigma_i) = 0_q$. Since $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)$, classical IV validity implies projected IV validity. The

converse need not hold when $q > p$, because components of $\mathbb{E}(z_i \varsigma_i)$ lying in $\ker(\Sigma_{xz}\Sigma_z^{-1})$ are removed by projection and therefore do not affect the projected restriction. Thus, the comparison between the two representations involves a validity–efficiency trade-off. The raw IV system retains all q moment restrictions and may exploit additional overidentifying information, but requires the stronger condition that all of these directions are valid. The projected system requires validity only after compression to the component relevant for predicting x_i, at the possible cost of discarding useful overidentifying information when the stronger classical restriction is in fact correct.

For the remainder of this section, suppose that $\mathbb{E}(z_i \varsigma_i) = 0_q$. Define the raw IV score by $\psi^{\mathrm{IV}}_{\mathrm{raw},i} := z_i \varsigma_i \in \mathbb{R}^q$, with $\Sigma^{\mathrm{IV}}_{\mathrm{raw}} := \mathrm{Var}(\psi^{\mathrm{IV}}_{\mathrm{raw},i})$, and define $T_{\mathrm{IV}} := \Sigma_{xz}\Sigma_z^{-1} \in \mathbb{R}^{p\times q}$. Under classical IV validity, the projected IV score satisfies $\psi_{\mathrm{IV},i} = T_{\mathrm{IV}}\psi^{\mathrm{IV}}_{\mathrm{raw},i}$, so that $\Sigma_{\mathrm{IV}} = T_{\mathrm{IV}}\Sigma^{\mathrm{IV}}_{\mathrm{raw}}T_{\mathrm{IV}}^\top$. The corresponding Jacobians are Σ_{zx} for the raw IV system and $Q_{\mathrm{IV}} = T_{\mathrm{IV}}\Sigma_{zx}$ for the projected system.

Let $\mathcal{I}^{\mathrm{IV}}_{\mathrm{raw}} := \Sigma_{zx}^\top(\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{-1}\Sigma_{zx}$ and $\mathcal{I}^{\mathrm{IV}}_{\mathrm{proj}} := Q_{\mathrm{IV}}^\top\Sigma_{\mathrm{IV}}^{-1}Q_{\mathrm{IV}}$ denote the efficient information matrices in the raw and projected IV systems, respectively. For the equality characterization, define

$$P_{T_{\mathrm{IV}}} := (\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{1/2}T_{\mathrm{IV}}^\top(T_{\mathrm{IV}}\Sigma^{\mathrm{IV}}_{\mathrm{raw}}T_{\mathrm{IV}}^\top)^{-1}T_{\mathrm{IV}}(\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{1/2}$$

.

Proposition 13 (Raw IV information dominates projected IV information)**.** Suppose the following conditions hold.

(i) The classical IV restriction is correctly specified, $\mathbb{E}(z_i \varsigma_i) = 0_q$.

(ii) The projected IV block is individually identified, $Q_{\mathrm{IV}} \in \mathbb{S}^p_{++}$, and the projection is well defined with $\Sigma_z \in \mathbb{S}^q_{++}$.

(iii) The regularity conditions required for the first-order expansions of the raw and projected IV systems hold, and $\Sigma^{\mathrm{IV}}_{\mathrm{raw}} \in \mathbb{S}^q_{++}$.

Then

$$\mathcal{I}^{\mathrm{IV}}_{\mathrm{proj}} \preceq \mathcal{I}^{\mathrm{IV}}_{\mathrm{raw}}, \qquad V^\star_{\mathrm{raw,IV}} := (\mathcal{I}^{\mathrm{IV}}_{\mathrm{raw}})^{-1} \preceq (\mathcal{I}^{\mathrm{IV}}_{\mathrm{proj}})^{-1} =: V^\star_{\mathrm{proj,IV}}. \tag{C.1}$$

More precisely,

$$\mathcal{I}^{\mathrm{IV}}_{\mathrm{raw}} - \mathcal{I}^{\mathrm{IV}}_{\mathrm{proj}} = \Sigma_{zx}^\top(\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{-1/2}(I_q - P_{T_{\mathrm{IV}}})(\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{-1/2}\Sigma_{zx} \succeq 0. \tag{C.2}$$

Hence, projection is information-lossless if and only if $\mathcal{C}\{(\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{-1}\Sigma_{zx}\} \subseteq \mathcal{C}(T_{\mathrm{IV}}^\top)$.

Proposition 13 separates preservation of identification from preservation of efficiency. By Proposition 12, the projected IV moment preserves exactly the same identifying directions as the raw IV moment. Hence, the additional directions available when $q > p$ are not required for point identification of β. They may nevertheless contain useful overidentifying information. When the stronger classical restriction $\mathbb{E}(z_i \varsigma_i) = 0_q$ is valid, retaining these directions can reduce asymptotic variance, whereas projection may discard part of this information.

The equality condition has a direct geometric interpretation. The space $\mathcal{C}(T_{\mathrm{IV}}^\top)$ contains the raw-IV directions retained by the projection, whereas $\mathcal{C}\{(\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{-1}\Sigma_{zx}\}$ contains the efficient raw-IV directions relevant for estimating β. Projection is therefore information-lossless precisely when every efficient raw-IV direction is already contained in the subspace retained by T_{IV}. If some efficient direction lies outside that subspace, the projected system still preserves identification but loses information that could otherwise improve precision.

This result clarifies the trade-off underlying the projected formulation. The raw IV system can be more efficient when all q classical IV moment restrictions are valid, but this gain relies on the stronger validity requirement. The projected formulation instead requires only the weaker projected restriction in the main analysis and provides an identification-preserving p-dimensional representation that can be placed on the same scale as the surrogate-control moment. Its role is therefore not to dominate the raw IV system in efficiency, but to provide the common representation required for the unified IV–SV analysis.

When $q = p$ and $Q_{\mathrm{IV}} \in \mathbb{S}^p_{++}$, T_{IV} is nonsingular, so the raw and projected IV moments are related by an invertible transformation. Consequently, $\mathcal{I}^{\mathrm{IV}}_{\mathrm{proj}} = \mathcal{I}^{\mathrm{IV}}_{\mathrm{raw}}$, and no IV information is lost through projection. The validity–efficiency trade-off is therefore relevant primarily in the overidentified case $q > p$.

C.1 Proof of Proposition 12

Proof of Proposition 12. We first show that the projected IV Jacobian and the raw IV Jacobian have the same null space. Recall that $Q_{\text{IV}} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$. For any $a \in \mathbb{R}^p$,

$$a^\top Q_{\text{IV}} a = a^\top \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx} a = \left\| \Sigma_z^{-1/2}\Sigma_{zx} a \right\|_2^2 .$$

Since $\Sigma_z \in \mathbb{S}^q_{++}$, the right-hand side is zero if and only if $\Sigma_{zx}a = 0_q$. Moreover, $Q_{\text{IV}} \succeq 0$, so $Q_{\text{IV}}a = 0_p$ if and only if $a^\top Q_{\text{IV}} a = 0$. Therefore,

$$\ker(Q_{\text{IV}}) = \ker(\Sigma_{zx}). \tag{C.3}$$

We next compare the identifying directions of the projected and raw combined systems. By definition,

$$J = \begin{pmatrix} Q_{\text{IV}} \\ Q_{\text{SV}} \end{pmatrix}, \qquad J_{\text{raw}} = \begin{pmatrix} \Sigma_{zx} \\ Q_{\text{SV}} \end{pmatrix}.$$

For any $a \in \mathbb{R}^p$, $Ja = 0_{2p}$ if and only if $Q_{\text{IV}}a = 0_p$ and $Q_{\text{SV}}a = 0_p$. By (C.3), this is equivalent to $\Sigma_{zx}a = 0_q$ and $Q_{\text{SV}}a = 0_p$, which in turn is equivalent to $J_{\text{raw}}a = 0_{q+p}$. Hence,

$$\ker(J) = \ker(J_{\text{raw}}).$$

Because both matrices have p columns, the rank–nullity theorem then gives

$$\operatorname{rank}(J) = \operatorname{rank}(J_{\text{raw}}).$$

Thus, replacing the raw IV block by the projected IV block does not remove any parameter direction that can be identified by the combined IV–SV system. In particular, $\operatorname{rank}(J) = p$ if and only if $\operatorname{rank}(J_{\text{raw}}) = p$.

It remains to establish the minimal dimension of an identification-preserving linear IV compression. Suppose $\operatorname{rank}(\Sigma_{zx}) = p$, and consider any k-dimensional linear compression of the raw q-dimensional IV moment. The Jacobian of the resulting compressed moment has rank at most k. Therefore, preserving all p identifying directions of Σ_{zx} necessarily requires $k \geq p$.

Finally, $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i \in \mathbb{R}^p$, so the projected IV moment has exactly p components. By (C.3), its Jacobian Q_{IV} has the same null space, and hence the same identifying directions, as Σ_{zx}. Therefore, the projected IV block preserves all IV identifying directions using the smallest possible number p of linearly compressed moments.

This completes the proof.

□

C.2 Proof of Proposition 13

Proof of Proposition 13. Recall that $T_{\text{IV}} = \Sigma_{xz}\Sigma_z^{-1}$, $\Sigma_{\text{IV}} = T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top$, and $Q_{\text{IV}} = T_{\text{IV}}\Sigma_{zx}$. The corresponding efficient information matrices are $\mathcal{I}^{\text{IV}}_{\text{raw}} = \Sigma_{zx}^\top(\Sigma^{\text{IV}}_{\text{raw}})^{-1}\Sigma_{zx}$ and $\mathcal{I}^{\text{IV}}_{\text{proj}} = Q_{\text{IV}}^\top\Sigma_{\text{IV}}^{-1}Q_{\text{IV}}$.

We first verify that Σ_{IV} is nonsingular. Since $Q_{\text{IV}} = T_{\text{IV}}\Sigma_{zx} \in \mathbb{S}^p_{++}$, the matrix $T_{\text{IV}} \in \mathbb{R}^{p\times q}$ has full row rank p. Together with $\Sigma^{\text{IV}}_{\text{raw}} \in \mathbb{S}^q_{++}$, this implies $\Sigma_{\text{IV}} = T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top \in \mathbb{S}^p_{++}$. Hence, Σ_{IV}^{-1} and $P_{T_{\text{IV}}}$ are well defined.

Using $Q_{\text{IV}} = T_{\text{IV}}\Sigma_{zx}$, we can write

$$\begin{aligned} \mathcal{I}^{\text{IV}}_{\text{proj}} &= \Sigma_{zx}^\top T_{\text{IV}}^\top \left(T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top\right)^{-1} T_{\text{IV}}\Sigma_{zx} \\ &= \Sigma_{zx}^\top(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2} P_{T_{\text{IV}}} (\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx}, \end{aligned} \tag{C.4}$$

where $P_{T_{\text{IV}}} = (\Sigma^{\text{IV}}_{\text{raw}})^{1/2}T_{\text{IV}}^\top(T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top)^{-1}T_{\text{IV}}(\Sigma^{\text{IV}}_{\text{raw}})^{1/2}$.

The matrix $P_{T_{\text{IV}}}$ is symmetric. Moreover, $P^2_{T_{\text{IV}}} = P_{T_{\text{IV}}}$, since the middle product $T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top$ cancels with its inverse. Hence, $P_{T_{\text{IV}}}$ is the orthogonal projector onto $\mathcal{C}\{(\Sigma^{\text{IV}}_{\text{raw}})^{1/2}T_{\text{IV}}^\top\}$. It follows that $I_q - P_{T_{\text{IV}}} \succeq 0$.

The raw information matrix can similarly be written as $\mathcal{I}^{\text{IV}}_{\text{raw}} = \Sigma_{zx}^\top(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}I_q(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx}$. Subtracting (C.4) therefore yields

$$\mathcal{I}^{\text{IV}}_{\text{raw}} - \mathcal{I}^{\text{IV}}_{\text{proj}} = \Sigma_{zx}^\top(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}(I_q - P_{T_{\text{IV}}})(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx} \succeq 0. \tag{C.5}$$

Thus, $\mathcal{I}^{\text{IV}}_{\text{proj}} \preceq \mathcal{I}^{\text{IV}}_{\text{raw}}$.

We next translate the information ordering into an asymptotic covariance ordering. Since $Q_{\mathrm{IV}} \in \mathbb{S}^p_{++}$, the projected Jacobian has full rank p. By $\ker(Q_{\mathrm{IV}}) = \ker(\Sigma_{zx})$ from Proposition 12, Σ_{zx} also has full column rank p. Together with positive definiteness of $\Sigma^{\mathrm{IV}}_{\mathrm{raw}}$ and Σ_{IV}, this gives $\mathcal{I}^{\mathrm{IV}}_{\mathrm{raw}}, \mathcal{I}^{\mathrm{IV}}_{\mathrm{proj}} \in \mathbb{S}^p_{++}$. Since inversion reverses the Loewner order on positive definite matrices,

$$V^\star_{\mathrm{raw,IV}} = (\mathcal{I}^{\mathrm{IV}}_{\mathrm{raw}})^{-1} \preceq (\mathcal{I}^{\mathrm{IV}}_{\mathrm{proj}})^{-1} = V^\star_{\mathrm{proj,IV}}.$$

It remains to characterize when projection is information-lossless. By (C.5), $\mathcal{I}^{\mathrm{IV}}_{\mathrm{raw}} = \mathcal{I}^{\mathrm{IV}}_{\mathrm{proj}}$ if and only if

$$(I_q - P_{T_{\mathrm{IV}}})(\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{-1/2}\Sigma_{zx} = 0.$$

Because $P_{T_{\mathrm{IV}}}$ projects onto $\mathcal{C}\{(\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{1/2}T^\top_{\mathrm{IV}}\}$, this is equivalent to

$$\mathcal{C}\left\{(\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{-1/2}\Sigma_{zx}\right\} \subseteq \mathcal{C}\left\{(\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{1/2}T^\top_{\mathrm{IV}}\right\}.$$

Applying the nonsingular transformation $(\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{-1/2}$ to both spaces gives the equivalent condition

$$\mathcal{C}\left\{(\Sigma^{\mathrm{IV}}_{\mathrm{raw}})^{-1}\Sigma_{zx}\right\} \subseteq \mathcal{C}(T^\top_{\mathrm{IV}}).$$

Hence, projection is information-lossless exactly when every efficient raw-IV direction relevant for estimating β lies in the raw-IV subspace retained by T_{IV}. Otherwise, $I_q - P_{T_{\mathrm{IV}}}$ removes a nonzero component of at least one such direction, so the projected system preserves identification but loses efficiency information.

This completes the proof.

□

D Feasible weighting for the projected GMM system

This appendix records the feasible covariance estimator used for the projected GMM system in Section 3.1. Throughout, the stacked population moment remains

$$h_i := \begin{pmatrix} \Pi_z x_i \\ \Pi_w^\perp x_i \end{pmatrix} \in \mathbb{R}^{2p}, \qquad J = \mathbb{E}(h_i x_i^\top) = \begin{pmatrix} Q_{\mathrm{IV}} \\ Q_{\mathrm{SV}} \end{pmatrix},$$

and the feasible sample moment remains

$$\widehat{g}(\theta) = \frac{1}{n}\begin{pmatrix} X^\top P_Z(Y - X\theta) \\ X^\top P_W^\perp(Y - X\theta) \end{pmatrix} \in \mathbb{R}^{2p}.$$

The first block is related to the classical q-dimensional IV moment through

$$\frac{1}{n}X^\top P_Z(Y - X\theta) = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left\{\frac{1}{n}Z^\top(Y - X\theta)\right\}.$$

Thus, the projected IV moment is a p-dimensional transformation of the classical IV moment. When $q > p$, replacing the projected block by the raw q-dimensional IV moment would define a different GMM problem rather than an alternative notation for the same one. We therefore retain the projected formulation throughout.

Proposition 14 (Consistency of the feasible projected-score covariance)**.** Suppose the following conditions hold.

(i) Assumption 3 holds with the fixed first-step weight $M = M_0 \in \mathbb{S}^{2p}_{++}$.

(ii) The quadratic terms entering the plug-in scores have finite second moments:

$$\mathbb{E}\|x_i x_i^\top\|^2_{\mathrm{op}} < \infty, \qquad \mathbb{E}\|z_i z_i^\top\|^2_{\mathrm{op}} < \infty, \qquad \mathbb{E}\|w_i w_i^\top\|^2_{\mathrm{op}} < \infty. \tag{D.1}$$

Let $\widehat{\beta}^{(1)}_{\mathrm{comb}} := \widehat{\beta}(M_0)$ and $\widehat{r}^{(1)}_i := y_i - x_i^\top\widehat{\beta}^{(1)}_{\mathrm{comb}}$. Define $\widehat{\Sigma}_z := n^{-1}\sum_{i=1}^n z_i z_i^\top$, $\widehat{\Sigma}_{xz} := n^{-1}\sum_{i=1}^n x_i z_i^\top$, and $\widehat{m}^{(1)}_z := n^{-1}\sum_{i=1}^n z_i\widehat{r}^{(1)}_i$. Similarly, define $\widehat{\Sigma}_w := n^{-1}\sum_{i=1}^n w_i w_i^\top$, $\widehat{\Sigma}_{xw} := n^{-1}\sum_{i=1}^n x_i w_i^\top$, and $\widehat{m}^{(1)}_w := n^{-1}\sum_{i=1}^n w_i\widehat{r}^{(1)}_i$.

For each observation, let $\widehat{p}_{z,i} := \widehat{\Sigma}_{xz}\widehat{\Sigma}_z^{-1} z_i$ and $\widehat{v}_i := x_i - \widehat{\Sigma}_{xw}\widehat{\Sigma}_w^{-1} w_i$. Define $\widehat{c}_z^{(1)} := \widehat{\Sigma}_z^{-1}\widehat{m}_z^{(1)}$ and $\widehat{c}_w^{(1)} := \widehat{\Sigma}_w^{-1}\widehat{m}_w^{(1)}$, together with $\widehat{e}_{z,i}^{(1)} := \widehat{r}_i^{(1)} - z_i^\top \widehat{c}_z^{(1)}$ and $\widehat{e}_{w,i}^{(1)} := \widehat{r}_i^{(1)} - w_i^\top \widehat{c}_w^{(1)}$.

The corresponding plug-in first-order scores are

$$\widehat{\psi}_{\mathrm{IV},i}^{(1)} := \widehat{p}_{z,i}\widehat{e}_{z,i}^{(1)} + x_i z_i^\top \widehat{c}_z^{(1)}, \qquad \widehat{\psi}_{\mathrm{SV},i}^{(1)} := \widehat{v}_i \widehat{e}_{w,i}^{(1)}.$$

Stack

$$\widehat{\psi}_i^{(1)} := \begin{pmatrix} \widehat{\psi}_{\mathrm{IV},i}^{(1)} \\ \widehat{\psi}_{\mathrm{SV},i}^{(1)} \end{pmatrix} \in \mathbb{R}^{2p}, \qquad \overline{\widehat{\psi}}^{(1)} := \frac{1}{n}\sum_{i=1}^n \widehat{\psi}_i^{(1)},$$

and define

$$\widehat{\Sigma}^{(1)} := \frac{1}{n}\sum_{i=1}^n \left\{\widehat{\psi}_i^{(1)} - \overline{\widehat{\psi}}^{(1)}\right\}\left\{\widehat{\psi}_i^{(1)} - \overline{\widehat{\psi}}^{(1)}\right\}^\top.$$

Then

$$\widehat{\Sigma}^{(1)} \xrightarrow{\mathrm{p}} \Sigma = \mathrm{Var}\begin{pmatrix} \psi_{\mathrm{IV},i} \\ \psi_{\mathrm{SV},i} \end{pmatrix}.$$

If, in addition, $\Sigma \in \mathbb{S}_{++}^{2p}$, then

$$\Pr\{\widehat{\Sigma}^{(1)} \in \mathbb{S}_{++}^{2p}\} \to 1, \qquad (\widehat{\Sigma}^{(1)})^{-1} \xrightarrow{\mathrm{p}} \Sigma^{-1}.$$

D.1 Proof of proposition 14

Proof of proposition 14. By Theorem 5, $\widehat{\beta}_{\mathrm{comb}}^{(1)} \xrightarrow{\mathrm{p}} \beta$. The law of large numbers gives $\widehat{\Sigma}_z \xrightarrow{\mathrm{p}} \Sigma_z$, $\widehat{\Sigma}_{xz} \xrightarrow{\mathrm{p}} \Sigma_{xz}$, $\widehat{\Sigma}_w \xrightarrow{\mathrm{p}} \Sigma_w$, and $\widehat{\Sigma}_{xw} \xrightarrow{\mathrm{p}} \Sigma_{xw}$. Since $\Sigma_z \in \mathbb{S}_{++}^q$ and $\Sigma_w \in \mathbb{S}_{++}^m$, the continuous mapping theorem further gives $\widehat{\Sigma}_z^{-1} \xrightarrow{\mathrm{p}} \Sigma_z^{-1}$ and $\widehat{\Sigma}_w^{-1} \xrightarrow{\mathrm{p}} \Sigma_w^{-1}$.

We next consider the estimated error-projection coefficients. Using $\widehat{r}_i^{(1)} = \varsigma_i - x_i^\top(\widehat{\beta}_{\mathrm{comb}}^{(1)} - \beta)$, we have

$$\widehat{m}_z^{(1)} = \frac{1}{n}\sum_{i=1}^n z_i\varsigma_i - \left(\frac{1}{n}\sum_{i=1}^n z_i x_i^\top\right)\left(\widehat{\beta}_{\mathrm{comb}}^{(1)} - \beta\right) \xrightarrow{\mathrm{p}} \mathbb{E}(z_i\varsigma_i),$$

and, by the same argument, $\widehat{m}_w^{(1)} \xrightarrow{\mathrm{p}} \mathbb{E}(w_i\varsigma_i)$. Recall that $c_z := \Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)$ and $c_w := \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)$, and define $e_{z,i} := \varsigma_i - z_i^\top c_z$ and $e_{w,i} := \varsigma_i - w_i^\top c_w$. It follows that

$$\widehat{\Sigma}_{xz}\widehat{\Sigma}_z^{-1} \xrightarrow{\mathrm{p}} \Sigma_{xz}\Sigma_z^{-1}, \qquad \widehat{\Sigma}_{xw}\widehat{\Sigma}_w^{-1} \xrightarrow{\mathrm{p}} \Sigma_{xw}\Sigma_w^{-1},$$

and $\widehat{c}_z^{(1)} \xrightarrow{\mathrm{p}} c_z$ and $\widehat{c}_w^{(1)} \xrightarrow{\mathrm{p}} c_w$.

These convergences first imply L_2-consistency of the estimated projected regressors and residuals. Indeed,

$$\widehat{p}_{z,i} - \Pi_z x_i = \left(\widehat{\Sigma}_{xz}\widehat{\Sigma}_z^{-1} - \Sigma_{xz}\Sigma_z^{-1}\right) z_i, \qquad \widehat{v}_i - \Pi_w^\perp x_i = -\left(\widehat{\Sigma}_{xw}\widehat{\Sigma}_w^{-1} - \Sigma_{xw}\Sigma_w^{-1}\right) w_i,$$

while

$$\widehat{e}_{z,i}^{(1)} - e_{z,i} = -x_i^\top\left(\widehat{\beta}_{\mathrm{comb}}^{(1)} - \beta\right) - z_i^\top\left(\widehat{c}_z^{(1)} - c_z\right),$$

and analogously

$$\widehat{e}_{w,i}^{(1)} - e_{w,i} = -x_i^\top\left(\widehat{\beta}_{\mathrm{comb}}^{(1)} - \beta\right) - w_i^\top\left(\widehat{c}_w^{(1)} - c_w\right).$$

Therefore, by the law of large numbers,

$$\frac{1}{n}\sum_{i=1}^n \left\{\|\widehat{p}_{z,i} - \Pi_z x_i\|_2^2 + \|\widehat{v}_i - \Pi_w^\perp x_i\|_2^2 + |\widehat{e}_{z,i}^{(1)} - e_{z,i}|^2 + |\widehat{e}_{w,i}^{(1)} - e_{w,i}|^2\right\} \xrightarrow{\mathrm{p}} 0.$$

It remains to show that this plug-in consistency carries over to the first-order scores. For the IV block,

$$\widehat{\psi}_{\mathrm{IV},i}^{(1)} - \psi_{\mathrm{IV},i} = (\widehat{p}_{z,i} - \Pi_z x_i)e_{z,i} + \widehat{p}_{z,i}(\widehat{e}_{z,i}^{(1)} - e_{z,i}) + x_i z_i^\top(\widehat{c}_z^{(1)} - c_z).$$

The first term is controlled by $\mathbb{E}\|z_i e_{z,i}\|_2^2 < \infty$ from Assumption 3(ii). For the second term, the component involving $x_i^\top(\widehat{\beta}^{(1)}_{\mathrm{comb}} - \beta)$ is controlled by $\mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$, while the component involving $z_i^\top(\widehat{c}_z^{(1)} - c_z)$ is controlled by (D.1). The final term is again controlled by $\mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$.

Similarly, for the SV block,

$$\widehat{\psi}^{(1)}_{\mathrm{SV},i} - \psi_{\mathrm{SV},i} = (\widehat{v}_i - \Pi_w^{\perp} x_i) e_{w,i} + \widehat{v}_i(\widehat{e}^{(1)}_{w,i} - e_{w,i}).$$

The first term is controlled by $\mathbb{E}\|w_i e_{w,i}\|_2^2 < \infty$. The second is controlled by $\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty$, together with the $x_i x_i^\top$ and $w_i w_i^\top$ conditions in (D.1). Consequently, the Cauchy–Schwarz inequality and the preceding convergences yield

$$\frac{1}{n}\sum_{i=1}^{n} \left\|\widehat{\psi}_i^{(1)} - \psi_i\right\|_2^2 \xrightarrow{\mathrm{p}} 0, \qquad \psi_i := \begin{pmatrix} \psi_{\mathrm{IV},i} \\ \psi_{\mathrm{SV},i} \end{pmatrix}. \tag{D.2}$$

Under Assumption 3, $\mathbb{E}(\psi_i) = 0_{2p}$ and $\mathbb{E}\|\psi_i\|_2^2 < \infty$. Hence, by the law of large numbers,

$$\frac{1}{n}\sum_{i=1}^{n} \psi_i \psi_i^\top \xrightarrow{\mathrm{p}} \mathbb{E}(\psi_i \psi_i^\top) = \Sigma, \qquad \frac{1}{n}\sum_{i=1}^{n} \psi_i \xrightarrow{\mathrm{p}} 0_{2p}.$$

Moreover, (D.2) and the Cauchy–Schwarz inequality imply

$$\left\| \frac{1}{n}\sum_{i=1}^{n} \widehat{\psi}_i^{(1)} \widehat{\psi}_i^{(1)\top} - \frac{1}{n}\sum_{i=1}^{n} \psi_i \psi_i^\top \right\|_{\mathrm{op}} \xrightarrow{\mathrm{p}} 0$$

and $\overline{\widehat{\psi}}^{(1)} \xrightarrow{\mathrm{p}} 0_{2p}$. Since

$$\widehat{\Sigma}^{(1)} = \frac{1}{n}\sum_{i=1}^{n} \widehat{\psi}_i^{(1)} \widehat{\psi}_i^{(1)\top} - \overline{\widehat{\psi}}^{(1)} \overline{\widehat{\psi}}^{(1)\top},$$

we obtain $\widehat{\Sigma}^{(1)} \xrightarrow{\mathrm{p}} \Sigma$.

Finally, if $\Sigma \in \mathbb{S}_{++}^{2p}$, continuity of the minimum eigenvalue gives $\lambda_{\min}\{\widehat{\Sigma}^{(1)}\} \xrightarrow{\mathrm{p}} \lambda_{\min}(\Sigma) > 0$. Thus, $\Pr\{\widehat{\Sigma}^{(1)} \in \mathbb{S}_{++}^{2p}\} \to 1$, and the continuous mapping theorem yields $(\widehat{\Sigma}^{(1)})^{-1} \xrightarrow{\mathrm{p}} \Sigma^{-1}$.

□

E Consistency of the feasible local-risk-optimal weight

This appendix establishes the consistency of the feasible local-risk-optimal weighting rule introduced in the local-misspecification analysis. Recall that, for an admissible weighting matrix $M \succeq 0$,

$$K_M := (J^\top M J)^{-1} J^\top M, \qquad V_{\mathrm{miss}}(M) := K_M \Sigma_{\mathrm{miss}} K_M^\top,$$

and the population worst-case local risk is

$$R_{\mathcal{C}}(M) := \sup_{\tau \in \mathcal{C}} \|K_M \tau\|_2^2 + \mathrm{tr}\{V_{\mathrm{miss}}(M)\}.$$

Its feasible analogue is constructed from

$$\widehat{J} := \frac{1}{n}\begin{pmatrix} X^\top P_Z X \\ X^\top P_W^{\perp} X \end{pmatrix}, \qquad \widehat{\Sigma}^{(1)}_{\mathrm{miss}},$$

where $\widehat{\Sigma}^{(1)}_{\mathrm{miss}}$ is the plug-in covariance estimator formed from the first-step local-misspecification scores. For each admissible M, define

$$\widehat{K}_M := (\widehat{J}^\top M \widehat{J})^{-1} \widehat{J}^\top M, \qquad \widehat{V}_{\mathrm{miss}}(M) := \widehat{K}_M \widehat{\Sigma}^{(1)}_{\mathrm{miss}} \widehat{K}_M^\top,$$

and

$$\widehat{R}_{\mathcal{C}}(M) := \sup_{\tau \in \mathcal{C}} \|\widehat{K}_M \tau\|_2^2 + \mathrm{tr}\{\widehat{V}_{\mathrm{miss}}(M)\}.$$

Because multiplying M by a positive scalar does not change K_M, the population and feasible risk criteria are invariant to the scale of the weighting matrix. It is therefore sufficient to work with normalized weights. To rule out sequences of normalized weights that approach loss of identification, let $\mathcal{M}$ be a nonempty compact subset of

$$\left\{M \in \mathbb{S}_+^{2p} : \operatorname{tr}(M) = 1\right\}$$

such that, for some constant $\underline{\kappa} > 0$,

$$\inf_{M \in \mathcal{M}} \lambda_{\min}(J^\top M J) \geq \underline{\kappa}. \tag{E.1}$$

The class $\mathcal{M}$ may contain singular weighting matrices; the condition in (E.1) requires only that the weighted Jacobian remain nonsingular.

Define the population minimizer set

$$\mathcal{M}_{\text{opt}} := \operatorname*{argmin}_{M \in \mathcal{M}} R_{\mathcal{C}}(M),$$

and let

$$\widehat{M}_{\text{opt}} \in \operatorname*{argmin}_{M \in \mathcal{M}} \widehat{R}_{\mathcal{C}}(M).$$

When $\mathcal{M}$ contains the normalized solutions of the unrestricted population problem, $\mathcal{M}_{\text{opt}}$ coincides with the corresponding normalized local-risk-optimal weight set.

Proposition 15 (Consistency of the feasible local-risk-optimal weight)**.** Suppose the following conditions hold.

(i) Assumption 4 holds.

(ii) The quadratic-moment conditions in (D.1) hold uniformly along the local sequence.

(iii) The local misspecification set $\mathcal{C} \subset \mathbb{R}^{2p}$ is nonempty and bounded, and the normalized weight class $\mathcal{M}$ is compact and satisfies (E.1).

Then

$$\sup_{M \in \mathcal{M}} \left| \widehat{R}_{\mathcal{C}}(M) - R_{\mathcal{C}}(M) \right| \xrightarrow{\text{p}} 0. \tag{E.2}$$

Consequently,

$$\operatorname{dist}\left(\widehat{M}_{\text{opt}}, \mathcal{M}_{\text{opt}}\right) \xrightarrow{\text{p}} 0, \tag{E.3}$$

where

$$\operatorname{dist}(M, \mathcal{M}_{\text{opt}}) := \inf_{\widetilde{M} \in \mathcal{M}_{\text{opt}}} \|M - \widetilde{M}\|_F.$$

If, in addition, the normalized population minimizer is unique, so that $\mathcal{M}_{\text{opt}} = \{M_{\text{opt}}\}$, then

$$\widehat{M}_{\text{opt}} \xrightarrow{\text{p}} M_{\text{opt}},$$

and

$$\widehat{R}_{\mathcal{C}}(\widehat{M}_{\text{opt}}) \xrightarrow{\text{p}} R_{\mathcal{C}}(M_{\text{opt}}).$$

E.1 Proof of proposition 15

Proof of proposition 15. We first establish consistency of the ingredients entering the feasible risk. By the same law-of-large-numbers and continuous-mapping arguments used for the projected GMM system,

$$\widehat{J} \xrightarrow{\text{p}} J.$$

Moreover, by Theorem 6, the fixed-weight first-step estimator satisfies

$$\widehat{\beta}^{(1)}_{\text{comb}} \xrightarrow{\text{p}} \beta.$$

Under Assumption 4(ii) and the uniform quadratic-moment conditions in (D.1), the same score-by-score plug-in argument used in the proof of Proposition 14 applies uniformly along the local sequence. In particular,

$$\frac{1}{n}\sum_{i=1}^{n}\left\|\widehat{\psi}^{(1)}_{\mathrm{miss},i}-\psi_{\mathrm{miss},i}\right\|_2^2 \xrightarrow{\mathrm{p}} 0.$$

Let

$$\overline{\psi}_{\mathrm{miss}} := \frac{1}{n}\sum_{i=1}^{n}\psi_{\mathrm{miss},i}.$$

By Cauchy–Schwarz and the preceding mean-square convergence, replacing the population local-misspecification scores by their plug-in counterparts has an asymptotically negligible effect on the centered sample covariance:

$$\left\|\widehat{\Sigma}^{(1)}_{\mathrm{miss}}-\frac{1}{n}\sum_{i=1}^{n}\left(\psi_{\mathrm{miss},i}-\overline{\psi}_{\mathrm{miss}}\right)\left(\psi_{\mathrm{miss},i}-\overline{\psi}_{\mathrm{miss}}\right)^{\top}\right\|_{\mathrm{op}} \xrightarrow{\mathrm{p}} 0.$$

Along the local sequence,

$$\mathbb{E}(\psi_{\mathrm{miss},i})=\frac{\tau}{\sqrt{n}}, \qquad \mathrm{Var}(\psi_{\mathrm{miss},i})=\Sigma_{\mathrm{miss},n}.$$

Hence the triangular-array law of large numbers implied by Assumption 4(iii) gives

$$\frac{1}{n}\sum_{i=1}^{n}\left(\psi_{\mathrm{miss},i}-\overline{\psi}_{\mathrm{miss}}\right)\left(\psi_{\mathrm{miss},i}-\overline{\psi}_{\mathrm{miss}}\right)^{\top}-\Sigma_{\mathrm{miss},n} \xrightarrow{\mathrm{p}} 0.$$

Finally, Assumption 4(iii) also gives

$$\|\Sigma_{\mathrm{miss},n}-\Sigma_{\mathrm{miss}}\|_{\mathrm{op}}\to 0.$$

Combining the preceding displays yields

$$\widehat{\Sigma}^{(1)}_{\mathrm{miss}} \xrightarrow{\mathrm{p}} \Sigma_{\mathrm{miss}}.$$

Thus, it remains to show that these plug-in convergences continue to hold uniformly after the weighting matrix is allowed to vary over $\mathcal{M}$.

Since $M \succeq 0$ and $\mathrm{tr}(M)=1$ for every $M\in\mathcal{M}$, we have $\|M\|_{\mathrm{op}}\le 1$. Therefore,

$$\sup_{M\in\mathcal{M}}\left\|\widehat{J}^{\top}M\widehat{J}-J^{\top}MJ\right\|_{\mathrm{op}} \le \|\widehat{J}-J\|_{\mathrm{op}}\left(\|\widehat{J}\|_{\mathrm{op}}+\|J\|_{\mathrm{op}}\right) \xrightarrow{\mathrm{p}} 0.$$

Together with (E.1), continuity of the minimum eigenvalue implies that, with probability approaching one,

$$\inf_{M\in\mathcal{M}}\lambda_{\min}(\widehat{J}^{\top}M\widehat{J}) \ge \frac{\kappa}{2}.$$

Hence all matrices $\widehat{J}^{\top}M\widehat{J}$ are invertible uniformly over $M\in\mathcal{M}$ with probability approaching one, and their inverses are uniformly bounded.

Using the identity

$$A^{-1}-B^{-1}=A^{-1}(B-A)B^{-1},$$

we obtain

$$\sup_{M\in\mathcal{M}}\left\|(\widehat{J}^{\top}M\widehat{J})^{-1}-(J^{\top}MJ)^{-1}\right\|_{\mathrm{op}} \xrightarrow{\mathrm{p}} 0.$$

Since

$$\widehat{K}_M=(\widehat{J}^{\top}M\widehat{J})^{-1}\widehat{J}^{\top}M, \qquad K_M=(J^{\top}MJ)^{-1}J^{\top}M,$$

the preceding convergence and $\widehat{J}\xrightarrow{\mathrm{p}}J$ yield

$$\sup_{M\in\mathcal{M}}\|\widehat{K}_M-K_M\|_{\mathrm{op}} \xrightarrow{\mathrm{p}} 0. \tag{E.4}$$

The same bounds imply

$$\sup_{M\in\mathcal{M}}\|\widehat{K}_M\|_{\mathrm{op}}=O_p(1), \qquad \sup_{M\in\mathcal{M}}\|K_M\|_{\mathrm{op}}<\infty.$$

We next consider the squared local-bias component. Because $\mathcal{C}$ is bounded,

$$\sup_{\tau\in\mathcal{C}}\|\tau\|_2<\infty.$$

For any $M\in\mathcal{M}$ and $\tau\in\mathcal{C}$,

$$\left|\|\widehat{K}_M\tau\|_2^2-\|K_M\tau\|_2^2\right|\le\|(\widehat{K}_M-K_M)\tau\|_2\left(\|\widehat{K}_M\tau\|_2+\|K_M\tau\|_2\right).$$

Therefore, (E.4) and the uniform boundedness of $\widehat{K}_M$ and K_M imply

$$\sup_{M\in\mathcal{M}}\left|\sup_{\tau\in\mathcal{C}}\|\widehat{K}_M\tau\|_2^2-\sup_{\tau\in\mathcal{C}}\|K_M\tau\|_2^2\right|\xrightarrow{\mathrm{p}}0. \tag{E.5}$$

For the variance component, write

$$\widehat{V}_{\mathrm{miss}}(M)-V_{\mathrm{miss}}(M)=\widehat{K}_M\left(\widehat{\Sigma}_{\mathrm{miss}}^{(1)}-\Sigma_{\mathrm{miss}}\right)\widehat{K}_M^\top+(\widehat{K}_M-K_M)\Sigma_{\mathrm{miss}}\widehat{K}_M^\top+K_M\Sigma_{\mathrm{miss}}(\widehat{K}_M-K_M)^\top.$$

Since $\widehat{\Sigma}_{\mathrm{miss}}^{(1)}\xrightarrow{\mathrm{p}}\Sigma_{\mathrm{miss}}$, together with (E.4), we obtain

$$\sup_{M\in\mathcal{M}}\left\|\widehat{V}_{\mathrm{miss}}(M)-V_{\mathrm{miss}}(M)\right\|_{\mathrm{op}}\xrightarrow{\mathrm{p}}0.$$

Because p is fixed, this further gives

$$\sup_{M\in\mathcal{M}}\left|\mathrm{tr}\{\widehat{V}_{\mathrm{miss}}(M)\}-\mathrm{tr}\{V_{\mathrm{miss}}(M)\}\right|\xrightarrow{\mathrm{p}}0. \tag{E.6}$$

Combining (E.5) and (E.6) proves (E.2).

It remains to translate uniform convergence of the risk criterion into consistency of its minimizer. Because $\mathcal{M}$ is compact and $R_{\mathcal{C}}(M)$ is continuous on $\mathcal{M}$, the set $\mathcal{M}_{\mathrm{opt}}$ is nonempty and compact.

Fix any $\epsilon>0$, and define

$$\mathcal{M}_\epsilon:=\{M\in\mathcal{M}:\mathrm{dist}(M,\mathcal{M}_{\mathrm{opt}})\ge\epsilon\}.$$

The set $\mathcal{M}_\epsilon$ is compact and contains no population minimizer. Hence continuity of $R_{\mathcal{C}}$ implies that there exists $\eta_\epsilon>0$ such that

$$\inf_{M\in\mathcal{M}_\epsilon}R_{\mathcal{C}}(M)\ge\min_{M\in\mathcal{M}}R_{\mathcal{C}}(M)+\eta_\epsilon. \tag{E.7}$$

By (E.2), with probability approaching one,

$$\sup_{M\in\mathcal{M}}\left|\widehat{R}_{\mathcal{C}}(M)-R_{\mathcal{C}}(M)\right|<\frac{\eta_\epsilon}{3}.$$

On this event, every $M\in\mathcal{M}_\epsilon$ has feasible risk strictly larger than the feasible risk at any population minimizer. Indeed, for any $M_{\mathrm{opt}}\in\mathcal{M}_{\mathrm{opt}}$,

$$\widehat{R}_{\mathcal{C}}(M)>R_{\mathcal{C}}(M_{\mathrm{opt}})+\frac{2\eta_\epsilon}{3},\qquad M\in\mathcal{M}_\epsilon,$$

whereas

$$\widehat{R}_{\mathcal{C}}(M_{\mathrm{opt}})<R_{\mathcal{C}}(M_{\mathrm{opt}})+\frac{\eta_\epsilon}{3}.$$

Therefore a minimizer $\widehat{M}_{\mathrm{opt}}$ of the feasible criterion cannot belong to $\mathcal{M}_\epsilon$. Since $\epsilon>0$ was arbitrary,

$$\mathrm{dist}\left(\widehat{M}_{\mathrm{opt}},\mathcal{M}_{\mathrm{opt}}\right)\xrightarrow{\mathrm{p}}0,$$

which proves (E.3).

If the normalized population minimizer is unique, so that $\mathcal{M}_{\mathrm{opt}}=\{M_{\mathrm{opt}}\}$, the preceding result reduces to

$$\widehat{M}_{\mathrm{opt}}\xrightarrow{\mathrm{p}}M_{\mathrm{opt}}.$$

Finally, choose any $M_{\text{opt}} \in \mathcal{M}_{\text{opt}}$. By optimality of $\widehat{M}_{\text{opt}}$,

$$\widehat{R}_{\mathcal{C}}(\widehat{M}_{\text{opt}}) \leq \widehat{R}_{\mathcal{C}}(M_{\text{opt}}).$$

Moreover, (E.2) gives

$$\left|\widehat{R}_{\mathcal{C}}(\widehat{M}_{\text{opt}}) - R_{\mathcal{C}}(\widehat{M}_{\text{opt}})\right| \xrightarrow{\text{p}} 0$$

and

$$\left|\widehat{R}_{\mathcal{C}}(M_{\text{opt}}) - R_{\mathcal{C}}(M_{\text{opt}})\right| \xrightarrow{\text{p}} 0.$$

Together with

$$R_{\mathcal{C}}(\widehat{M}_{\text{opt}}) \geq R_{\mathcal{C}}(M_{\text{opt}}),$$

we obtain

$$\widehat{R}_{\mathcal{C}}(\widehat{M}_{\text{opt}}) \xrightarrow{\text{p}} R_{\mathcal{C}}(M_{\text{opt}}),$$

which completes the proof.

□